Organ der Gesellschaft für Informatik e.V.
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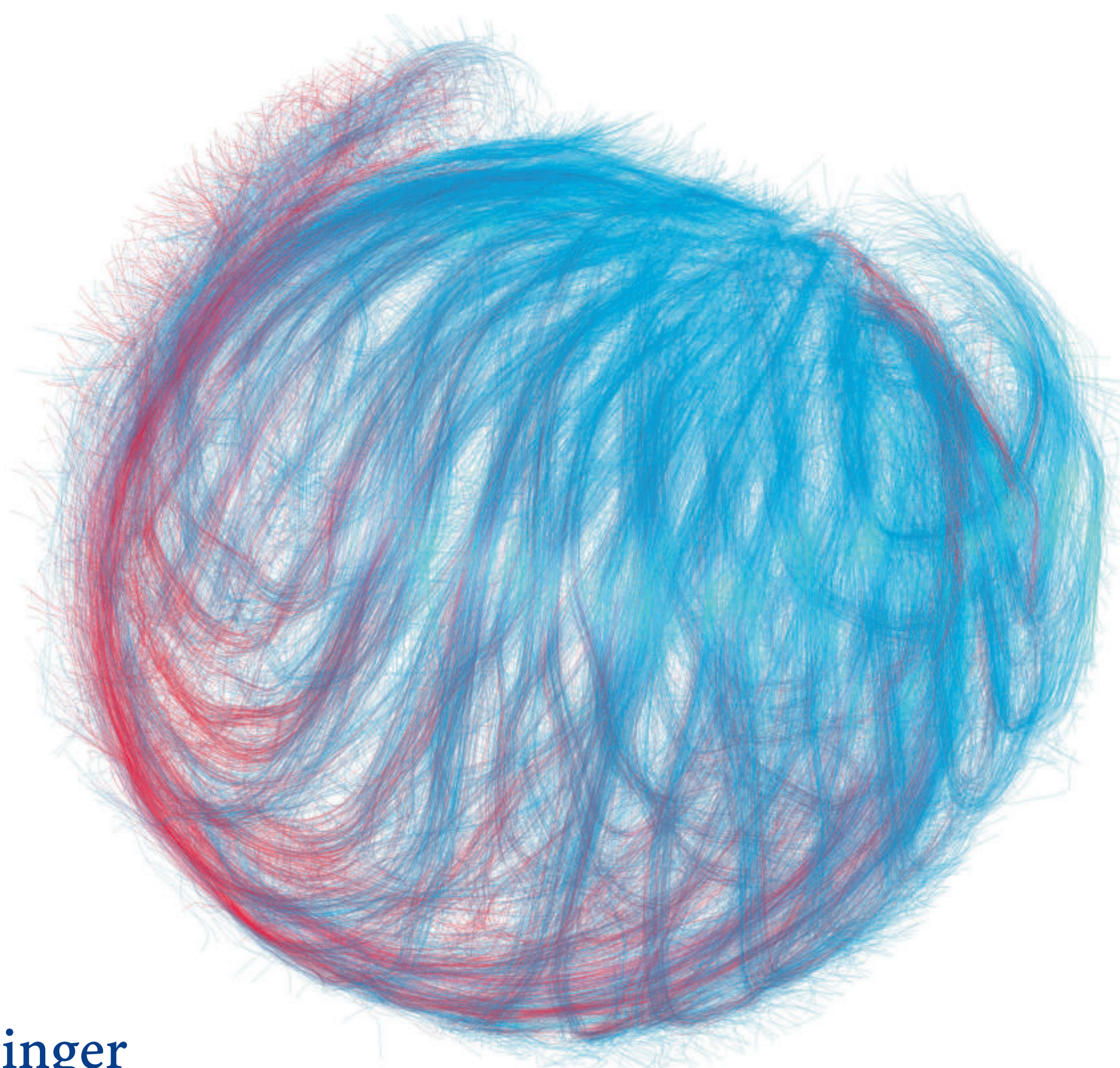
Informatik Spektrum

Demokratie in Zeiten des Internets



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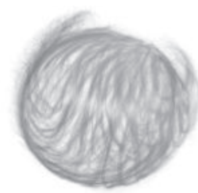

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Redaktion:
Peter Pagel, Vanessa Keinert
Tel.: +49 611 787 8329
e-mail: Peter.Pagel@springer.com

Herstellung:
Philipp Kammerer,
e-mail: Philipp.Kammerer@springer.com

Redaktion GI-Mitteilungen:
Cornelia Winter
Gesellschaft f¨ur Informatik e.V. (GI)
Wissenschaftszentrum,
Ahrstraße 45, D-53175 Bonn,
Tel.: +49 228-302-145, Fax: +49 228-302-167,
Internet: http://www. gi.de,
e-mail: gs@gi.de

Wissenschaftliche Kommunikation:
Anzeigen: Eva Hanenberg
Abraham-Lincoln-Straße 46
65189 Wiesbaden
Tel.: +49 (0)611/78 78-226
Fax: +49 (0)611/78 78-430
eva.hanenberg@springer.com

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EDITORIAL

Editorisches Interview: Demokratie in Zeiten des Internets

Peter Pagel[1] · Edy Portmann[1] · Joël Vogt[1]

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Das Internet prägt heute unser aller Leben, im Privaten wie im Beruflichen. Chefredakteur Peter Pagel sprach mit den Gastherausgebern dieses Schwerpunkts, Edy Portmann und Joël Vogt.

Peter Pagel: Welche Auswirkungen hat das Internet auf Demokratien?

Edy Portmann: Diese Frage adressiert unsere englischsprachige Mitherausgeberin Patricia Hetter Kelso, als US-Politikwissenschaftlerin und Witwe des berühmten US-Nationalökonomen Louis Kelso vom Kelso Institut, aus welchem viele der Texte in dieser Ausgabe des Informatik Spektrums ihre Kraft ziehen, in ihrem Essay auf sehr überzeugende Art und Weise. Auf Louis Kelso, den Pionier des Employee Stock Ownership Plans (ESOP), der es arbeitenden Menschen ermöglicht, Aktien ihrer Firma zu kaufen und diese aus zukünftigen Dividendenrendite zu bezahlen, geht auch der zweite Mitherausgeber, Joël Vogt, einem großen Kenner des Internets der Dinge sowie Experte für Beteiligungsformen im Internet, in seinem Artikel ein.

Joël Vogt: Ich habe diese Frage oft mit Patricia Kelso diskutiert. Dabei hat sie auch die Frage in den Raum gestellt, ob „Internet“ und „Demokratie“ überhaupt vereinbar sind. Ich denke schon, aber nicht in der aktuellen Form. Wir dürfen nicht vergessen, dass das Internet ein US-Militär Projekt zur Informationssammlung ist – Verteilung und Verwaltung. Und es wird heute mehr als je dazu verwendet; und nicht nur von Nachrichtendiensten, wo es immerhin oft Checks and Balances gibt, sondern auch von Privatunternehmen wie zum Beispiel Facebook, wo jedoch nicht nach nationaler Sicherheit optimiert wird, sondern nach Profit und Marktdominanz. Wir haben lange den Wert unserer persönlichen Daten nicht verstanden und vielleicht auch nicht realisiert, dass die digitale und physische Welt immer mehr zusammenwachsen. Daten und Information sind Macht, sie ermöglichen oder zerstören ein demokratisches System.

Es ist kaum zehn Jahre her, als sich das Internet als Plattform für unsere Demokratie bewiesen hat. Die flexible und robuste Architektur unseres Internets, sowie die Benutzerfreundlichkeit des Webs (inkl. Tools und Protokolle) erlaubte etwa Aktivisten im Nahen Osten, sich zu organisieren und so Diktaturen zu stürzen. Ein paar Jahre später stellen wir aber mit Schrecken fest, dass ein Laissez-faire System auch eine grosse Bedrohung für die Demokratie und Menschenrechte sein kann: Brexit in Großbritannien, Trump in USA, der Genozid in Myanmar (Burma) wäre ohne das Internet, oder ohne gewisse Plattformen, nicht passiert. Wir dürfen nicht vergessen, dass unser Internetarchitektur zwar dezentral ist, aber Dienste sich, dank dem Netzwerkeffekt, sehr schnell auf wenigen Plattformen konzentrieren. Das heisst, Plattformen haben reellen Nutzen. Das Problem ist aber, dass anders als in einer Demokratie, Plattformen keine Gewaltentrennung kennen. Und wie zum Beispiel Mark Zuckerberg immer wieder zeigt, keine Rechenschaft ablegen müssen. Und obwohl grosse Plattformen oft zu öffentlichen Vorsorgeunternehmen wurden, haben wir Nichts davon.

Louis und Patricia Kelso haben genau zu diesem Zweck demokratisch-kapitalistische Finanzierungsmethoden entwickelt, welche dies ermöglichen. Wir dürfen aber auf keinen Fall das Internet überregulieren – dann ist es kein Internet mehr – aber auch auf keinen Fall es sich selbst überlassen. Denn die freie Marktwirtschaft blüht nur, wenn faire Rahmen- und Wettbewerbsbedingungen herrschen. Dazu braucht es aber klare und verständliche Regeln. Und da der Wert des Internets sowohl von Entrepreneuren und Steuergelder wie auch von Benutzern generiert werden, müssten Nutzer auch für ihren Beitrag entlohnt werden. Daten sind digitales Gold sowie der Rohstoff für maschinelles Lernen und künstliche Intelligenz. Wir haben also alle ein Interesse daran, dass diese Quelle sprudelt. Dienste werden dank KI besser, Wähler können sich informieren, Kunden werden für die Verwendung ihrer Daten entlohnt, Innovatoren und Forscher können ihre Ideen als Startups verwirklichen – und so weiter. Ich bin optimistisch, dass

✉ Peter Pagel
peter.pagel@springer.com

1 Springer Fachmedien Wiesbaden GmbH, Wiesbaden, Deutschland

wir das Internet so ausbalancieren können, dass alle davon profitieren. Und ich muss optimistisch sein, denn die Alternativen machen mir Angst.

Edy Portmann: So geht es mir auch. Heutige System können (noch) tendenziös sein, was unterschiedliche Ursachen haben kann: Künstliche Intelligenz und ihr zugehöriges maschinelles Lernen benötigen eben Trainingsdaten, um Vorhersagen machen zu können. Aus dem Internet gesammelte Daten können aber befangen, verzerrt, unvollständig sowie ungenau – also unscharf – sein. Zum Beispiel klassifizierte vor ein paar Jahren ein Google-Algorithmus schwarze Menschen als Gorillas. Trotz grossen Anstrengungen war Googles bedrückende Lösung schlussendlich, den Bilderkennungsalgorithmus daran zu hindern, Gorillas zu erkennen.

Peter Pagel: Sharing Economy ist eines der gängigen Schlagworte, die mit dem Internet verknüpft sind, wie unterscheidet sich die Fiktion von der Wirklichkeit?

Edy Portmann: Das Konzept der Sharing Economy ist bestechend einfach: Alles, was nicht permanent genutzt wird, kann von uns Benutzern vermietet werden. Über eine Online-Plattform wird der Kontakt zwischen Anbieter und Nachfrager hergestellt. Die dem Konzept potentiell inhärente künstliche Intelligenz wird dabei immer stärker als Allzweckmittel und -technologie begriffen. Ähnlich der Elektrizität kann ihre Auswirkungen über alle Aspekte unseres Lebens verfolgt werden: von Use Cases im Gesundheitswesen, in der Industrie, im Finanzwesen und vielen mehr. Deshalb forscht beispielsweise Uber daran, wie sie ihre Daten nutzen könnten, um ihre Fahrer auch zu weniger rentablen Fahrten zu bewegen, um uns damit ein besseres Benutzererlebnis zu bieten.

Joël Vogt: Zurück zu Ihrer Frage von Fiktion oder Vision: So wie ich das verstehe, ist die Sharing Economy noch in den Kinderschuhen. Airbnb oder Couchsurfing sind erfolgreiche Beispiele; Uber wurde noch unter dem Vorwand der Sharing Economy lanciert, ist aber mehr ein Taxi-Unternehmen. Ich bin von der Sharing Economy begeistert, aber wir müssen sicher sein, dass wir nicht einfach Subscription-Modelle aus der digitalen Welt in die physische Welt übertragen. Für mich geht es bei einer Sharing Economy um Bürger, die ihre Ressourcen teilen und dafür entlohnt werden wolle. Als Beispiel: wenn ich mein Fahrzeug einer Sharing Economy zur Verfügung stelle, werde ich als Bürger und Kapitaleigentümer „empowered“. Wenn ich aber ein selbstfahrendes Fahrzeug von Uber zum Bahnhof verwende, ist dies keine richtige Sharing Economy, sondern eine sehr gefährliche Konzentration von Kapital in den Händen eines Großunternehmens. Kapitalismus funktioniert nur, wenn wir alle Kapital haben. Konzentration von Kapital ist für mich kein wirklicher Kapitalismus – so war die Situation vor Adam Smith.

Edy Portmann: Angesichts der angestrebten Datenmonopole fragte sich der Kolumnist Philip Stephens einmal, wie man den Kapitalismus vor den Kapitalisten schützen könnte. Seine Erkenntnis: Kapitalismus erfordert Legitimität, er kann sich langfristig nur entfalten, wenn er das Wohlergehen von uns Bürgern im Auge hat.

Tech-Utopisten, die etwa in der künstlichen Intelligenz eine Neuauflage der unsichtbare Hand Adam Smiths sehen, weisen häufig darauf hin, dass überhaupt erst aus der Digitalisierung eine Sharing Economy gewachsen sei – und das stimmt! Es ist jedoch sehr wichtig, die Dynamik dahinter zu verstehen, die der modernen Sharing Economy inhärent ist. Wir stellen dabei fest, dass das Internet eine immer wichtigere Rolle in der Informations- und Machtasymmetrie spielt, die zwischen den Organisationen, die diese Dienste anbieten, sowie den Nutzern, die sie befüllen, bestehen. In einer meiner letzten Kolumnen in der Luzerner Zeitung, für welche ich regelmäßig schreibe, beschäftigte ich mich mit der digitalen Metamorphose unserer öffentlichen Infrastruktur. Dabei fragte ich mich, wie es wohl wäre, wenn Airbnb- oder Uber-ähnliche Plattformen im Besitz der Netzwerkindustrien wie der Post oder Telekommunikation wären und diese zum Wohle von uns verwalten würde.

In diesem Schwerpunkt des Informatik Spektrums präsentiert der Medienwissenschaftler Nathan Schneider Lösungen, indem er Louis Kelsos ESOP-Gedanken mit dem Internet erweitert. Ihm gemäss suchen heutige Onlineplattformen und auch Startups nämlich nach intelligenten Antworten auf Eigentumsfragen. Sein Artikel legt Strategien dar, welche den Sharing-Plattformen wie Airbnb, Lyft oder Uber als mögliche Vertrauensmechanismen respektive Enabler einer umfassenden, demokratischen Verantwortung uns Benutzern gegenüber dienen können. Zu diesem Zweck stützt er sich auf gemachte Erfahrungen mit (C)SOP-Mitarbeiterbeteiligung sowie weitere, innovative Arten von Eigentumsstrukturen. Nutzerorientiertes Vertrauen im Internet kann eine Governance und Gewinnbeteiligung unter den Beteiligten ermöglichen.

Peter Pagel: Vertrauen ist ein wichtiger Aspekt bei allem, was Menschen im Internet tun, kann die Blockchain-Technik da hilfreich sein?

Edy Portmann: Als möglichen Vertrauensmeachanismus fürs Internet stellt Nathan Schneider eben Alternativen, wie (C)SOP vor. Eine weitere Möglichkeit eruiert er zudem auch in der Blockchain-Technologie. Toni Caradonna von der Swiss Blockchain Federation, die sich mit der Schaffung von Rechtssicherheit sowie guten Voraussetzungen für das Schweizer Blockchain-Ökosystem einsetzt, geht in seinem Artikel genau dieser Technologie und deren Einsatz für die Gesellschaft auf den Grund. Dazu beleuchtet er deren Ursprung und stellt sie in den Kontext von drei Digitalisierungswellen: In der ersten geht es um die Digitalisierung von Informationen, die mit dem Aufkommen des Internet verdeutlicht wird, in der zweiten, manifestiert in und durch Social Media, um deren Communities und, in der dritten um Werten, repräsentiert durch Blockchain-Technologie. Er zeigt auf, wie Blockchain als neue Internettechnologie die Eigenschaften der Digitalisierung der ersten Wellen, wie etwa die Zentralisierung, die durch technologieunabhängige Geschäftsmodelle entstand, überwinden kann.

Peter Pagel: Könnte man sich Online-Marktplätze vorstellen, die besser zu Demokratien passen, als das bei den heutigen der Fall ist?

Joël Vogt: Absolut! Ich würde behaupten, heutige Online-Marktplätze kreieren zu schnell Machtverhältnisse, welche den Verkäufer und auch den Kunden schaden können. Und es ist, wegen dem Netzwerkeffekt, eine Illusion, dass man einfach einen Marktplatz wechseln kann. Amazon ist zum Beispiel Marktplatz, Katalog, Verkäufer, Marktforschungsinstitut und Aufsicht in einem. Es ist bekannt, dass Amazon mit anderen Verkäufern konkurrenziert. Das können sie, da sie einfach mögliche Konkurrenten überwachen können und diese dann unterbieten. Diese haben aber keine Alternativen, Amazon ist ein zu wichtiger Kanal im westlichen Teil des Internets. Klagen können sie nicht – bei wem auch? Amazon?

Da Online-Marktplätze ein unentbehrlicher Teil der Geschäftstätigkeiten von Verkäufern sind, frage ich mich in meinem Artikel, wieso diese Unternehmen nicht gleich Eigentümer werden sollten. Louis Kelso hat dazu die CSOP erfunden. Bemerkung: Bei ESOP werden Angestellte Teilhaber ihrer Firma, bei CSOP können Benutzer und Kunden Teilhaber eines wichtigen Lieferanten, oder in unserem Fall eben eines Online-Marktplatzes werden. Verkäufer müssen sowieso Mitgliedergebühren zahlen, in einer CSOP würden diese mit der Zeit zu Teilhaber-Wertpapiere führen. Ich würde aber nicht einzigen bei einer Plattform bleiben, sondern eine Föderation von verknüpften Online-Marktplätze bilden, damit beispielsweise Unternehmer in China auf ihre Art und Weise die Plattform gestalten, aber trotzdem Kunden in den USA beliefern können. Der Verkauf könnte dabei über die US-Plattform abgewickelt werden. Somit profitieren die Eigentümer der US-Plattform, also die amerikanischen Verkäufer.

Ich erhoffe mir durch solche Maßnahmen eine Wirkung gegen Protektionismus. Vielleicht fragen sie sich nun, wieso ich nicht zuerst an Konsumenten gedacht habe. Das hatte ich eigentlich, aber nachdem Facebook (zusammen mit Cambridge Analytica) unsere Wahl gehackt hat und danach auch diese in den USA, stellte ich fest, dass selbst in London, wo ich wohne, viele den Wert von persönlichen Daten (noch) unterschätzen. Mir wurde mir klar, dass die Zeit für eine CSOP für Konsumenten noch nicht reif ist.

Edy Portmann: Meines Erachtens ist Joël Vogts Beitrag für uns eine praktikable Form, auf welche ich in einer anderen LZ-Kolumne aufmerksam machte, in der ich mich fragte, wie wir uns vor Überwachungskapitalisten aus dem Silicon Valley schützen könnten. Ihr Kapitalismus ruft laut US-Ökonomin Shoshana Zuboff in uns das ungute Gefühl aus, mit welchem uns die Silicon Valley-Giganten als Nutzer ihrer Plattformen heute eben überwachen. Heutige KI-Systeme profitiert von der Digitalisierung unserer Lebenswelt – dem Internet der Dinge sei etwa Dank –, die uns dazu gebracht hat, den Überwachungsapparat überhaupt erst zu entwickeln und welcher sowohl aus dem privaten wie auch aus dem öffentlichen Sektor gespeist wird. Immer mehr Organisationen verlassen sich darauf, Informationen zu sammeln, um ein scheinbar besseres Erlebnis zu ermöglichen. Zudem haben auch Regierungen im Namen der nationalen Sicherheit angefangen, gegen unsere Freiheiten und auf unser Recht auf Privatsphäre vorzugehen, wie etwa Aussagen von Informanten wie Edward Snowden, einem US-amerikanischer Whistleblower und ehemaligen CIA-Mitarbeiter, belegen.

Peter Pagel: Brauchen wir einen Systemwechsel?

Joël Vogt: Ja, wenn wir weiterhin in einer Demokratie leben wollen, müssen wir endlich verstehen, was Louis Kelso als Binary-Economics bezeichnet. Wir schaffen Vermögen nicht nur durch unsere Arbeitsleistung, sondern auch – und dass immer stärker und schneller –, durch Kapital. Wenn wir unser Vermögen nur durch Arbeitsleistung (er)schaffen wollen, werden wir immer ärmer, denn produktives Kapital verbessert sich exponentiell. Eine gesunde Demokratie braucht aber Wählerinnen und Wähler, welche wirtschaftlich eine solide Basis haben. Unsere Daten sind, wie gesagt, der Rohstoff des 21. Jahrhunderts, das digitale Gold. Daten ermöglichen eben maschinelles Lernen und künstliche Intelligenz. Je mehr Daten, desto fähiger – und produktiver – wird künstliche Intelligenz. Wenn wir unsere Daten einfach gratis weggeben, vergeben wir auch ein Teil unserer Stimme in der Demokratie. Facebook hätte nicht Kundendaten für politische Zwecke missbrauchen können, wären „ihre Kunden“ auch Inhaber ihrer eigenen Daten. Aber wenn wir unsere Daten als Kapital verwalten und investieren können, geht es noch weiter als individuelle Wahlen. Wir können sicherstellen, dass Startups nicht von Datensätze ausgeschlossen werden und dadurch Innovation fördern. Oder wir können auch die Evolution des Internets mitgestalten indem wir entscheiden, in welche Unternehmen oder Themen wir unsere Daten investieren wollen.

Edy Portmann: Louis Kelso soll einmal gesagt haben, dass „die Menschen sich weigern, ein Problem zu erkennen, bis sie eine Lösung zu diesem Problem sehen". So ging es mir hinsichtlich der in diesem Schwerpunktheft dargelegten Themen auch. Deshalb lege ich zum Abschluss dieses Heftes zur Demokratien im Zeitalter des Internets, als Abrundung, meinen Reifungsprozess als Schlusspunkt dar. Als ich mich nämlich vor ungefähr einem Jahr mit meinen beiden Mitherausgebern unterhielt, wusste ich noch nicht, welche Prozesse dies in mir auslösen würde. Als Informatiker kümmerte ich mich nämlich bis anhin noch relativ wenig um Technologie-Regulation. Das hat sich unterdessen dahingehend verändert, als dass ich mich auf europäischem Level mit der Regulation von Daten und zugehörigem maschinellem Lernen beteilige. Dies beleuchte ich in meinem Schlusspunkt.

Peter Pagel: Herr Portmann, Herr Vogt, besten Dank für das Gespräch. Es bleibt also spannend.

Anhang

Zum Titelbild

Die Abbildung zeigt rund 12.000 Zell-Tracks, welche die Entwicklung eines Zebrafisch-Embryos in den ersten zwölf Stunden nach Befruchtung darstellen. Neben der Bildaufnahmetechnik (Lichtblattmikroskopie), welche hochauflösende Einblicke in biologische Prozesse ermöglicht, ist besonders interessant, dass die Ergebnisse mit aktuellen Web-Technologien für jedermann erkundbar sind. Dank unseres Visualisierungstools kann jeder interessierte Wissenschaftler die biologischen Resultate der Publikation selbst im Browser seines Computers oder Smartphones betrachten und wird dabei durch verschiedene Methoden der Visualisierung, bspw. Edge Bundling, unterstützt.

Multi-scale imaging and analysis identify pan-embryo cell dynamics of germlayer formation in zebrafish Gopi Shah, Konstantin Thierbach, Benjamin Schmid, Johannes Waschke, Anna Reade, Mario Hlawitschka, Ingo Roeder, Nico Scherf & Jan Huisken.https://www.nature.com/articles/s41467-019-13625-0

Peter Pagel

Edy Portmann

Joël Vogt

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HAUPTBEITRAG

Foreword

Patricia H. Kelso[1]

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Abstract
Marshall McLuhan thought that Johannes Gutenberg's invention of movable type created the modern world. What kind of society will the Internet create? Will it be a democracy or a surveillance state, like Eastern Germany under the Stasi? We cannot have both! The Internet was developed by the U.S. Government during the Cold War to spy on foreign governments and to identify potential dissidents at home. It was explicitly designed as an instrument of espionage and control. But that was not what its early-stage originators intended, before the government took over. The original inventors were romantic peaceniks who believed that they were creating a good thing for humankind. The powerful in every age have tried to control access to information. Can the new generation of electronic technology experts devise ways to realign the Internet so that it can serve democracy?

Democracies are notoriously slow in responding to danger. They dither and dawdle until an advancing peril is not only at the door but ringing the bell. When they finally get around to it, however, democracies can mobilize quickly, efficiently and carry the day.

The Orwellian tendencies of electronic technology, particularly the Internet, have long been evident to a few. Their early warnings were ignored by a public fixated on the next new thing from Silicon Valley and the euphoric hype emanating from the industry and its boosters. However, now we are having second thoughts. We are asking questions about the dark side of the electronic future. We are not sure whether this future will include democracy.

True, western political systems, although called democracies, are combinations of political democracy and economic oligarchy. They could be described as systems where the many have the vote and the few have the money. The citizens of a quasi-democracy, however, may especially need accurate facts and information in order to retain the modest political power they possess.

From Gutenberg's invention of movable type in the mid-sixteenth century to the computer and the Internet of today, technology has been improving the means of disseminating information. Throughout this history, powerful individuals and institutions have tried to control the information communicated—the content of the information and those who have access to it. To control the dissemination of information is to control what people know, learn, and think. Thought control is not only a perk of power but a vital means of holding on to it.

As the Gutenberg Revolution began to replace handwritten manuscripts with relatively cheap, printed books, the dominant power of the day, the Roman Catholic Church, redoubled its efforts to control the dissemination of knowledge and information. Yet, despite the Index, persecution of heretics and clerical threats of hell, books proliferated throughout Europe and the world. The Renaissance happened, then the Enlightenment, then the Reformation. Revolutions revamped the political order. The Common Man emerged from the long feudal night. Democracy was rediscovered. Voilá, the modern world! Marshall McLuhan credits Johannes Gutenberg's invention of movable type for creating it.

If printing created the modern world, what world will the computer and the Internet create? How will electronic technology affect democracy and the constitutional rights and freedoms, which we take for granted, but which history warns us are rare and brief?

Printing and the Internet are radically different media. They have different histories, tendencies, and social effects. Printing is decentralized; its products flow outward to who knows where. It is democratic. Anyone with the requisite skills can set up a printing press or access one. The Internet, however, is centralized. It processes and disseminates information through gatekeepers, the powerful companies, espe-

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✉ Patricia H. Kelso
kelsoinstitute@aol.com

[1] Kelso Institute for the Study of Economic Systems, 872 North Point, San Francisco, CA 94109, USA

cially social media. Its users, unlike the readers of books, have names and identities. These users leave tracks in the form of data. Describing the Internet as a web suggests a spider at the center, as does its product, Spidergram, the surveillance chart. The spider owns the web and the insects caught in it.

Printing and the Internet differ in another important respect. Printing is a mechanical process. Press, ink, and paper are the product of human brains and hands. However, only the infrastructure of the Internet is attributable to humans. It is designed to access and utilize a subparticle of electricity to transmit information. Electricity is an element of nature. In a democracy, natural elements essential to life—water, fire, gas, electricity, atoms—may be utilized, subject to regulation, but not exclusively owned or monopolized.

Only the infrastructure itself, invented and created by humans, is *ownable*. The electronic infrastructure is a physical construction. It is designed to access, capture, and utilize a natural element. It is like the pipes and storage tanks of, say, a municipal water system, or the cables and wires of a telephone exchange. There is another similarity: the computer and the Internet have become essentials of modern life; they are now a perquisite of citizenship and of civilization itself. Not to have access to the electronic world is to be denied participation in one's society. It is felt to be a form of deprivation, like being without indoor plumbing, a telephone, radio, or television.

In short, the Internet now provides an essential *public service*. It has become a *utility*. Access to utilities is a democratic human right.

Electronic technology, however, was not invented to bolster either democracy or constitutional rights. The investigative reporter Yasha Levine reminds us in his book *Surveillance Valley* that the Internet was developed by the American military in the aftermath of World War II. The Cold War was heating up; the military wanted new technology in order to spy on foreign governments and possibly subversive US citizens. The military wanted to find out what people all over the world were thinking, doing, and might do. The rationale, as usual, was national security.

The Internet took years to develop at the cost of many billions of dollars. American taxpayers paid this cost through contracts approved by the US Congress. Under Capitalism, if you own the apple tree, you also own the apples the tree produces. Moreover, if you finance the development of a thing, you own the thing developed. So, the American public ended up owning the Internet, right? Wrong! The public was not even aware that the Internet was being privatized. The Government did it on the sly. The beneficiaries of taxpayer largesse included International Business Machines Corporation (IBM), the Massachusetts Institute of Technology (MIT), and AT&T Corporation (AT&T); also several new companies that the Government founded as privatization vehicles for the new technology. These companies are now the darlings of Wall Street. We consumers know their names by heart. We should, because we pay them every month in order to utilize electronic services, including the Internet.

The surveillance state

Marshall McLuhan gave us a metaphor for the electronically-connected world of the future: *global village*. In a small community, everyone knows everyone, and lives are an open book. Neighbors know whose bathroom light was switched on at 3 a.m., and who does not show up at church. Secrets are few. Everyone lives in fear of the mythical gossipmonger, Mrs. Grundy.

Yet what if Mrs. Grundy were a government agency equipped with the latest see-all, hear-all snoop technology?

In East Berlin there is a small museum devoted to life as it was when East Germany was part of the Soviet empire and controlled by the Stasi. American visitors might laugh at the crude listening devices, as large and intrusive as telephone receivers, which the Stasi inserted into walls, furniture, automobiles, and restaurant cubicles in order to eavesdrop on the private conversations of East German citizens. The German film, *The Lives of Others*, depicts the personal and social costs of living in a surveillance society, where the Government uses your deepest, most personal secrets to dominate and manipulate you. However, electronic technology infinitely multiplies the surveillance power that the Stasi had then.

Jean-Louis Gassée, a former Apple computer executive, describes the social media platform Facebook as a "surveillance machine." In an article on Peter Thiel and his dark web Palantir, *Forbes* magazine comments: "All human relations are a matter of record, ready to be revealed by a clever algorithm. Everyone is a spidergram now."

The surveillance state is on a collision course with democracy. We cannot have both.

Is democracy necessary?

We humans did not evolve in democracies. Our instincts, habits, and moral sense were formed over eons in small authoritarian hierarchies organized from the top down. This is the kind of society most of the world's people live in today. Democracy is an acquired taste.

The now emerging Global Village will be composed of billions of inhabitants, a community of strangers electronically connected. Eventually, it will include almost everyone on the planet: young and old, literate and illiterate, educated and unschooled, city-bred and tribal, rich and poor. This new electronic world will in theory be value-free. The Internet has no cultural, racial, sexual, religious, or political bias. It is a machine to which Krantzberg's First Law of Technology applies: "Technology is neither good nor bad. Neither is it neutral." In other words, technology is inseparable from the human beings who designed it and make use of it.

The inhabitants of the Global Village are humans. Most are under authoritarian rule. They represent many different interests, traditions, and customs, which they cherish or consider divinely ordained. They also have long histories of wars, conflicts, feuds, and rivalries with other members of the electronic community with whom they are now in instant and intimate contact.

So, how will diversity on a global scale work out?

In the Surveillance State there is no such thing as privacy. Today's world is only partially connected. Westerners who value privacy and can afford it may still flee to big cities like New York, London, or Paris. Yet most of the world's billions live in villages under repressive governments. They have never experienced the privacy which the Industrial Revolution gave to the West and that the electronic revolution now threatens to take away. Privacy, like democracy, may be an acquired taste, but once acquired it is hard to give up.

How will privacy fare when the Global Village encompasses the entire planet? When big cities, small towns, rural farmsteads, trailer parks, and homeless camps are all connected by electronic technology? The Global Village will have no privacy, no place to escape Big Brother.

Yet what if electronic technology has made Big Brother redundant? Your cell phone monitors you 24 h a day; your computer likewise. Street cameras photograph you as you pass by. Facial recognition technology identifies your unique visage from caches of official photographs going back to when you were a kid at camp.

As for social media companies, surveillance is their reason for existing. Scooping up your most intimate data to sell to advertisers is their business model, the source of their profits and their founders' wealth.

The US Government developed the Internet to surveil. S*urveil* is a euphemism for *spy on*. Why, then, are we surprised that the Internet is being used for the purpose for which it was explicitly designed? Or that American citizens are among those who are surveiled?

There is another social problem created by the Internet: anonymity. Users may conceal their identity or assume any identity they choose. Imagine the Internet as a masked ball to which the entire world is invited. People in masks behave differently from those with open faces. This is a sensitive problem, particularly for Americans, because the first amendment of the US Constitution protects anonymous writing and speech.

How will people earn a living when artificial intelligence and its robot offspring finish the job the Newcomen engine began, namely, substituting a force of nature for the labor power of beasts of burden and human beings? In the mid-seventeenth century, the Newcomen engine, the first steam engine, was set to work draining water from English copper mines. We do not know the fate of the 500 men who previously did that job with buckets. Were they deployed in other jobs or did they attack the Newcomen engine like the Luddites did the power loom when it destroyed their elite crafts in the textile industry?

These concerns are not theoretical. They are already affecting our economic and political life. They will be addressed, but not solved, by governments and the courts.

Might these problems have technical solutions that could align them with democracy?

Hackers, cyberpunks, and other romantics—the pioneers of electronic technology before the government took over—believed that they were working for the common good. They were anti-authoritarian peaceniks, spiritual siblings of those who had opposed the Vietnam War. They were dedicated non-conformists and philosophical anarchists, who believed that the best government was none or that which governed least. They dismissed society's power hierarchies as "The Man." So how did these free spirits end up working for their anathema? How were they co-opted by the Pentagon?

The short answer is economic power. Brilliant and original ideas require money to morph into commercial products. Those who have brilliant ideas are usually poor. Enter the outside investor! The angel! However, as the Internet had no practical potential except as a surveillance tool for the government, the government took on the role of venture capitalist. Money was no problem—taxpayers have deep pockets. The new companies created by the government through selective privatization collected the personal information that the government could access at will. In a way, they served as the government's front men. Moreover, there was frosting on this cake! *Individuals would subsidize their own surveillance!* As users and customers, they now pay the government-created front companies to collect the personal data they themselves supply!

Yet, who owns this data? The individual who provides it or the company that collects it? Our answer to this question will determine the future of democracy in the electronic age. Did the Stasi "own" the information its agents collected from spying on East German citizens? It certainly made use of it! However, thieves make use of the property they steal. Ownership is *legal* possession. What is the legal status of spying? Do people as individuals own their personal information, i.e., is it *intellectual property*? Does selling data without permission of its owner constitute theft?

What would the Internet pioneers—those free spirits who disdained both The Man and the military—think of the Internet as it exists today? What if their second-generation descendants could reprivatize the Internet? Might they invent a way to do it, so that it serves a democratic society? Is it possible to rescue our democracy from the Surveillance State?

Democracies are notoriously slow in responding to danger. However, when they finally get around to it, democracies can mobilize quickly and efficiently. Democracy can carry the day.

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HAUPTBEITRAG

User trusts: broad-based ownership for online platforms

Nathan Schneider[1]

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Abstract
This essay introduces what promise a novel broad-based capital strategy—trusts serving platform users—might hold for the online economy, especially as an enabler of more widespread, organized, and democratic user accountability. It draws on lessons from the experience of employee ownership alongside emerging opportunities for other kinds of broad-based ownership structures. User-oriented trusts could enable meaningful co-governance and profit sharing among essential stakeholders, a prospect that merits research and experimentation.

Crisis and craving

The companies formerly known as the "sharing economy"—a term mostly abandoned for want of actual sharing—have begun sharing ownership. As early as 2017, executives at the ride-sharing platform Uber reportedly met with US Securities and Exchange Commission officials about granting company stock to their independent-contractor drivers [1] (Current policy enables stock distributions mainly to formal employees.) Airbnb, the home-rental giant, filed a formal request to that effect with the SEC the following September [2], and Uber did the same just weeks later [3]. By February 2019, the *Wall Street Journal* reported that, with their initial public offerings approaching, Uber and its competitor Lyft had found a way to work around SEC rules by issuing their drivers cash bonuses that recipients could choose to invest in stock [4, 5].

I confess to having felt a sense of vindication from these developments. In May 2017, I helped bring a grassroots shareholder proposal to Twitter's annual meeting, which invited the then-embattled company to study possibilities for user ownership [6–8]. At the time, the company tried to persuade the SEC to bar the proposal outright [9] and, upon being overruled, advised shareholders to vote against it [10]. However, less than 2 years later, suddenly, some of Silicon Valley's biggest companies were trying to convince the SEC to let them turn users into owners.

✉ Nathan Schneider
nathan.schneider@colorado.edu

1 University of Colorado, Boulder, USA

The direction in which Airbnb, Lyft, and Uber have been gesturing could amount to much more than a scheme for boosting their initial public offerings, especially if they end up competing to offer ever-better deals. An accountability crisis has been ricocheting across the online economy in recent years; platform CEOs have been summoned to Washington for congressional hearings and reprimands, while politicians threaten leading companies with antitrust action, regulation, and even nationalization. Prominent scholars have documented causes for concern about surveillance [11, 12], unjust discrimination [13, 14], monopolistic practices [15, 16], and labor abuses [17, 18]. The proposed remedies generally involve either demanding better behavior within existing company structures, or government regulation along the lines of Europe's General Data Protection Regulation. Comparatively few proposals have sought to change to whom these companies are structurally accountable and who shares in their profits.

The Twitter shareholder campaign emerged out of a phenomenon known as "platform cooperativism," which calls for transforming the online economy through the cooperative business tradition [17, 19, 20]. Platform cooperativism has generated enthusiasm but encounters considerable challenges, as cooperatives are incompatible with much of the existing financial and legal infrastructure in tech hubs like Silicon Valley [21]. However, I will suggest here that, alongside cooperativism, those seeking to shift the accountability of the online economy should draw from the legacy of broad-based capital ownership through employee stock-ownership plans, which during the last quarter of the twentieth century were responsible for a rapid expansion of rank-and-file employee ownership in the United States [22]. While cooperatives seek to bypass the capitalist corporate structure altogether, broad-based ownership schemes devise means of enabling widespread, equitable participation within the capitalist corporate paradigm.

Evidently, the online economy needs better options to take it through this well-earned accountability crisis. With the promise of government regulation comes the danger of compliance burdens that could crowd out newcomers, especially as multiple jurisdictions simultaneously attempt to regulate the intrinsically transnational Internet. User ownership could be a means, alongside appropriate regulation, of ensuring democratic self-governance and equitable profit-sharing from within companies themselves. However, techniques for achieving this are lacking. The platform companies that sought SEC approval for even modest distributions of equity to their contractors had to devise workarounds short of their goals. Earlier-stage tech startups, meanwhile, have been calling for means of accessing capital that do not demand exponential returns for outside investors [23]. Both the biggest platforms and the newest startups are looking for more creative answers to the question of ownership.

This essay introduces what promise one such broad-based capital strategy—trusts serving platform users—might hold for the online economy, especially as an enabler of more widespread, organized, and democratic user accountability. It draws on lessons from the experience of employee ownership alongside emerging opportunities for other kinds of broad-based ownership structures. User-oriented trusts could enable meaningful co-governance and profit sharing among essential stakeholders, a prospect that merits research and experimentation. The time may be coming for a renaissance of ownership designs [24] to help address the crises of accountability in the online economy, which otherwise have no end in sight. Like the proposal for Twitter shareholders, this is a call for further study toward that renaissance.

"Extending the ESOP revolution"

Lawyer, investment banker, and polemicist Louis O. Kelso began honing his lifelong economic theory while serving as a navy intelligence officer in Panama during World War II [25]. Nonhuman capital, he concluded, contributes to production alongside labor, and thus full citizenship in the economy should involve not just employment but capital ownership as well. As capital increases its share in the productive matrix through technology, he reasoned, ownership can begin to eclipse people's dependence on employment altogether [26–28]. Kelso and his collaborators—notably, popular philosopher Mortimer Adler and political scientist Patricia Hetter Kelso—used various terms for this proposition and what it entails, such as two-factor theory, universal capitalism, democratic capitalism, and binary economics. However, regardless of its name, with Kelso's idea came an agenda for dramatically expanding participation in capital ownership.

"The practical implication of Kelso's insights," Patricia Kelso explained after her husband's death, "is that we cannot efficiently produce goods and services through two factors while distributing income through just one" [29].

Kelso proposed a series of means by which the income generated through capital ownership could become more widely available. The proposals generally involved trust entities that could manage capital on behalf of beneficiaries and acquire new capital, such as company stock populis, with financing based on the expectation of future productivity. It was a populist kind of leveraged buyout, before the technique became popular among corporate raiders. The most successful version of this gambit, and the only one Kelso was able to significantly advance before his death in 1991, was the employee stock-ownership plan, or ESOP. Kelso first demonstrated the model in 1956, and after he secured its enshrinement in the US employee-benefits law in 1974, the number of companies with an ESOP rapidly grew into the thousands. As a retirement plan, employee-owned stock is paid for by the employer, not out of the employee's pocket. Boards, bankers, retiring owners, and employees alike found reasons to embrace the model, particularly in certain types of firms best poised to benefit from the mix of costs and incentives. The National Employee Ownership Center now counts more than 14 million ESOP participants, through around seven thousand companies, in the United States [30].

Kelso's achievement suggests how, with a relatively minor policy accommodation, a market can emerge and spread to a potentially transformative practice within capitalist companies. ESOPs were able to proliferate much faster and grow much larger, for instance, than worker cooperatives, which today in the United States number just several hundred, with several thousand employees among them [31]. The ESOP's trust structure enables its parent company to obtain financing on its behalf, while consolidating oversight and most governance powers in a single trustee [32]. The company's ability to make tax-deductible contributions to an ESOP, and the ability to defer taxes on the stock sold to it, helps make it cost-competitive even for directors who do not value employee ownership for its own sake. ESOPs have been used for such purposes as ownership succession, increasing employee engagement, self-financing firm growth, private equity turnarounds, and bolstering share value. ESOPs can operate within several kinds of existing capitalist company forms, both closely held and publicly traded.

The ESOP formula, however, comes with a set of shortcomings, which have contributed to the leveling-off of ESOP growth since the 1990s [33]. The balance of costs and benefits has contained ESOPs for particular kinds of corporate contexts (such as medium-sized, consistently profitable engineering firms) and kept them out of others (such as venture-backed tech startups, where stock-option plans are more common) [32]. Although ESOP law includes provisions designed to protect employees' interests, the employer-managed nature of the model has introduced opportunities for abuse [33]. Even those protections tend to privilege maximizing financial returns over other values that employees might hold, including the continued existence of the ESOP itself.

The fact that the ESOP came into law through the Employee Retirement Income Security Act of 1974 introduced certain limitations. It consigns the model to service primarily as a means for retirement savings, with features that make it cumbersome and unattractive in many contexts. Perhaps most significantly for the context of the online economy, it presumes the relevant stakeholder class to be formal employees. That is, it cannot be applied to the Airbnb, Lyft, and Uber user-contractors, much less the unpaid, value-creating users of a platform like Twitter. Uber, for instance, claims to contract with 3 million drivers, compared to only 16,000 formal employees [34].

The ESOP was never meant to be the only sort of broad-based ownership on hand. Louis and Patricia Kelso [27] referred to Louis' most famous invention as "the Trojan Horse for democratizing American capitalism"—a starting point, not the destination. In their final book, on "extending the ESOP revolution," they proposed eight distinct kinds of capital ownership plans. Given the degree to which the ever more pervasive online economy relies on nonemployee drivers, hosts, content moderators, custodians, warehouse staff, and content creators [35]—along with the rise of contingent work across the economy at large [36]—it may be time to move beyond the Trojan Horse, beyond the ESOP.

User trusts

Louis Kelso's second best-developed proposal—the consumer stock ownership plan or CSOP—would be a plausible next step. Kelso and Kelso refer to it as "the capitalist equivalent of the socialist-derived producer cooperative" [27], with features of a consumer cooperative as well. He developed his primary CSOP experiment starting in 1957, with a farmer-owned fertilizer provider in California called Valley Nitrogen. The primary usefulness of CSOPs would be in cases of natural monopolies, such as power companies, or long-term consumer relationships, like grocery stores [27, 37]. They could operate in tandem with an ESOP for the company's workers. The Valley Nitrogen experiment ran aground due to changes in tax law, but with proper enabling legislation, Kelso envisioned that it could function, and access financing, much like an ESOP. Jens Lowitzsch, who holds the Kelso Professorship at Viadrina European University, has proposed CSOPs for municipal water systems [38] and renewable energy [39].

Although the conflation of consumer and producer stakeholders might seem to muddy the model, it proves prescient for the online economy. Multisided online platforms frequently blur the roles of consumer and producer in the ambiguous designation of the user—who might be an app-based delivery worker, an amateur video creator on a social-media platform, a resident of an area with only one broadband provider, or a smartwatch wearer whose biometric data is collected in a profitable database. The CSOP logic could have transformative relevance for each of these. The delivery worker could accumulate a long-term capital stake alongside irregular gigs; the video creator might weigh in on decisions that affect the craft; the monopoly-broadband customer might have a way of demanding fair terms of service; the smartwatch wearer could have additional assurance, as a co-owner, that the company will not misuse personal data. User trusts of these sorts would take into account that our economic lives do not begin and end in fixed employment relations. Such trusts could become the sole owner of a company, a majority shareholder, or the holder of just a small minority of shares.

When we consider a broad-based ownership trust separate from the confines of employment law, we lose the current tax advantages of the ESOP, but we gain considerable opportunities—opportunities that may be deserving of their own legal recognition and incentives to encourage their use. For instance, a trust that is not primarily a retirement plan allows more leeway for participants to intentionally craft its mission. Developments in trust legislation since Kelso's heyday enable perpetual purpose trusts that serve a stated purpose beyond value maximization for a set of beneficiaries [40, 41]. Some user trusts, such as those for gig workers or broadband customers, could emphasize economic returns for participants through stock ownership. For startups, these might resemble stock options as a means of incentivizing early adoption. However, for trusts whose participants may be making smaller or hard-to-calculate economic contributions to the business, such as the social-media personalities or smartwatch users, the trust could primarily be an advocate for user voice and values in corporate governance, devoting stock proceeds to fund such advocacy rather than distributing them among participants. Trusts oriented toward governance might be issued special classes of shares that have only voting rights, with or without financial rights.

Although Kelso imagined future dividends as the main financing mechanism for his plans, ESOPs have typically depended on tax-advantaged employee-benefit contributions [42]; especially since tech companies have tended to direct resources to growth rather than dividends, other means of funding a user trust are likely necessary. This could occur as a form of equity compensation or through a modified stock buyback. Like stock options in a startup, the trust might be oriented toward a future liquidity event, enabling users to benefit alongside founders and early investors. In some cases, again like stock options, there may be an expectation that users participate in the purchase financially themselves.

One way or another, a user trust should hold a stake sufficient to make a difference in the lives of its participants. It should also matter in a way distinct from simply owning shares in a public company. One of Twitter's arguments against our shareholder proposal was that users could (and already do) buy shares in the company [10]. We wanted something else, however—a relationship tied to our actual use of the platform, a kind of ownership not merely of individuals but of a collective able to wield collective influence. As a tool for delivering economic returns, a user trust could purchase shares on behalf of the vast majority of users who never would or could invest in the company with their own resources. As a mechanism of governance, such a trust could enable otherwise-atomized users and shareholders to speak with one voice on issues of common concern.

ESOPs have the capacity to involve employee participation in corporate governance, especially for decisions about liquidating the trust. However, to allay management's fears of meddling underlings, and to keep focus on the task of running a retirement program, active employee governance (such as in regularly electing board members) has been relatively rare. Such a conservative approach may serve some user trusts well. Already, Google's parent Alphabet proposes to use a "data trust" vehicle for stewarding data gathered by its "smart city" urban sensors, with the intent of distancing control from both the company and the population subject to its surveillance [43, 44]. The founder of the Chinese electric vehicle company NIO has put 50 million shares into a user trust that can deliver economic returns but no governance powers [45]. However, in other cases, companies and employees have come to see active participation in governance as something to be encouraged; research on ESOPs, for instance, suggests that combining ownership and inclusive governance can have positive effects on productivity [22], and ESOP companies have sought to reap those benefits through practices like open-book management. Platform companies might similarly see value in cultivating highly participatory user trusts. Large user communities could draw on methods such as quadratic voting [46], sortition [47], and liquid democracy [48] to enable efficient, inclusive participation.

Consider Facebook, for instance. As a company now habitually embattled in accountability crises, it could be a valuable sign of goodwill to adopt a user-governance regime with power at the level of a large bloc of shares and even board representation. CEO Mark Zuckerberg himself has written about the need for forms of user governance on the platform [49]. He has also committed to relinquishing nearly his entire stake in the company to fund charitable endeavors. Perhaps a leveraged user trust would be a convenient vehicle for addressing the challenges of accountability, governance, and divestment all at once.

User trusts of various kinds could become a new norm in the online economy—for startups seeking to draw early adopters and buy out their investors, or for larger platforms maturing into roles as near-monopoly utilities. Companies would then have to compete with each other to offer more and more attractive, responsible trust arrangements to win the loyalty of potential users. A market could emerge for private-equity firms and others to finance CSOP formation. As with the ESOP, all this could take place within conventional capitalist companies, in tandem with existing financial infrastructure. Through both shared wealth and shared governance, online platforms could operate less like the investor fiefdoms that are so common today and more like laboratories of democracy.

Further action-research

Louis Kelso's crusade to bring capital ownership to more people's economic lives may be even more relevant today than in his time. The divide between those who do and do not own capital is an important driver of widening wealth inequality [50]. Forms of accessible co-ownership could be an antidote to chronic wage stagnation and anxieties about job-replacing automation. Strategies beyond the ESOP, like the CSOP and other "SOPs" [37], do not assume that fixed employment is our sole form of economic life, or the sole form relevant to ownership. At a time when the former expectation (at least among privileged classes) of a lifelong salary with benefits has frayed, Kelsoism presents a schematic for new mechanisms of widespread economic security, even economic democracy. This legacy bears relevance for models gaining momentum today, such as social wealth funds, universal basic income, platform cooperativism, and benefit corporations. Yet there is much still to be learned about whether and how Kelso's sketches might someday come to life on the platforms that orchestrate the twenty-first-century economy.

The idea of user trusts for online platforms remains speculative, in need of further investigation and experimentation. For instance, some areas of further inquiry should include:

- Applicable trust structures and other vehicles (such as stock-holding cooperatives [51, 52] and blockchain networks [53]), across various jurisdictions and legal contexts

- Policy interventions to encourage the development of a market for user trusts, attractive to both platform companies and their users
- Additional policies that might obligate certain kinds of utility-scale companies to include user trusts (similar to laws requiring worker codetermination for corporations over a certain size)
- Challenges of and opportunities for serving the multinational user bases of transnational platforms
- Necessary protections, including legal constraints and rights to organize, that shield user-trust participants from deception, manipulation, and abuse
- Business models for financing CSOP formation, such as through private-equity and other investment firms
- Relevant guardrails for startups seeking to retain user ownership as an exit option for early investors and founders.

One thing the ESOP legacy suggests is that research alone will not be nearly as instructive as conducting such investigations in the field of practice, the way Louis Kelso and his associates did. Trying out new experiments with company equity might seem like a lot to ask, except that even companies with much to lose like Airbnb, Lyft, and Uber already seem willing to attempt programs that do so. Regulators, meanwhile, are grasping for better ideas about how to bring online platforms to account. Users deserve better options, too.

According to Patricia Kelso, Louis Kelso frequently observed a pattern among those around him: "People refuse to recognize a problem, he said, until they see its solution" [29]. Opinion research has found widespread feelings of powerlessness and resignation with how online platforms gather, monetize, and manipulate personal data [54]. Users have little choice but to accept platforms' terms of use as given. User trusts, or any other structural contrivance, cannot pretend to be *the* solution to the bewildering array of problems that the online economy presently poses. Democracy is never a drop-in fix for anything or everything. It is, rather, a manner of taking responsibility, and of wielding the power that responsibility requires. Having a tool of this sort on hand might allow more people to finally regard the online economy's problems with agency rather than resignation, because now they can be part of finding solutions.

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Democracy and the language of power

Rob Stone[1]

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Abstract
Data is a language spoken and transcribed by those without literacy. A form of capital produced by those without knowledge of its production. Our data has become a language owned as a capital. The concentration of ownership over "data capital" has created an unprecedented asymmetry of information and economic power; which calls into question the viability of a political and economic democracy. Meanwhile, we are presented with a singular yet temporary opportunity to take back ownership and control over the new language of power by claiming ownership over the productive power of the data we collectively serve to create. The following essay seeks to examine (a) the distinct nature of current technology platforms (b) the network effects at work within such platforms (c) the implications of replacing a platfrom owned by "absenstee" shareholders with one owned by those who comprise the platform/network.

Introduction

The following article examines and introduces the prospects for an actionable framework to create a more rational, productive, and democratic political economy—one that calls for fundamental *system change*. The next system can be built on top of the prevailing logic of the current system through its redirection and reorientation. That is to say, the current system has come to produce all the factors required for its succession.

The degree and distribution of individual and relative power determines the nature of a society's or civilization's democracy. To *increase* democracy is to increase the degree of power and agency held by its citizens *while* concurrently maintaining (controlling for) the relative equality of power citizens wield *vis-à-vis* one another.

Data is *the language of power* of the 21st century and will define and dictate the future course of our democracy. Data is at once a new form of language, capital, *and* power, the ownership of which conveys (and will in the future increasingly convey) a degree of power upon its proprietor(s) that is unparalleled in human history. The *Age of the Internet* will only be an *Era of Democracy* if we individually and collectively reclaim the rational and rightful ownership over our data *as a form of personal capital.*

✉ Rob Stone
rob@datavest.org

1 800 Sawmill Road, Cedar Grove, NC 27231, USA

The formation of capital, language, and history

History depends upon storytelling. The nature of our power and the character of our politics are products of the stories we tell. Mankind's first formation of capital was human language—the first and most foundational means of production that has ever been developed, a form of capital wielding the power and *the means to produce history itself*. With a simple story, told *and* recorded, the *historical arrow of time* of our civilization was able to commence.

We define a civilization's stage of development based upon its application of language: prehistoric–protohistoric–historic. Civilizations with stories unrecorded are considered *prehistoric;* civilizations with stories told and transcribed only by others are *protohistoric*; while a *historic* civilization is one that had/has the knowledge and power to tell and record their own story.

Modern (human) civilization faces the prospect of devolving back into a protohistoric state of existence. Today, the explicit ownership over data as a capital asset is unprecedented, representing the first language capable of being owned as a revenue-generating capital asset; a language poised and perfectly positioned to become the prepotent language of the future; a process driven by the fact that data conceptualized as language will soon dwarf the collective capacity of all current and previous languages in terms of the sheer quantity of information it maintains the ability to collect, record, and communicate.

The ability to author our story and therefore direct our future will depend upon our individual and collective ability to obtain ownership and control over the new language of power—ownership over the data capital we currently produce.

The economic function and political purpose of capital

The distribution of capital ownership is a useful proxy for the distribution of social and political power within a society. As such, the economic returns and political influence capital ownership affords necessitates that any proposed remedy deal primarily with the medium and mechanism(s) of its accumulation. Understanding the fundamental nature and purpose of capital is of critical importance, if we are to direct its productive capacity in a way that maximizes the value it is capable of creating [4].

The information economy continues to magnify the return and productivity attributable to capital relative to labor. The degree of today's technological disruption is unprecedented and only accelerating. The power and productivity attributable to capital has, by now, safely solidified its preeminent position as the primary factor of production: AI, machine learning, big data, exponential technologies, automation. The list goes on, all converging, all compounding the potency of their respective productive capacities.

The accelerating returns attributable to technology are currently accruing at a non-linear and therefore unintuitive pace. The profound speed of change has diminished our perceived ability to understand the implications of current economic trends, as well as the perceived agency we have to direct their course. However, amidst the noise and increasing complexity, we have collectively lost sight of more foundational matters related to the actual social purpose we have come to ascribe to technology.

Capitalism is capable of defining the economic function of technology, but not its intended social or political purpose. Its economic function is quite clear: to increase the productive capacity of capital.

We must depend upon our politics if we intend to attach purpose to economic function. Doing so requires that we address, ask, and answer normative questions; questions that remain largely by design beyond the scope of conventional economic models and modes of inquiry; questions that reflect upon what ought to be—political questions, democratic questions, if we so choose.

Democracy is most often defined as a system of government where power is either held by elected representatives or directly by the people themselves, formed for the purpose of increasing the degree of power and agency that individuals have to direct their lives and influence their environment.

If we can agree upon the general tenor of the definition above, the demands placed upon us by democracy requires that we reconcile the social purpose of technology with its economic function; more broadly, that we reconcile the social purpose of capital with its objective function within our economy.

Own the residual

Capitalism rests upon the premise that an individual's ownership over capital reflects the distribution of his or her contribution(s) to the formation of capital. However, the *technological residual* represents the increase in aggregate productivity not statistically explained by traditional capital and labor inputs alone. This "residual" often constitutes the core and most predictive variable, representing the laboriously built scaffolding of all accumulated knowledge and technological advancement achieved and bestowed upon us by our progenitors, *which no one person today can claim just credit for creating*. Similar to the "capital" inheritance and development of language evading our ability to ascribe individual credit for its formation, so it is with other intangibles similar in nature; including the *data residual* [1].

Despite the broad contributions responsible for driving the productive capacity of the residual, the result has been a singularly concentrated distribution of ownership over the benefits of the productive returns they have enabled. The residual is based upon the original promise of capitalism that claimed to be capable of accurately ascribing credit through ownership, that those who currently have ownership, now claim credit—credit not only for the capital they create but for the magnified returns that the data and technological residual imbues upon it. Factors that remain "residual" by nature ultimately subsume to capital.

Own the capital

> *"The basic moral problem that faces man as he moves into the age of automation, the age of accelerating conquest of nature, is whether he is really fit to live in an industrial society; whether his institutions will adjust rapidly enough; whether he will rivet himself with an absurd institution like full economic order when it is not only unnecessary but unadministratable in anything but a slave society; whether freed from the necessity to devote his brain to the production of goods and services, he can address himself to the work of civilization itself."*

The reality of the economic trends outlined inform a critique and suggest a solution centered upon capital and the nature of its ownership. Current labor creates technology capable of reducing future labor. This capacity is the product of technology's economic function. The application and unfolding of this capacity will be the product of its politically informed purpose—democratic, or otherwise. Should we decide to choose democracy, the productive capacity of Civilization must serve the purpose of enhancing the power, agency, and freedom of its citizens. Democracy cannot survive unless labor replaced by capital can acquire capital capable of replacing income.

Kelso believed that by increasing the number of capital owners, and therefore the number of individuals entitled to the productive returns generated by such capital, individuals would over time be freed from the necessity of undertaking work for the sole purpose of ensuring their basic subsistence, and free to engage in what Louis Kelso poetically referred to as *the work of civilization itself.*

While the current policy debate remains focused on determining the most effective means to increase employment, the primary challenge we face today is not one of figuring out how technology can make more work for more people and for higher pay. The challenge and opportunity is to identify the means by which individuals can gain ownership over the capital and technology that is gradually reducing or replacing the requirement of their labor [2].

Own the machine

In 1965, Louis and Patricia Kelso posed the following (yet to be adequately addressed) question:

"Are machines supposed to make work?"

Knowledge and technology acquire economic form as the machines of our productive enterprises. Machines are owned as capital; therefore, those who own capital, own the machines, including the full potency and productivity of the knowledge and technology contained within them—ownership that necessarily conveys not only economic power but also social agency upon its proprietor(s).

Historically, machines have been physical extensions of the human mind, products of the human intellect, the corporeal scaffolding of all knowledge and technology, both acquired and inherited, while human language and knowledge previously coalesced to become the physical machines and tools of industry, i.e., capital. These machines are now in the process of dematerializing back, this time—into a digital existence still subsumed under capital. The machine becomes data, and data—a form of capital.

Imagine a hypothetical future where a fully automated machine operating within a "zero marginal cost society" has been developed that is capable of producing and delivering all goods and services mankind could ever need or want [6]. Within this theoretical world, algorithms fueled by Big Data *could* be calibrated to precisely allocate resources across the economy so as to maximize the utility and well-being of each person based upon their distinct needs, wants, and preferences.

Without the requirement of labor and the absence of recompense via wages and salaries, the relative allocation of goods and services to which a given individual within the economy is entitled would be determined based upon their relative ownership stake in the machine. The nature of an automated and data driven economy, therefore, would depend upon the degree of ownership that individuals were able to accrue over the capital (over the machine) prior to the full replacement of the requirement of their labor [3].

Own the data

The monetary returns attributable to personal data are accruing not to the individuals producing it, but to the shareholders of the enterprises that are able to extract, aggregate, and therefore capitalize upon its value. Currently, the largest and most profitable enterprises are actively pricing the future value of our data into the current value of their corporate share prices. Big data, i.e., *our* data, while nonrival and nearly non-finite in nature, are nonetheless made excludable to those who have collectively served to produce the asset in the first place.

Data-dependent technology platforms have retained for themselves the near-exclusive position of being able to not only successfully *monetize* but also *capitalize* the future value of our personal data. This distinct ability and position is principally the product of the following factors:

- The corporate capitalization of data is made possible by means of the (current and future) monetizable applications that depend/will depend upon the **aggregation of our data**.
- Platforms become "natural" monopolies due to both the positive and negative network effects at work, thereby significantly limiting the prospect of competition by newentrants.
  - Positive network effects increase the value of certain networks at a non-linear (roughly quadratic) rate as additional users join the network.
- The ability to limit the portability of the data shared by individuals using the platform.
- Machine learning and predictive analytics enhance the ability of platforms to retain users and predict and direct their future behavior.
- Platforms **own the algorithms** that access to the aggregated data enable/have enabled.

- The value of an individual's unaggregated raw data is not significant. In other words, a given user's data increases in value as it is increasingly aggregated with the data of other users.
- The lack of knowledge among users as to the true value of their data. That is to say, users currently overpay for access to "freemium" services.
- Obscure terms of service limit the ability of platform users to understand the full implications of using the service.

The successful capitalization of personal data is not a new phenomenon for corporations. Therefore, the methodology, framework, and protocols of a viable solution should take note of the key factors and processes that have enabled corporations to capitalize on the value of personal data. The conversion of corporate capital (ownership by absentee shareholders) to personal capital requires that we retain the elements responsible for driving the productive power of our aggregated data assets, while also maintaining its excludability. Personal data can only be turned into personal capital by similarly retaining its excludability—its excludability *to* the platform monopolies that are currently dependent upon it. The solution is to place Zuckerburg on his head [8].

Defining the nature of a "new asset class"

- Personal data (when shared/aggregated) mostly resembles a form of capital.
- Equivalent data (post aggregation) is a fungible asset.
  - For example, when aggregated, a data point I have shared about my coffee preference should be considered equivalent in value to others sharing their preference.
  - Note: I will have to leave it to a subsequent paper to provide the rationale behind this.
- Data has consumable information value as a commodity (unaggregated) and a revenue generating value as a capital asset (when aggregated).
- The value of data *increases* when aggregated.
- Data is only capital when exhibiting the qualities of a *non-rival* and *excludable asset*.
- Presently, data owned as corporate capital is made excludable to the individuals who produce it.
- Data owned as personal capital is made (could be made) excludable to corporations/organizations that depend upon it.
- Should data become a non-rival and non-excludable good, it would cease to be a form of capital.
- Making data non-rival and non-excludable (a "public" good) would prevent personal data from becoming a personal capital asset.
- Data as a "public good" would *remain* capital for existing platforms.

The network effects of democracy

My personal and professional interest in data is largely accidental.

I had come to identify the ownership over capital as the primary means by which political and social power and agency is currently both held and expressed. The nature of the problem seemed quite clear: *If* (a) political democracy did indeed depend upon economic democracy and (b) economic democracy was dependent upon capital ownership being broadly held, we were not (are not) in any objective sense living up to even the most basic demands dictated to us by the ideals of the democracy we currently claim to possess? The character that any proposed solution would therefore need to assume also seemed quite apparent. Simply *democratize the ownership over capital*.

Nonetheless, any attempt to properly identify and outline a viable and practical path towards this solution *based upon where we find ourselves today* (as is often the case) appeared as unlikely to me as it proved elusive. Could the seemingly relentless forces responsible for driving both the current concentration of capital ownership and the future concentration of capital accumulation be stopped? If so … could they be reversed? If so … how?

What previously seemed to be impossible, however, occurred to me as obvious when I considered the profound implications of a simple thought experiment. During the midnight musing of a sleepless night, half asleep–half awake, I considered the following scenario and asked myself:

1. ***What if the users of a given network or platform were also its owners?***
2. ***What if each network user maintained an equal ownership share over the platform itself?***
3. ***Assuming (1) and (2), how then would the value of existing shares change as more users/owners joined the platform?***

If a network is owned by those who comprise it, the value of the shares/ownership held by its users would be expected to increase following the issuance of additional shares (!) (the entry of additional users to the network) due to the *non-linear* marginal increase in value we would expect to accrue to the platform (driven by network effects), yet the *linear* dilution of existing shares due to increasing the quantity of outstanding shares. Within this scenario, issuance (of shares) *is* acquisition (of capital).

As an example, based upon the simple assumptions listed, we can calculate the relative share of a network's total value that is attributable to a given user (owner). Consider a hypothetical network that increases from four to five users. The total "value" of the network has now increased quadratically from 16 to 25 following the new user's entry into the network. Concurrently, the percentage share of existing users will have decreased from 25% to 20%. Per the example, the value attributable to each user, when the network was comprised of four users, was $25\% \times 16 = 4$. Following entry of the fifth user, the attributable value per user increases to $20\% \times 25 = 5$.

Lev's law: network value per user = (N^2)/S

This article proposes the term *Lev's Law* in order to characterize the nature and value of a "user" owned network or platform. Lev's Law states that, ceteris paribus, the network value attributable to a particular user is equal to the number of individuals comprising the network. If a network is owned equivalently by those who comprise the network, the value of each individual's share of the network will *increase* as the network grows. The addition of a new user/shareholder to a network will increase the value of the network at a greater-than-linear rate while diluting the value of existing network shareholders at a linear rate.

The opportunity

The opportunity available to us right now cannot be overstated. The majority of the population currently produces a new and novel asset that, for the most part, they do not even know exists. The means of production within the age of the internet increasingly depends upon the data and information we ourselves produce. Data that is both broadly and equitably held and also happens to be the essential productive element fueling the most innovative advances in technology. In short, when networked (aggregated), our data is translated into a potent form of productive capital.

We currently live in a historical yet temporary moment that has put the prospect of economic democracy within our reach. To the majority of people, productive capital has remained out of reach. Today, however, a new and powerful asset is in our hands. If we collectively aggregate our data and individually claim an ownership share over the aggregated data, we can convert our personal data into a personal and productive capital asset. Moreover, per Lev's Law, we can leverage the network effects of democracy to create an economy of abundance rather than scarcity.

To the great majority of people, past and present, power has been concentrated at an inaccessible distance. However, in contrast to remote mines or oil fields, the capital of the future is close at hand. While its value remains concentrated, power over its production is collectively within our reach. We are the data mines that are replacing the gold mines. If we collectively aggregate our data *and* individually claim an ownership interest over the aggregated data, we can convert our personal data into a personal and productive capital asset.

A proposed solution

We propose the creation of a data platform owned by data producers [7]—a platform that would operate as a cooperatively owned **International Central Bank**. The bank or "International Data Fund" ("IDF") would issue a decentralized digital currency (ERC-20 *security* token) directly to individuals in exchange for the data they optionally choose to share. Each token or unit of currency would represent fractional ownership over the platform itself (e.g., the IDF). The platform would enable individuals to direct their data (and receive currency) towards specific purposes, companies, or sectors with full and transparent knowledge of how it will be used and who will be making use of it.

The IDF would therefore highly intuitively operate similarly to an investment platform, where individuals would be able to direct and receive dividends (denominated in the IDF's data backed currency) based upon where they choose to share/invest their data. The fund would subsequently aggregate the data shared based upon the type of investment and monetize its value on behalf of the individuals who have shared the data, by leasing organizations, research institutions, or companies' temporary subscription based access to the aggregated data via a data as a service model (DaaS).

As revenue is received, the funds would immediately go towards the repurchase of outstanding tokens/units of currency on the secondary market, which would serve to support both the value of tokens issued and the maintaining of a liquid marketplace. The value and price stability of the currency issued would be supported by policies adopted that would make it so that each unit of currency was equivalent to one US dollar for DaaS subscription payments.

The design of the platform would be informed by the logic of Lev's Law and would therefore not cap the lifetime quantity of tokens that would be available to issue. Due to both the data *and* currency network effects would result from broad adoption; if executed properly, we would *not* expect the inflationary effects traditionally observed from increasing the money supply of a fiat currency. While an in depth outline would not be possible within one essay, the following sections will introduce the key monetary policies that would inform the proposed data platform.

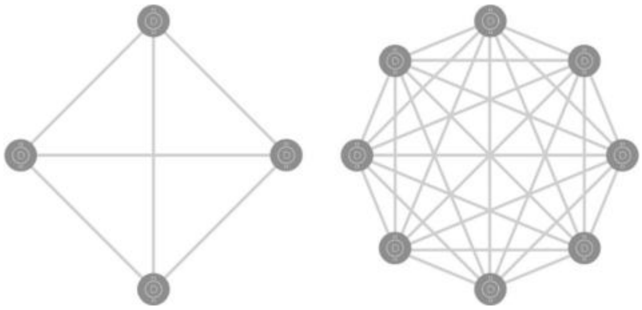

Fig. 1 Currency network effects

Monetary policies and protocols

The monetary objectives of the proposed creation of the IDF and their corresponding protocols provide the parameters and rules of governance for the prudent management of the money supply.

Triple mandate

1. Capitalize the value users receive for their investment/sharing of data through enabling shared ownership over the aggregated data held by the IDF.
2. Promote price stability.
3. Ensure individuals maintain permanent ownership and control over the IDF.

A data capital backed currency

In order for an asset to function effectively as a form of currency it must be capable of serving as: (1) a store of value, (2) a unit of account, and (3) a medium of exchange.

The revenue *from* DaaS and the redeemability of the currency *for* DaaS would directly support the token as a "store of value." However, this in itself does not satisfy all the basic elements of a currency. In order for an asset to function successfully as a currency, it would need to be broadly enough held to have utility as a medium of exchange and commonplace enough for the value of other assets, prod-

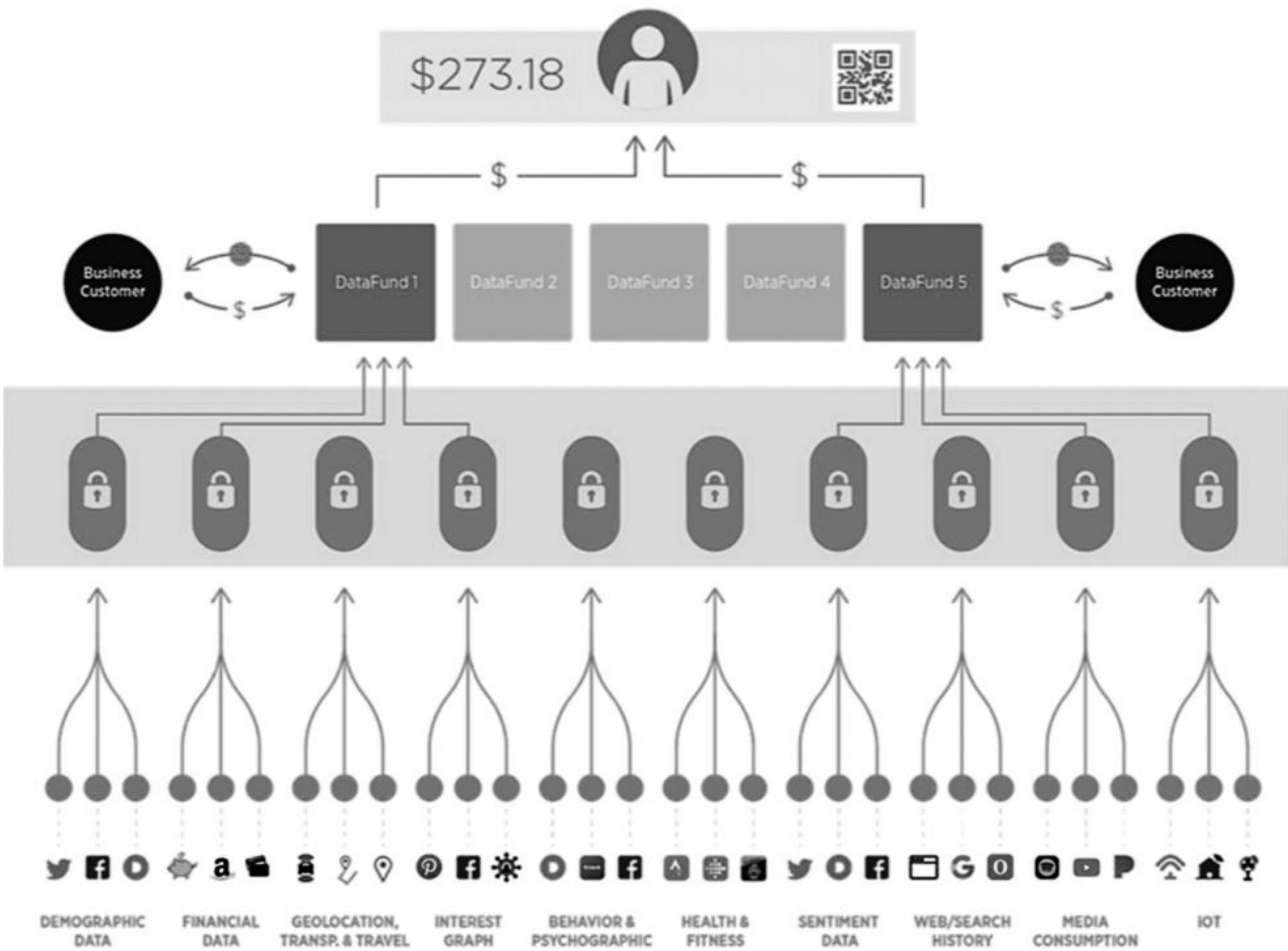

Fig. 2 Data investment platform

ucts, and services to be priced and understood in its terms, and therefore have application as a standard unit of account.

The framework would therefore be designed to maximize its currency value (the degree of its adoption) as well as its capitalized ownership value ([5]; Fig. 1).

Issuance of a data backed currency

Newly minted currency notes would *only* be issued following an individual's investment of data. This ensures that an increase in the money supply will always follow an increase in the supply of data-capital held by the IDF. Individuals would therefore represent the exclusive conduit by which new money can enter the monetary system.

Currency repurchase protocol

1. DaaS revenue will go towards repurchases of notes on the secondary market.
2. Revenue received is applied via a smart contract upon receipt towards the repurchase of outstanding and actively traded notes.
3. Secondary market purchases will be executed at the current market price.
4. The value of notes will be directly and unambiguously supported by platform revenue, thus providing a recurring, consistent, and increasing source of demand (and therefore liquidity) for previously issued notes.
5. The reduction of the money supply from currency repurchases increases the availability of additional units for future issuance and supports price stability.

Data fungibility

1. When equivalent bits of data are of equivalent quality but provided by different individuals, the data contributed will be considered fungible in value with regard to the relative quantity of notes issued.
2. The analysis of large data sets often leads to a large number of unanticipated/monetizable applications and analytical insights, which precludes the rational attribution of differential pricing.

Data redemption protocol

1. When redeeming data through the IDF's DaaS platform, one note would equal one USD. When the price of currency notes depreciates (appreciates) relative to USD, demand for the notes would be expected to increase (decrease), since the currency note's price for DaaS subscriptions relative to USD will have decreased (increased).
2. Allowing DaaS redemption with issued notes will, pending broad adoption, serve to incentivize companies to purchase and/or offer good/services in order to accrue the IDF issued notes (Fig. 2).

Conclusion

Political democracy depends upon economic democracy. Economic democracy depends upon an equitable distribution of capital ownership, while a more equitable distribution of capital ownership is obtainable through the equitable distribution and ownership over data and the value it creates when aggregated.

Democracy therefore depends upon a *Kelsonian Revolution* if it is to persist in the 21st century—a revolution that calls upon us to democratize capitalism through the democratization of capital ownership. Together, we currently hold all the tools we require to assert our rightful ownership over our democracy and the new and prepotent language of power by reclaiming and asserting ownership over the data we produce. If we choose to act on the opportunity that has been made available to us, the formation of a beautiful future awaits, one where the productive capacity of technology and capital find alignment with their democratic purpose.

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HAUPTBEITRAG

What if we had a democratic, capitalistic online marketplace?

Joël Vogt[1]

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Abstract

Can consumer ownership schemes provide a more democratic financing mechanism for the online marketplace? When it comes to being successful as an online platform, size matters. The network effect, which tells us that the value of a platform increases exponentially as a function of the user base, creates a natural concentration of power, if you're a vendor trying to sell or advertise products, you will pick the platform with the largest audience. The network effect also creates an exit barrier, as moving to a competing platform without the same reach and user base means fewer potential customers. The consumer ownership scheme that I will be focusing on is the Consumer Stock Ownership Plan pioneered by Louis Kelso. The CSOP is a method of corporate finance that enables consumers to buy an ownership interest in a company on which they heavily rely. It is a democratic capitalistic tool to mitigate the harm caused monopolies or oligopolies. A CSOP does not require consumers to buy the company's shares in the stock market. Instead, the company acquires its shares in a trust on behalf of consumers and uses dividends to amortise the investment. Shares that are fully amortised are transferred to their owners. In this essay, I make the case that CSOP financing for online marketplaces makes sense for three reasons.

A CSOP does not discriminate against those who lack the capital to acquire shares in the first place;

Users are an integral part of the value proposition of an online marketplace. A CSOP acknowledges this fact by making users owners.

Users' data generates value, which indirectly flows back to users in a CSOP.

Introduction

Democracy is at a crossroads. One direction leads to a future in which innovation, democracy and the economy will prosper, unleashing human and social development to an unprecedented degree. The other direction leads to a darker place characterised by the concentration of power and capital in the hands of a few organisations and individuals. Although this economy might still be called capitalism, it is actually a revival of feudalism on a global scale and with totalitarian power. The key ingredient to making capitalism work is wide-spread ownership of capital! In a truly capitalistic society, everyone would be a capital owner. Do you own capital? Or do you rely on your labour as a source of income and neglect capital ownership, even when the value of that capital is inherently based on you as a consumer?

The Internet introduced a new form of capital: information. Information is the fuel of the digital economy and of the artificial intelligence revolution. Yet to grow and expand the economic pie, to actually unleash the fourth industrial revolution, all of us need to be able to participate in the economy as owners, producers and consumers of the products of capital. The marketplace is the heart of the capitalist economy. The Internet promised to create not only a global but a more inclusive marketplace. By lowering barriers to entry and bypassing intermediaries, the Internet should allow everyone to freely trade with one another. Alas, in reality, former intermediaries were replaced by a few dominant online platforms. This is not to say that we should not be grateful to these companies for driving and fuelling the development of the Internet. Thanks to Amazon, Google, eBay, and Facebook we have instant access to information, faster and cheaper access to physical goods, and the ability to directly connect with people from all walks of life. Yet at what cost? Information is power. These oligopolies are acquiring a lot of information about us, shifting the balance of power in their favour. Moreover, as global platforms, these Internet giants are mostly left to regulate themselves. Facebook's behaviour should be a stark warning of what happens when consumers are downgraded to unpaid workers (e.g. by tagging pictures or posting messages) or as raw data sold to brands and political parties to market products and candidates.

✉ Joël Vogt
joel.vogt@evrythng.com

[1] EVRYTHNG, Avenue d'Ouchy 4, 1006 Lausanne, Switzerland

Fig. 1 Business processes in the digital marketplace. Based on Menychtas et al. [5]

Thanks to the open and unregulated nature of the Internet, we have experienced an unprecedented growth of commercial and social possibilities on a global scale. However, hidden in plain sight, the Internet has at the same time created a platform for a new asset, personal data, e.g., a platform for the creation, collection, processing and trading of information about us as the Internet users. In the wake of recent large-scale privacy violations, an increasing number of citizens and lawmakers are reluctantly confronting the fact that "free" services are not free after all. Users pay for purportedly "free" services with their own personal data, without knowing what they have paid for or how much. However, it does not have to be like this. We have sound economic theories and technologies to build online platforms that are not only inclusive and profitable but democratic.

In this essay, I ask myself, "What if we had a democratic capitalistic marketplace?". I will propose a framework, based on the Consumer Stock Ownership Plan, to organise and finance this new marketplace.

Online marketplaces

The Internet revolutionised the way we conduct business, enabling people to directly engage in business transactions with one another, independent of time and location. The by-product of these transactions is metadata information about the context in which transactions occur. Data trains machine-learning models, informs advertisers about whom to target and to enable operators to streamline their processes. Unlike physical products, data does not necessarily lose value by being utilized. Data is the new digital gold and a key factor of production in the digital age (Fig. 1).

Let us view the online marketplace from the perspective of processes that create and use data. The model by Menychtas et al. [5] identified four business processes in the online marketplace and the exchange of information between these business processes:

- Information: making product information available to consumers. This includes information about the products, such as what products are made of, where they were made, how products can be recycled, etc.
- Negotiation and price setting: buyers add products to their "baskets" and agree to pay the price set by sellers.
- Contracting and settlement: completing the transaction with the product, price, method of payment, means of shipment, etc.
- Post-purchase: buyers takes possession of their products and start using them. Products that have a digital identity create additional data throughout their life. Finally, products are or should be recycled.

At each stage, information is created by different actors. This information can be used to optimise tasks, create more relevant recommendations and increase recycling, to mention a few examples. In today's marketplaces, data management is primarily controlled and owned by the operator, not by consumers or sellers. Amazon understood early on the power of artificial intelligence to constantly improve its processes and operations. It also gives Amazon an edge over other sellers. Unfortunately, information producers have no way of being compensated for their data, nor do other actors have the opportunity to access these data pools and innovate on a level playing field with Amazon. Participants have no choice but to renounce the right to own their own data, the new form of productive capital. If most participants in the digital economy, who are also the majority of data producers through their online activities, cannot exercise their right to owning their personal capital, we are undermining the foundation of a capitalist market economy.

Binary economics: the economic paradigm made essential by the Digital Revolution

Louis and Patricia Kelso, two of the great thinkers of our time, proposed solutions for fixing the problem. The problem originates in the obsolete belief that labour is the only factor that creates wealth. In reality, both labour *and* productive capital create wealth [2, 3]. Why is this so relevant today? Because the productiveness of capital instruments is growing exponentially and thus eroding the productive power of human labour. For those like me who grew up with computers, Moore's Law [6] may help to explain the rapidly narrowing productive gap between people and computers. Moore's Law tells us that computing power doubles roughly every 18 months. As a result, artificial intelligence automates tasks that were once creative in nature and thus considered safe from automation. It is not all doom and gloom, however. Some technological advances do create employment but not enough to maintain "full employment". The democratisation of big data and artificial intelligence

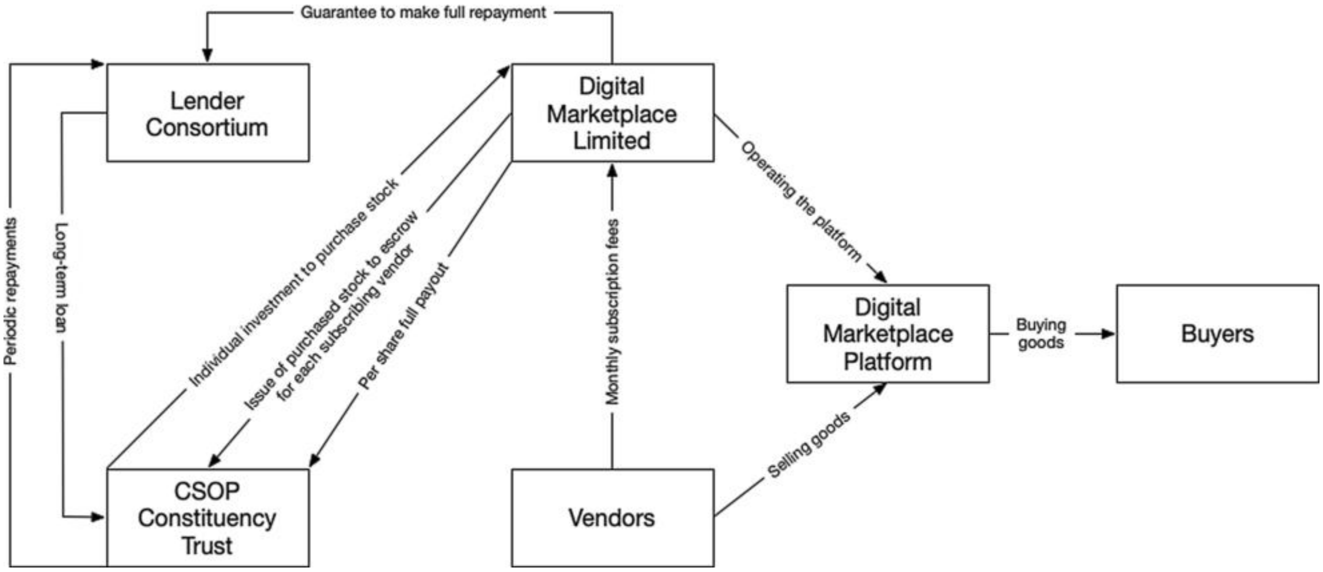

Fig. 2 Design of a CSOP for a digital marketplace based on [3]

gave birth to new roles of information workers, such as data scientists or machine learning engineers. These professionals are not your classic statisticians or data analysts. They operate in a business environment and rely on very large data sets and powerful computing systems to automate the extraction of information with the aim of improving business processes and automate repetitive tasks. However, technology tends to replace or reduce human labour, it is an illusion to think that everyone can be magically retrained. Think about it! If a computer were able to acquire skills that an individual had acquired over years of hard work and experience, what are the odds that retraining would result in a job that a machine would not soon replace?

If we think in terms of human versus machine, this dilemma has no solution. However, binary economics gives us a different way to visualise the problem: ownership of the machine. Allow me to illustrate this with my taxi example. In a few years, autonomous cars will replace taxi drivers. If taxi drivers own their car, nothing would make them happier than to send their car to work every day while they engage in leisure (as defined by Aristotle). However, if cars are owned by a large taxi cooperation, then, suddenly, wealth that could once be generated by a skilled driver is being transferred to that cooperation. Retraining unemployed drivers can hardly be seen as a solution. Perhaps some niche jobs will be left, but most commercial vehicle drivers, without either a job or investment capital, will be excluded from participating in the economy.

As I mentioned previously, Louis and Patricia Kelso identified and explained not only the problem but the solution: enabling people to become capital owners [3]. Starting with the Employee Stock Ownership Plan, a financial model that allows employees to buy stocks of their company over time—not by paying for the stock from their paycheck but from the "wages" of their "newly acquired capital" [3]. Louis and Patricia intended for the ESOP to be applied to different types of companies and organisations, financing stock purchases not from past savings but from current stock dividends. The Consumer Stock Ownership Plan was developed for companies where consumers have a long-term service relationship, such as a utility. This makes the CSOP an ideal financing model for the Digital Age. To succeed, Internet companies need to achieve a critical mass. Then the network effect applies, which says the value of a company increases exponentially, the more users participate [1]. This is why most activities online congregate around a few platforms, Google for search, Twitter and Facebook for social interaction and Amazon for retail.

Capitalism in the Digital Age

Internet giants are monopolies that grew organically. Users flock to Google, Amazon and Facebook, because they offer great services and/or convenience. Breaking up platforms in the hope that, magically, their components will agree on open standards to ensure that no dominant player will emerge after that breakup is unrealistic.

This raises the question: if we are already tied to these platforms, why not become their (partial) owners? When you think about it, it is our right[1]. By being successful, Internet giants also lock their users in, since the value they offer to users depends on how many other people use that plat-

[1] Well at least the right of the American people, whose tax dollars funded the research and investment that created the Internet [4].

form. Unlike public utilities, such as roads or public transportation systems, these platforms are privately owned, have no independent oversight, and their business models rely on users not reaping the benefits of their personal data. Instead, we are witnessing a massive redistribution of wealth online of personal data. What makes more sense, I believe, for everyone in the long run, are platforms where personal ownership of data is maintained, and mechanisms are in place for a personal data marketplace. There are promising initiatives to democratise personal data ownership, e.g. in Switzerland, the cooperative MIDATA. COOP is pioneering health data ownership.

An e-commerce platform with democratic capitalism by design

We touched on five problems with the value chain in the Internet

- Concentration of wealth and the resulting erosion of our democracy
- Lack of transparency and accountability
- Lack of due process
- Lack of control over personal information.

I find the first the most important. It forms the basis for others' issues to be addressed. I will, therefore, focus on the aspect concentration of capital and democracy. To recap, Louis Kelso invented the Employee Stock Ownership Plan, a financial innovation that enabled employees over time gain stock ownership in their own company. The interesting aspect of this idea is that an employee's ability to buy shares does not depend on their personal wealth. Louis Kelso's idea overcomes this barrier to acquiring capital ownership by creating a mechanism, the Employee Stock Ownership Plan (ESOP), in which a company takes a loan to finance employee's stocks and pledges to pay off the loan on behalf of the Employee Stock Ownership Plan. Before employees receive dividends from their shares, dividends are used to pay off employees' shares. The beauty of this financing model is that salaries are not affected in the process because stocks pay for themselves.

When reading Louis and Patricia Kelso's book "Democracy and Economic Power: Extending the ESOP Revolution Through Binary Economics" it struck me that the Consumer Stock Ownership Plan could be a pillar in financing a more democratic and economically free Internet. ESOP and ESOP-derived financing tools are designed for a long-term relationship between the employee shareholder and the company. The Internet, and Internet companies, are essentially utilities, because users cannot easily move from one successful Internet platform to another, even if the new service is technologically better. This is a consequence of the network effect, which says the value of a company depends exponentially on its number of users. The CSOP was invented in the middle of the previous century to allow farmers in California gain ownership of a new chemical fertiliser plant, enabling them to not only receive dividends but to buy fertiliser at little over production cost [3].

I outline the CSOP for the online marketplace in Fig. 2, which is mostly based on the CSOP model in [3]. The CSOP online marketplace is initially intended for vendors only. I am not excluding consumers also becoming owners, but I do not yet see sufficient awareness of users about the value of data, nor are there any real financial incentives for consumers to become share owners in the marketplace.

A vendor subscribes to the service of the online marketplace at a price determined by projected trading volume. The vendor pledges to pay regular subscription fees while also acquiring shares that earn profits. This is no different from existing online marketplaces, except that through a subscription, a vendor also becomes a shareholder. The CSOP stocks are managed by the CSOP Constituency Trust. They are held in escrow until the dividends have fully paid for them, where upon the vendor receives the entire dividend yield. Vendors may cancel their subscription at any time. However, partly paid off stocks will not be reimbursed. I suggest this as an incentive to encourage long-term engagement, especially in the creation phase of the online marketplace. Once the online marketplace has penetrated the market sufficiently, the network effect will create a virtuous cycle in that vendors and buyers will increasingly use the platform.

It would be natural for "online marketplace company" to also implement an ESOP for its employees. Ultimately, both employees and consumers will be owners. Moreover, once the online marketplace is sufficiently established, I would consider opening up the CSOP to buyers.

The lender consortium should be open to a variety of investors. Venture capitalists for one, but also banks and large corporations that benefit from e-commerce. Obvious candidates would be postal services, whose loss in mail delivery has been more than offset by parcel delivery.

Discussion and next steps

Writing this essay was a way to collect my thoughts about the things that have troubled me for a while: the erosion of privacy, disparagement of private market economy by using it as a euphemism for failed government policies (e.g. the privatisation of trains here in the UK), corporate greed and ignorance of how automation will impact our society and economy. At first, I wanted to write about a CSOP-based platform for consumers. Over time, I came to realise that it would be very difficult to make a case strong enough for consumers to invest in such an enterprise. The attitude towards the value of data is changing. However, I believe that the personal data revolution is only beginning. For this reason, I focused on commercial actors who are forced to use online platforms that are quasi monopolies. Louis Kelso invented the ownership tools that will enable us to keep the advantages of scale while mitigating the negative effects of monopolies.

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From Intelligent to Wise Machines: Why a Poem Is Worth More Than 1 Million Tweets

Davide Picca[1]

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Abstract

In recent years, the interest in artificial intelligence and big data has grown exponentially, and the amount of data produced every day is truly staggering. Data are considered to be the "new oil" making algorithms capable of delivering meaningful information, which makes them more "intelligent." In this position paper, we review the DIKW pyramid model, shedding a new light on each component. In particular, examining the engineering point of view, we focus on the definition of *information*, giving it a new conceptual structure. If tradition has always considered information as data-bearing meaning, in this paper, we argue that information is not meaningful. In fact, from the analysis of Shannon's studies in communication engineering, we highlight how the notion of meaning is not necessary for the definition of information. It follows that we need to explore other paths in order to find a sustainable conceptual theory able to provide a new insight. Therefore, we show how it will not only be necessary to carry out a semiotic revolution to be able to introduce meaning into the communicative act, but it is also necessary to introduce the figure of an interpreting agent. Thanks to the interaction between such interpretative acts, which take place in conscious freedom *à la Eco*, cultural units emerge. Thus, we should address part of today's research into new forms of data in order to facilitate a semiotic revolution. In particular, digital humanities and cultural heritage can funnel a new type of data for which semiotic representativeness has a greater degree of quality. Knowledge and wisdom are the next steps to truly craft intelligent machines.

Keywords DIKW pyramid · Information systems · Knowledge engineering

✉ Davide Picca
davide.picca@unil.ch

1 Language and Information Sciences, University of Lausanne, Lausanne, Switzerland

Von intelligenten zu weisen Maschinen: Warum ein Gedicht mehr wert ist als 1 Million Tweets

Zusammenfassung
In den letzten Jahren ist das Interesse an künstlicher Intelligenz und Big Data exponentiell gewachsen. Die Menge an Daten, die tagtäglich produziert wird, ist wirklich erschütternd. Daten werden als das „neue Öl" bezeichnet und erlauben es den Algorithmen, aussagekräftige Informationen zu liefern und dadurch „intelligenter" zu werden. In diesem Thesenpapier überprüfen wir das DIKW-Pyramidenmodell und werfen ein neues Licht auf jede einzelne Komponente. Aus der Perspektive der Technik konzentrieren wir uns insbesondere auf die Definition von Information und geben ihr eine neue konzeptionelle Struktur. Obwohl in der Tradition Information immer als bedeutungsträchtige Daten angesehen wurde, argumentieren wir in diesem Artikel, dass Information nicht an und für sich aussagefähig ist. Aus der Analyse von Shannons Studien in der Kommunikationstechnik zeigen wir sogar, dass der Begriff „Bedeutung" für die Definition von Information nicht notwendig ist. Daraus folgt, dass wir andere Wege erforschen müssen, um eine tragfähige konzeptuelle Theorie zu finden, die neue Erkenntnisse liefern kann. Wir zeigen, dass eine semiotische Revolution nicht genug ist, um Bedeutung in den kommunikativen Akt einzuführen. Es ist hingegen notwendig, die Figur eines interpretierenden Agenten einzuführen. Nur durch die Interaktion zwischen solchen Deutungsakten, die in bewusster Freiheit à la Eco geschehen, ist das Entstehen kultureller Einheiten möglich. Daher sollte sich ein Teil der heutigen Forschung nach neuen Datenformen richten, um die semiotische Revolution zu unterstützen. Insbesondere die Digital Humanities und das kulturelle Erbe können eine neue Art von Daten hervorbringen, mit einer höheren Qualität der semiotischen Repräsentativität. Wissen und Weisheit sind die nächsten Schritte, um wahrhaft intelligente Maschinen herzustellen.

Schlüsselworte DIKW-Pyramide · Informationssysteme · Wissenstechnik

Machine learning and the information era

In recent years, the interest in artificial intelligence and big data has grown exponentially, and the amount of data we produce every day is truly staggering. If we look at the growth rate of Google Trends, searches for the topics "Machine Learning" and "Big Data" have been growing since 2012.[1]

The amount of data produced daily does not tend to decrease; currently, we create 2.5 quintillion bytes of data every day, and this rate is accelerating with the growth of the Internet of Things (IoT). 90% of worldwide data have been generated in the last 2 years alone. This considerable increase in data raises some questions about their nature and the type of information they convey. Bernard Marr for Forbes [21] identifies at least six channels from which this deluge of information originates. Based on a previous work entitled "Data never sleeps 5.0" [34], Marr highlights the following six main axes through which data are created: the Internet, Social Media, Communication, Digital Photos, Services, and IoT. Examining this investigation, we quickly realize that the totality of generated data comes from social media services, where with social media we identify any means of democratic communication in a strictly etymological sense, "[...] in which the sovereign power is vested in the people as a whole exercise power directly." [11]

This large amount of data is generated without control by anyone. If we take the example of the communication axis as mentioned by Marr, and we analyze in greater detail the nature of communication produced every minute, we notice that people send 16 million text messages, 990,000 Tinder swipes, 156 million emails; 15,000 GIFs are sent via Facebook messenger, 154,200 calls are made via Skype, and 103,447,520 spam emails are sent [38]. This mass of data is extraordinary for the creation of information considering that data are the bricks with which information is built. Moreover, it justifies the characterization of data as the "new oil" [11, 34] that fuels today's economy. Data are fundamental for machine learning. In fact, machine learning algorithms could not be trained without data, nor would they be able to perform the intelligent tasks that we have become used to. As reported in the literature (Sect. 2), data are the building blocks for producing information and, in turn, information is commonly defined as the result of data processed in a specific context in order to produce meaningful knowledge.

In this paper, we will dissect the DIKW (data-information-knowledge-wisdom) pyramid and we will perform a detailed analysis of the constituting elements defining data, information, knowledge, and wisdom. In the course of this work, we will review the definitions coined by other scholars from the field of information systems and knowledge management, highlighting the aporia and contradictions arising therefrom. We would like to stress the importance that the opacity surrounding the DIKW concepts has in the impact on daily life and socio-political events. For

[1] https://trends.google.com/trends/explore?date=all&q=%2Fm%2F01hyh_,%2Fm%2F0bs2j8q accessed on 28/11/2018.

example, consider the recent headlines and facts reported in the newspapers regarding Tay, Microsoft's A.I.-powered bot, which was able to tweet and chat on GroupMe and Kik, and which was recently shut down due its inability to recognize when it was making offensive and racist statements after 16 hours of chats [22]. If we validate the hypothesis that algorithms produce meaning-bearing information, as widely advocated by the community (Sect. 2, then the type of embarrassing situations involving Tay should not happen. Obviously, Microsoft did not train the bot with the purpose to be racist; however, the data were biased by users with whom Tay was interacting, leading it to learn how to be racist. Another example is represented by Amazon's AI Recruiter, when "by 2015, the company realized that its new system was not rating candidates for software developer jobs and other technical posts in a gender-neutral way. That is because Amazon's computer models were trained to vet applicants by observing patterns in resumes submitted to the company over a 10-year period. Most came from men, a reflection of male dominance across the tech industry" [5]. A clearly serious issue emerges from a recent mistake made by Facebook's automatic translation mechanism, which erroneously translated a post picturing a Palestinian man leaning against a bulldozer with the caption "good morning" with "attack them." The man wa arrested and detained for several hours before being released.

Thus, if information is data bearing meaning, how can this kind of event occur? Should not machines be able, on the basis of information alone, to assess the risks and consequences of such inappropriate behavior in relation to the specific context? If, in practice, an information technology system is unable to discern and apply the right meaning in the right context, it means that information can not be meaning-aware. We will, therefore, try to shed light on these impasses by arguing that information machines are not machines bearing meaning, since information does not intrinsically carry any meaning. We will see how meaning cannot be confined in a state of facts but is an emerging phenomenon taking place in a relation perspective. If we consider the previous example of the Facebook translator and look at information as "what is useful and meaningful for a receiver" as some authors seem to suggest, then should not the Facebook translator bot be "useful and meaningful" for the Palestinian man leaning against a bulldozer? In the light of what actually happened to the protagonist of this misadventure, maybe we are prone to affirm that information has not only been useless and meaningless but overall dangerous! Information is not sufficient to code meaning-aware machines. We are likely to need knowledge and even wisdom to move towards a truly meaningful device and we should go even further to have context-and-meaning-aware machines.

In the next sections, we will argue that knowledge and wisdom are actually the only semiotic component carrying meaning. Nonetheless, we will highlight how it is not sufficient to make machines aware of the surrounding contexts in order to avoid uncomfortable or even dangerous situations such as those described above. In fact, in order to produce context and meaning-aware machines, they should not only *know* what knowledge and wisdom are but they should also *truly behave* as knowledgeable and wise machines. This means that knowledge and wisdom are not just intellectual concepts, but they are *embodied experiences*. As a result, knowledge and wisdom are not reducible to just a *bunch of bits* encoded in a hardware; they cannot be reduced to a pure mathematical representation but belong to the sphere of action and interaction with the world because, as we will widely discuss in Sect. 3.4, they pertain to the sphere of the *praxis*.

As it can be noticed from these introductory remarks, the analysis we are about to conduct is complex and requires a preliminary study of the state of the art on the different definitions that have been provided with respect to the components of the DIKW pyramid. This will allow us to point out the various contradictions and inconsistencies that emerge from the studies in the field. So, we begin with outlining the state of the art with respect to the definitions of data, information, and knowledge in the previous literature. Subsequently, we introduce a new perspective on these concepts in order to demystify and give to the notion of information a new capacity. Thanks to this new stance, we can move towards a clear statement about the role played by knowledge and wisdom leading us to recognize that digital culture and, more specifically, digital humanities, are assuming a central function for the future of machine learning and artificial intelligence in general. In fact, if literary production and cultural heritage are the results of manipulation and construction of artifacts issued from human creativity, then it can be assumed that they play a fundamental role in the understanding of human action in the world; they help us understand how the world is shaped and modified by the poetic and poietic action of men. Action from which machines could draw a great advantage in order to improve their ability to get a meaningful and wise understanding.

DIKW: Some definitions

Defining what data, information, and knowledge are constitutes a challenge. If we consider several authors who previously worked in this field, we rapidly observe that there is no consensus on a unique definition [10].

In this regard, Tom Wilson, endeavoring to define the notion of "information," writes that

> It is such a widely used word, such a commonsensical word, that it may seem surprising that it has given 'information scientists' so much trouble over the years [40].

In light of the intricacy surrounding the notion of information, for our investigation, we have relied on the instructive paper written by Rowley [29].

Rowley revisits the DIKW hierarchy by examining the articulation of the hierarchy in a number of widely-read textbooks and analyzing their statements about the nature of these four concepts. He takes into account 15 textbooks, of which 8 originate from the information systems field and 7 from the knowledge management field. What this paper reveals is a quite large disagreement among scholars on their definitions.

Although most scholars agree on the explanation of data, as soon as researchers move towards the definition of information, many discrepancies arise. If, subsequently, we look at the interpretations of knowledge and wisdom, the matter becomes increasingly intricate and finding a univocal definition proves even more difficult. For example, let us consider Rowley's affirmation about wisdom:

> Arguably the most significant observation is that only three of the books seek to define wisdom, despite its position at the pinnacle of the wisdom hierarchy. Such an omission might lead to a deduction that many authors in information systems and knowledge management view wisdom as being beyond their remit for some reason.[...] On the other hand, they may agree with Jashapara [18], when he suggests that wisdom is a very elusive concept [29].

For the sake of completeness, let us now examine in greater detail the definitions of each element of the DIKW pyramid as provided by Rowley.

Data is the only notion on which most of the scholars agree. It is usually defined as meaningless or valueless because it is without context and interpretation. Data are discrete, objective, and unprocessed facts. All researchers examined by Rowley share the same point of view when they observe and characterize data as the primitive on which information is founded; data are an unorganized and unstructured flow of primitive items. Interestingly, Jashapara [18] considers data as a physical signal coming from an external world "through our senses and try to make sense of these signals through our experience" [29]. In our view, this approach, stemming from the knowledge management field, should be adopted in the information systems field as well, as we will widely discuss in Sect. 3.2.

The definition of information raises several disagreements among scholars. Although they all see a strong connection between data and information, scholars from the knowledge management field tend to define information in terms of "context" and decision-making purposes. Therefore, information is what sustains decisions in a specific context. To the contrary, in the information systems field, researchers place emphasis on the concepts of "meaning" and "value to the recipient," thereby viewing information as what is useful and meaningful for a receiver. At first glance, the two communities do not seem to provide the same definitions; however, they converge to a same conclusion, if we consider the context at the base of what determines meaning [30, 41]. In this regard, Bocij *et al.* [9] clearly state that information is data that have been processed so that they are meaningful and understood by a recipient. Hence, information as *data that are meaning-bearing* is the common ground of both definitions. Information is seen, thus, as data carrying meaningful messages to the receiver.

As we discussed in Sect. 1, this approach can be problematic and can lead to incoherence that is difficult to resolve. In fact, once context and meaning are involved in such a definitional negotiation, new elements come into play, as we will extensively consider in Sects. 3.3 and 3.4. The discrepancies between the knowledge management field and the information systems field with respect to the definition of information are highlighted in Table 1, which is adopted from Stenmark's work [33].

It appears that the definition of information is surrounded by opacity. However, when scholars attempt to

Table 1 Data, information, and knowledge definition as viewed by several authors (adapted from [33])

Author	Data	Information	Knowledge
[12]	Facts and messages	Data vested with meaning	Justified, true beliefs
[15]	Not yet interpreted symbols	Data with meaning	The ability to assign meaning
[14]	Simple observations	Data with relevance and purpose	Valuable information from the human mind
[13]	A set of discrete facts	Data with relevance and purpose	Text that answers the questions why and how
[28]	Text that does not answer questions to a particular problem	Text that answers the questions who, when, what, or where	Text that answers the questions why and how
[24]	–	A flow of meaningful messages	Commitments and beliefs created from these messages
[39]	–	Facts organised to describe a situation or condition	Truths and beliefs, perspectives and concepts, judgements and expectations, methodologies and know-how

define knowledge, it is even more problematic. As Rowley declares: "*Definitional statements on knowledge are often much more complex than those for data or information*" [29]. The notion of knowledge has been contended for many years but as a final outcome,

> these debates make it more difficult to distill the essence of the statements on the nature of knowledge than it is to capture and represent the definitional statements that relate to data and information [29].

Defining knowledge is a knotty problem that is not easily solved and, to date, there is still no consensus on the definition of knowledge. As Jashapara [18] states, knowledge is "*an intrinsically ambiguous and equivocal term*" [8]. Most of the definitions provided in books on information systems are, at most, a great example of acrobatic intellectualism. We state hereunder the main definitions reported in Rowley's paper.

For example:

- *"Knowledge is the combination of data and information, to which is added expert opinion, skills, and experience, to result in a valuable asset which can be used to aid decision making."*
- *"Knowledge is data and/or information that have been organized and processed to convey under-standing, experience, accumulated learning, and expertise as they apply to a current problem or activity".*
- *"Knowledge builds on information that is extracted from data [...] While data is a property of things, knowledge is a property of people that predisposes them to act in a particular way" [29].*

The difficulties encountered in defining knowledge are further exacerbated when we attempt to characterize wisdom. It is remarkable that out of 15 books examined by Rowley, only three of tackle the problem of wisdom.

> Such an omission might lead to a deduction that many authors in information systems and knowledge management view wisdom as being beyond their remit for some reason. [...] It perhaps has more to do with human intuition, understanding, interpretation and actions, than with systems [29].

For the majority of authors, wisdom is an intuitive notion that escapes the human rational cognition and belongs to an individual's belief system more than to an information system [18].

A new attempt at a definition for DIKW

In the light of what has been said so far, researchers agree that the DIKW pyramid should be evaluated in the context of communicative acts, so that, arguing on the topic of information mainly means discussing on the nature of communication.

Given the diversity and discordance of opinions on the definition of information (and, consequently, of the other components of the pyramid), it is important to establish a focal point to start off the discussion. It is also important that this point is based on studies that are as objective as possible and that gather the greatest number of consensus in the information engineering community. One of the most important authors investigating communication and its components from a technical point of view is undoubtedly Shannon [32]. Thanks to his perspective, we are able to provide a novel way to define the DIKW pyramid in the context of information engineering. In fact, we will see that communication in information systems should mainly be treated from a technical standpoint, since machines are, first of all, technical items.

Thus, it is first of all important to clarify the notion of communication and reconstruct its genesis in order to define information in the field of engineering and information systems. Following Shannon's investigation on communication, the transfer of a message from a human being to another human being necessarily requires the presence of a communication tool by means of which the message is transferred. In human communication the tool we use is language. Fig. 1, taken from Shannon's article, provides a visual representation of what we mean.

Fig. 1 illustrates the structure of communication, which is composed of five essential elements:

- *Information source*: it produces a message or a sequence of messages to be communicated to the receiver.
- *The transmitter*: it receives the message and elaborates it so as to produce a "signal" for transmission.
- *The channel*: it represents the medium of communication.
- *The receiver*: it is the entity that performs the reconstruction of a message from the "signal."

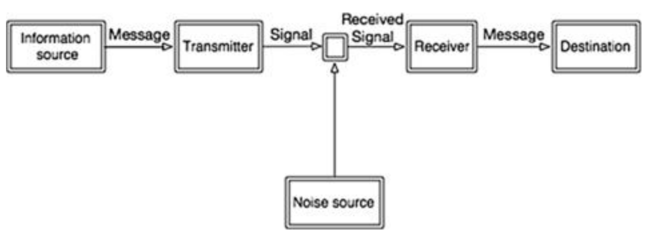

Fig. 1 The communication diagram illustrating the structure of communication as it was introduced by Shannon [32]

- *The destination*: it is the unit that ultimately receives the message.

Shannon recognizes the paramount importance of the *signal* in the communication chain. In Shannon's graph, the element *signal* is considered essential for the transmission of information from the source to the destination channel. A signal can take any form; it can be a word, images, icons, or an act, but it does not have a meaning, and it becomes a *sign* in the semiotic sense only when we attribute to it some meaning. This aspect will be widely discussed in the next section.

Definition of data and information

As discussed in Sect. 2, data is generally defined as unprocessed information and a representation of objective facts. Data is considered to be a solid, physical object with an objective existence. Thus, if we want to attempt a clear definition of data based on Shannon's work and in line with the opinion shared by most of the existing literature, we can state along with [18] that *data is the flood of signals coming from the external world without being preprocessed*. Conversely, the definition of information is not so unambiguous based on the literature tradition. Nonetheless, as reported in Sect. 2, information is generally seen by scholars as structured data bearing meaning [12, 29]. Moreover, if we consider the equality: *information = data bearing meaning* as widely suggested in literature [3, 12, 28, 29], then, from a technical point of view and following the information system's tradition initiated by Shannon, the properties we just attributed to information could easily be transposed to meaning.

If information systems are proper *information* systems, then according to the mainstream in the literature, it means that they are devices able to carry data along with a way to semiotically *signify* them. To illustrate what is meant by this, let us consider Fig. 1 reported in Shannon's article. Aside from Shannon's technical description of information, we can also derive some interesting properties of information from his paper. In particular, the rigorous mathematical infrastructure provided by Shannon allows us to deduce some more general observations:

- Information can be kept entirely controlled.
- The role of information consists in the transmission of *data* (= *flood of signals coming from the external world without being preprocessed*).
- Information can be easily formalized using mathematical and statistical methods.
- Information is strictly dependent on codification methods.

According to Shannon, the above definitions of information are meant to be applicable to information systems as well. In this fashion, if the equivalence *information = data bearing meaning* holds true, as stated by a large number of scholars in the field, then we would be able to apply most of the features listed above to the notion of meaning. However, a careful consideration of the characteristics outlined above reveals that those definitions do not fit very well the notion of meaning. In fact, in the context of the information system field, we can mention at least three points for which the assimilation of information to meaning is misleading.

First of all, meaning is not entirely controllable, because if it were, we would have been able to forecast meanings produced by machines and avoid unpleasant events such as those of Tay or Facebook ML.

Secondly, if meaning were controllable, it would have been easy to formalize it with statistical paradigms. In addition, there exists a clear confusion between the two definitions of data as a *flood of signals coming from the external world without being preprocessed* and *data as bearing meaning*. Effectively, if it is true that information is a sort of juxtaposition of data with meaning, the notion of meaning holds still undefined.

Finally and most importantly, if information systems are mainly engineering devices, it is also true that the *semantic aspects of communication are irrelevant to the engineering problem* [32]. In fact, it is not a coincidence that Shannon intentionally chose to ignore the aspect of meaning in communication when suggesting that:

> The fundamental problem of communication is that of reproducing at one point either exactly or approximately a message selected at another point. Frequently the messages have meaning; that is they refer to or are correlated according to some system with certain physical or conceptual entities. These semantic aspects of communication are irrelevant to the engineering problem. The significant aspect is that the actual message is one selected from a set of possible messages [32].

Hence, from Shannon's point of view, an information system does not necessarily involve the meaning aspect. As we will argue in the next sections, this is a very insightful intuition that seems to be neglected by the community working in this field.

At this point, the fundamental question we should ask is: What does the notion of *information machines* stand for? What is *information* at least from an engineering and computer science point of view? These questions suggest a new paradigm for many reasons.

A new attempt at a definition for information

Considering the nature and characteristics of information, we would like to suggest a new way of interpreting it. This novel viewpoint should take into account and involve a semiotic dimension of information. From an engineering point of view, we posit that it is misleading and unnecessary to involve the semantic dimension, but it would be enough to dwell and base the information on the notion of semiotics.

For the reasons discussed so far, *information should no longer be considered as data bearing meaning but rather as signs bearing data*. Although meaning and sign are tightly connected, as they support each other in a mutual information exchange [16], the sign, i.e., the *carrier*, cannot be reduced to being considered the *carried* [25]. Thus, we are able to produce meaning through specific signs *evoking* some meaning. Or, stated in Saussurean terms [31], the *signifier* is the element by means of which the *signified* is given. It is particularly important to understand the distinction between signifier and signified for the purpose of the unlimited semiosis [17] as, in this framework, each sign *signifies* a *signified (meaning)*, which, in turn, constitutes the *signifier* for another *signified* . The role of the *signifier* is most accurately described in Atkin's book [7]:

> Consider, for instance, a molehill in my lawn taken as a sign of moles. Not every characteristic of the molehill plays a part in signifying the presence of moles. The color of the molehill plays a secondary role since it will vary according to the soil from which it is composed. Similarly, the sizes of molehills vary according to the size of the mole that makes them, so again, this feature is not primary in the molehill's ability to signify. What is central here is the causal connection that exists between the type of mound in my lawn and moles: since moles make molehills, molehills signify moles. Consequently, primary to the molehill's ability to signify the mole is the brute physical connection between it and a mole [7].

The sign is thus a sign-vehicle granting access to the next dimension called "meaning" but it does not contain or bring meaning in itself. A message is composed of a set of symbols (or signs in the semiotic sense), such as the letters of the alphabet, but also any other type of sign. Hence, we can paraphrase the notion of *information as symbols bearing a flood of sensation representable in bits, where a bit is a unit of measure of the freedom of choice in selecting a message* [36]. Since data are defined as the flood of signals coming from the external world without being preprocessed, we argue that *information measures the degree of freedom in selecting data*. The degree of freedom is completely meaningless unless data reach some *agent*. The *agent* obtains this freedom by *interpreting* the data it receives. This procedure, which has been extremely well-documented in cognitive psychology [19, 20, 36], leads the agent to *transform* information into knowledge.

As Weaver clearly states: *Information must not be confused with meaning* [37]. This point of view is shared by the majority of authors belonging to Shannon's line of research, and it could be summarized with the following quotation:

> Information is converted to knowledge once it is processed in the mind of individuals and knowledge becomes information once it is articulated [4, 33].

If we assume that information is the measure of the degrees of freedom in selecting data, this is precisely what happens in machine-learning techniques. In fact, in order to train an algorithm, a matrix of examples/features is given as raw data input, i.e., a flow of signals encoded in bits. Then, the agent, by means of a machine-learning algorithm, "chooses" among these data and selects which data are useful based on the most discriminating features with the objective of approximating the target value. This selection process leads to a rate of information, which gives a measure of the degrees of freedom that the agent practices in selecting the best data for a specific task.

From information to knowledge

In the light of what has been argued so far, information appears to be completely meaningless unless it reaches an interpreting agent. By interpreting the data, the agent exercises his freedom and thereby creates knowledge. This is a complex process, and even though many scholars have been involved in exploring it, it is still far from being fully understood. Information deals with the transmission of data, while knowledge is more than a simple transmission of data and mainly involves the organization of information. Knowledge concerns what we can create and organize with information, and it allows us to model the world internally. It should also be recognized that every interpreting agent belonging to the category of human agents has a distinctive element known as "tacit knowledge." The most famous aphorism by Polanyi [27], *we know more than we can tell*, essentially contains the meaning of the notion of tacit knowledge. Tacit knowledge cannot be codified mathematically as information but involves the acquisition of personal experience. It is entirely embedded in culture (for instance, a regional culture, organizational culture, or social culture), and it is very unlikely for it to be shared and understood by people not embedded in that culture.

Knowledge is often considered to be a shared context for people in which ideas and experiences can be organized. Moreover, knowledge and meaning are two faces of the same coin: both contribute to the communication chain, in a way that referring to knowledge implies referring to meaning, and vice versa. As argued in the previous section, knowledge is an emerging relation among three elements:

- The world from which experiences come.
- Thoughts by means of which an agents can internally organize experiences and create meaning.
- *Signs* used by interpreting agents for expressing the relationship between the world and thoughts.

Thus, knowledge is an emerging phenomenon in which an interpreting agent and a set of symbols interact in order to create meaning. One of the most interesting studies examining this phenomenon is represented by Eco's "The open work" [17]. In this essay, Eco argues that a work is open when it does not convey a closed message, that is, when the meaning of the transmitted message is not entirely defined. Interpretation is the process by which a cognitive system produces a meaning-bearing structure from a set of symbols. Performing an interpretation of a set of symbols in a semiotic context involves interpreting systems of signs as they are used by a community of interpreters (culture, group, or organization) in a given place and time, and who share a specific "tacit knowledge." Therefore, the meaning of a sign varies considerably if we change the interpretative circumstances surrounding it. As Umberto Eco states [17]:

> Pousseur has observed that the poetics of the "open" work tends to encourage "acts of conscious freedom" on the part of the performer and place him at the focal point of a network of limitless interrelations, among which *he chooses to set up his own form without being influenced by an external necessity* which definitively prescribes the organization of the work in hand. At this point one could object (with reference to the wider meaning of "openness" already introduced in this essay) that any work of art, even if it is not passed on to the addressee in an unfinished state, demands a free, inventive response, if only because it cannot really be appreciated unless the performer somehow reinvents it in psychological collaboration with the author himself.

Unfortunately, machines are not (yet) able to perform this "free" act. Although, technically, the machine receives the information "as the active center of a network of inexhaustible relations," it is not able to process this information "without being determined by a necessity that prescribes the definitive ways of organizing the work." An information system is *by necessity* a determined system without any degrees of *freedom of consciousness*. All it is allowed is to do is to *determinedly choose* the degrees of freedom in selecting the most informative data for a specific task. Meaning is an extraneous element for information systems. In fact, it is an emerging phenomenon where cultural units, that become contents of possible communications and that are organized, emerge into structures (fields and semantic axes) that follow the same semiotic rules identified for systems of signifiers.

> A cultural unit cannot be isolated merely by the sequence of its interpreters. It is defined in as much as it is placed in a system of other cultural units which are opposed to it and circumscribe it. A cultural unit exists and is recognized insofar as there exists another one which is opposed to it [16].

From this quote, we can infer that every cultural unit is an element belonging to a system of other cultural units, which define themselves precisely by means of these relationships. In other words, and in order to give a clear example, consider the cultural unit "car." "Car" is not just a semantic entity when it is related to the signifier *car*; it is also inasmuch as it is arranged on an oppositional axis with other semantic units such as "chariot," "bicycle," or "foot." This is more than just a distributional semantic phenomenon [30], it is also a fully-fledged *distributional semiotic phenomenon*. As Umberto Eco argues extensively in his book [16], the object *car* becomes the signifier of a semantic unity that is not only suggestive of "car" but can be, for example, suggestive of "richness" or "wealth." In the same way, the object *car* becomes the signifier of its possible uses. Thus, the object holds a significant function at *both social and functional levels* to make every cultural phenomenon investigatable in its functioning as a signifying artifice. Therefore, culture can be fully studied under a semiotic profile and not just as a semantic phenomenon as has been done so far.

When we refer to a communication system, we basically refer to a language game [41] that is imposed by a speaking community, belonging to a particular cultural unit in which the *interpreter acts in relation with a system of cultural units in a conscious freedom*.

Only the day in which machines will reach such a degree of acting *in relation with a system of cultural units in a conscious freedom,* will we be allowed to truly speak about an artificial intelligence. What we have reached so far is an artificial optimization of human intellectual or practical activities.

Definition of wisdom

As we widely argued in Sect. 2, defining wisdom in the information system field, as well as in the knowledge management field, is a knotty problem to the extent that many authors do not even attempt to define it. We recognize the difficulties in exploring its nature but, in the light of the analysis in the previous sections, we can affirm that wisdom is tightly related to the notion of knowledge, although it cannot be reduced to it.

In order to clarify the notion of wisdom and give a perspective definition of it in the field of information systems, we should refer to the philosophical tradition. Wisdom has been the subject of discussions and explorations for many years throughout the history of philosophy; even the etymology of the word philosophy itself has a clear reference to the concept of wisdom. Therefore, it would be odd not to refer to the philosophical heritage.

Although the theme of wisdom can be found in almost the entire history of philosophy, Aristotle's reflections are of particular importance to our discussion. Aristotle was a polyedral thinker who investigated several topics, among them the notion of wisdom. From a high-level perspective, the philosophy of Aristotle can be divided into two parts: on the one hand, theoretical philosophy relating to the knowledge of the external physical and metaphysical world and, on the other hand, practical philosophy having as its purpose the reflection on everything concerning the human being: his actions, his productions, and his social and political relations. For the purposes of this paper, we will specifically focus on the aspects concerning Aristotle's practical philosophy.

Aristotle's practical philosophy is essentially *"he peri ta anthropeia philosophia,"* i.e., the philosophy of human affairs, which takes into account the freedom of deliberation of each individual on his behavior. In this regard, a practical philosophy is a search for practical truth, which for ancient Greeks was equivalent to looking for "wisdom". Thanks to it, human agents are able to decide which are the most suitable means to reach a certain purpose. In other words: wisdom moves with respect to *contingency*, taking into account that the same "right and good" scopes cannot be achieved by the same means and behaviors at all times and in all places. In this sense, wisdom seems to depend very much on cultural units and the tacit knowledge in which it is embedded, as discussed in Sect. 3.3.

In particular, at this point, it is necessary to mention Aristotle's work *Nichomachean Ethics* [6], in which Aristotle makes an important difference with respect to ethics, as relevant to our discussion. In fact, he distinguishes between action (*pràxis*) and production (*poíesis*), where the *poíesis* is the action directed to the production of an object that remains independent and alien to the one who produced it, whereas *pràxis* does not involve an action directed to the ultimate end of production; instead, it is the action carried out for its own sake; it is the action that is good just for itself. *Pràxis* is supported by *phronesis,* which was regarded by the Greeks as wisdom, in the sense of intellectual faculty.

The field of philosophy, wanting to deal with the good of mankind and named wisdom, guides men in taking care of their actions and politics; wisdom helps men in the search for good and good living in the dimension of the city and the state.

With respect to our topic and adopting the Aristotelian vocabulary, artificial intelligence machines are merely *poietic* production, supported by technical aspects (*techne*). Artificial intelligence algorithms are produced based on the advancements of *technology;* hence, these algorithms are not able to take into account the same "right and good" objectives, as they totally lack the "wisdom" (*phronesis)* enabling them to assume the "right and good" behaviors at all times and in all places.

As we discussed in Sect. 1, intelligent machines are, after all, not so intelligent. We will likely have to be more patient in order to see a proper intelligent machine. In fact, a real intelligent machine should not only be a technical product in the proper Aristotelian sense but an in-and-out wise machine. We should concentrate our research efforts on creating machines that are aware of *peri ta anthropeia,*, i.e., aware of human behavior as it is guided by beliefs resulting from the cultural and environmental situations that humans have faced throughout time, immersed in contextual and contingent situations issued from a network of cultural units (Sect. 3.4). At the same time, beliefs and experience lead us to form and conceive systems of values that are intrinsically oriented to our needs. Values are intimately related to our needs, and they change along with our psychological development. Values are human-oriented, and they guide us in identifying the *right means* [6] to obtain well-being. Based on the contents presented above, the main issue we are facing in the creation of intelligent machines is not the availability of more and larger pools of data; rather, it is the need to shift towards a new approach for training inputs. Nowadays, the most powerful machine-learning algorithms are able to learn complex nonlinear relationships between input and output characteristics. This greater flexibility and power has the cost of requiring more training data, often much more data. Wisdom is still out of reach today; nevertheless, scientists should broaden their view and stop focusing exclusively on data (like social-media data). Conversely, they should adopt a more comprehensive approach that integrates also data related to *homogeneous* and *uniform cultural units*.

The wisdom era where a poem is worth more than 1 million tweets

At this point, in order to shift towards *distributional semiotics,* we need to seriously consider other forms of data that would be able to represent *both social and functional* levels and not only the semantic level.

We are aware that the technological advances to make a machine knowledge-aware and wise also depend on the development of hardware in future years and, in particular, on an in-depth understanding of the psychological aspects that underlie the poetic and poietic production of human interpreters. Although research in this sense still has wide limits and aporia, we should at least begin to study more carefully the level of representativeness of the data, so that they are increasingly close to the description of elements useful and significant to capturing cultural units and not only semantics. While we are aware that they are not sufficient, we can still speculate that digital humanities and digital heritage initiatives play a fundamental role in achieving this goal. In fact, the link and interconnection between cultural heritage and cultural units is extremely tight. Any production made by human poetic or practical activities represents and creates our cultural heritage. Cultural heritage and humanities are, thus, the imaginary places where cultural units are born and grown. They are the best approximation to represent cultural units *à la* Eco. Hence, the digitalization of productions in humanities, is, from our perspective, the next step to enhance information systems and guide machines to become truly wise machines.

In fact, the digital humanities are, for the moment, the only means that could allow us to capture the elements of meaning not only as semantic units but as real cultural units inscribed in a network of "open works." This shift from semantic to cultural units brings with it a greater degree of semiotic representativeness. In fact, it is important to focus on the *degree of semiotic representativeness* of data.

If, for example, we consider the use of tweets as semantic units widely used to create corpora in order to train intelligent agents for sentiment analysis [1, 2, 23, 26], we can easily observe how the degree of representativeness of some cultural units, such as the feeling of depression, is expressed in a qualitatively different way in tweets compared to the literature. So, for instance, let us consider just a few tweets like the following:[2]

- Sleeping for 16 hours and still being tired.
- Not wanting to get up, bathe, or talk to anyone.
- Remembering your adventurous self yet not being able to imagine going on any adventures in the future.
- Agonizing pain, physical & emotional. Extreme sadness, worthlessness, severe agitation, irritability.

Although these tweets about the desperation of human beings carry a certain degree of semiotic message, they hardly compare with the degree of deepness and representativeness expressed, for example, in Thoreau's work, *Walden*:

> The mass of men lead lives of quiet desperation. What is called resignation is confirmed desperation. From the desperate city you go into the desperate country, and have to console yourself with the bravery of minks and muskrats. A stereotyped but unconscious despair is concealed even under what are called the games and amusements of mankind. There is no play in them, for this comes after work. But it is a characteristic of wisdom not to do desperate things [35].

The concepts expressed by Walden yield much more semiotic power on us because of the variety of the semiotic dimensions mediated by the poetic vehemence. By means of its powerful existential contingency, we can get his meaning not just by a *cognitive* but a *sympathetic* learning process. A machine could certainly learn many facts about this book and its author. It could *know* about the transcendentalism movement in the USA or about the influence of Emerson's thought on such a movement; nonetheless, a computer would miss the *practical wisdom* embedded in Thoreau's message, because it is mainly addressed to mankind as moral advice. The poem carries a message to tell us a way to *feel* and *behave* with respect to the cross-time society.

> Most men, even in this comparatively free country, through mere ignorance and mistake, are so occupied with the factitious cares and superfluously coarse labors of life that its finer fruits cannot be plucked by them. Their fingers, from excessive toil, are too clumsy and tremble too much for that. Actually, the laboring man has not leisure for a true integrity day by day; he cannot afford to sustain the manliest relations to men; his labor would be depreciated in the market. He has no time to be anything but a machine [35].

The concepts expressed in this excerpt cannot just be learnt and known, they have to be *felt* and *experimented* by a sensitive-embodied agent in order to be fully understood. In these lines, the author does not just want to convey knowledge but a real wisdom, that is, a warning about the awareness and the practical way in which we lead our lives in order to avoid to "be anything but a machine."

Cultural heritage and masterworks from humanities are not just pieces of information. They are cultural-unit carriers grasped by means of a practical behavior that allows us to register sensations and feelings through our bodies. Thanks to this embodied process of ethical conduct, we are capable of understanding Thoreau's caveats and feel them as true and valid (at least for some part of the interpreting agents).

As we have seen, poems and literary works as well as paintings bring about several semiotic dimensions able to capture *both social and functional levels;* as such, they con-

[2] These tweets are retrieved using the hashtag #ThisIsWhatDepressionFeelsLike at https://twitter.com/hashtag/ThisIsWhatDepression FeelsLik?src=hash.

tain and represent the *praxis* dimensions in a truly Aristotelian manner (Sect. 3.4). In fact, the social and functional levels take into account the analysis of Aristotle's actions and politics, which is the search for good and good living in the dimension of the city and the state. In this sense, this is precisely what Thoreau is suggesting. He encourages us to be aware of our daily life, as there exists a meaningless and contentless way of living and behaving. While tweets are the expression of an individual or, at best, the expression of a particular social community, literature is the expression of a cultural heritage belonging to mankind. It is the patrimony of an entire species since it touches values and beliefs belonging to all human beings. A poem is worth more than 1 million tweets simply because a poem conveys a profounder *degree* of cultural semiotic values than a tweet.

Hence, since intelligent machines such as Microsoft Tya and the Amazon Recruiter Robot are, first of all, *behaving* machines, they must take into account a *practical* dimension. As long as we keep neglecting the semiotic dimensions that are intimately intertwined with behavior, we will continue to face the ambiguity that it is typical of contemporary science, in which paradoxical behaviors rest on assumptions that are based on purely rational processes and analyses, such as computers and bots manifesting some inappropriate, if not irrational, attitudes. Therefore, in order to avoid unpleasant behaviors such as those manifested by Microsoft Tya and the Amazon Recruiter Robot, we should seriously consider to advance research in a new direction that looks at methods able to capture and model semiotic aspects and cultural units. This approach will consequently allow us to modify machines' behavior in order for them not to make frantic decisions because *it is a characteristic of wisdom not to do desperate things*.

Conclusion

In this paper, we have reviewed the DIKW pyramid model, shedding new light on each component. In particular, examining the engineering point of view, we have focused on the definition of *information*, giving to it a new conceptual framework. If tradition has always considered information as data bearing meaning, in this paper, we have argued that information is not meaningful. In fact, from the analysis of Shannon's studies in communication engineering, we highlighted how the notion of meaning is not necessary for the definition of information. It follows that we have to direct our research in other directions in order to find a sustainable conceptual theory able to provide new insight into the matter. We have shown how it will, therefore, not only be necessary to carry out a semiotic revolution to be able to introduce meaning into the communicative act, but it will also be necessary to introduce the figure of an interpreting agent. Thanks to the interaction between such interpretative acts, which take place in a conscious freedom *à la Eco*, cultural units emerge.

In other words, any verbal or written expression never speaks of a specific object but always conveys a larger cultural content. As far as we want to conceive and describe the meaning of a sign, Eco believes that the problem of the referent, that is, the states of the world that sustain the content of the signic function, is not relevant within the semiotic theory. Instead, it is better to think about meaning in the light of cultural units. The semantic system is, therefore, configured as a set of units that stand out from their positions, their oppositions, and their differences.

We should thus address part of the research to new forms of data in order to facilitate such a semiotic revolution. In particular, digital humanities and cultural heritages can funnel a new type of data for which representativeness has a greater degree of quality.

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HAUPTBEITRAG

Blockchain and society

Toni Caradonna[1]

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Abstract
In this text, the origin of blockchain technology is explained and put into the context of three waves of digitalization: digitalization of information represented by the internet, digitalization of communities and community building tools represented by social media and digitalization of value represented by blockchain. The text will show that blockchain can bring back promised features of digitalization of the first two waves, features that were lost with centralization tendencies created by business models that are independent of the underlying technology. After a short introduction to the terminology, a map of the potential blockchain has for society is sketched. The text also debates the decentralization aspects of the technology and the challenges it poses for individuals and organizations. Later, the storytelling aspects, as well as some religious aspects within and around the blockchain community are analyzed. Finally, some thoughts about adoption scenarios are laid out, and some ideas why blockchains have become very important in Switzerland are formulated. The focus of this text is on the opportunities and strengths of the technology. It deliberately oversimplifies the technological aspects in order to focus on the implication of the technology on society.

Blockchain—etymology

The term “blockchain” and Satoshi Nakamoto’s whitepaper on bitcoin[1] are often associated with each other. Yet the term “blockchain” never appears in the paper that sketches the characteristics of the data structure associated with blockchains. The bitcoin white paper refers to a “chain of blocks”, and only in late 2008 was the term “block chain” in two words used in a cryptography mailing list to refer to the data structure Satoshi Nakamoto had described. However, the term “block chain” was used in a similar way as early as in 1997 in mailing lists, but it did not relate to the data structure bitcoin represented. These mailing lists were connected to the cypherpunk movement in the bay area near San Francisco. The topic of the discussion was a tool that was created to reduce spam mails. The cypherpunk movement can be seen as the community that generated the debates about society and technology that made blockchain technology possible in the first place.

In this context, it is worth mentioning the cypherpunk manifesto, written by Eric Hughes in 1993. It shows that the origins of the cypherpunk movement were related to privacy in an open society in the digital age and about the possibility of creating transparency.

“Privacy is necessary for an open society in the electronic age. Privacy is not secrecy. A private matter is something one doesn’t want the whole world to know, but a secret matter is something one doesn’t want anybody to know. Privacy is the power to selectively reveal oneself to the world.”[2]

Blockchain—the context

The blockchain as a phenomenon can be seen and interpreted in the light of the digitalization process of the world. Since the 1950s the planet has been getting more and more digital. There are many ways to structure and enumerate these waves of digitalization. We propose to enumerate the waves of digitalization relevant in the blockchain context as follows:

The first wave is represented by the digitalization of information facilitated through the introduction of the internet. The second wave is the digitalization of community building and communities through social media. Finally, the third wave is the digitalization of value with the introduction of blockchain and distributed-ledger technologies.

[1] https://bitcoin.org/bitcoin.pdf.

[2] https://www.activism.net/cypherpunk/manifesto.html.

✉ Toni Caradonna
toni@caradonna.ch

1 Bern, Switzerland

Looking at waves 1 and 2, I will identify similarities and compare them with the development of the current third wave that is represented with blockchain. I will show that this third wave could bring back some features relevant to society that have been washed away by the first two waves.

Digitalization, decentralization and disintermediation

The narratives around blockchain technology have similar traits to the narratives and developments in the early days of the Internet and of social media. It is the narration of a new promising technology with an enormous potential for disruption that enters the world to make it a better place.

The elements that are relevant and similar in all three digitalization waves are: decentralization and intermediation. In the three digitalization waves, internet, social media and blockchain, we can see a first phase of digitalization that is connected to a physical disintermediation, which leads to a decentralization. Later the pendulum swings back, bringing a centralization element based on a business model that is built on top of the technological innovation. Usually the added value that the centralized platform models brings is a combination of convenience and added value through networking scaling effects. While it is possible to have a decentralized lifestyle in a centralized environment, users usually have to make certain tradeoffs: privacy and autonomy for convenience, freedom for the access to mediation services. This dynamic nature of the digitalization processes in the centralization versus decentralization spectrum depends on a variety of factors, which will be highlighted later in this text.

Digitalization of information—the Internet

The first wave of digitalization that hit mainstream society heavily was the digitalization of information. With digitalization of information and the availability of the Internet, information has been disintermediated from books, teachers, universities, libraries, institutions and authorities of all sorts. Information can be multiplied, altered, shared and distributed with very few transactional costs. The digitalization of information and the disintermediation from the physical representation of information enables users to act independently from the origin or the creator of the information. Access to information and the possibility of trading and creating commerce has improved drastically.

Back in the 90s the internet was a big promise of a better world. The utopia is probably best condensed in the declaration of independence of cyberspace[3]. In its early days, the Internet was the promise of a shared network of individually owned servers and domains in which everyone participates equally, a decentralized web that would empower people. There were promises, dreams and expectations of freedom, independence and user-centric lives. Promises over ownership over data. However, we have to face the reality that we woke up in a cyber-dystopia, a quasi-totalitarian digital world that is dominated by companies that are more powerful and have more influence on society than many governments. We must face the possibility that technological advancements have made the world worse. In cyberdystopias, individuals lose control, become dependent and are unable to stop the change that is happening. Even Tim Berners-Lee, the inventor of the Internet, is calling for action and working on a project to bring back the decentralized aspect of the Internet that should empower users.

This first wave of digitalization where the Internet increases access and proliferation of information is creating a tendency towards information symmetry. On the other hand, control over digital information is very hard to handle. It is difficult to keep ownership over data, e.g. copyright, user-generated data and sensitive data that is meant to be private. It is also difficult to keep track of data iteration and the origin of data, and it is difficult to keep data access restricted. Last but not least, the disintermediation of data from the physical world creates a challenge when we talk about identity. In a globalized and digitized world, we depend heavily on trusted third parties to navigate, act and interact in the digital world, and we must trust those parties when we organize cooperation in our society.

In terms of adoption of this first wave of digitalization, we can see parallels to what is happening with the adoption of the blockchain technology. The innovation arrives, but nobody knows how to use it or how to generate business with it. Typically, a business model is not necessarily connected to a feature of the technology it uses. In the first wave of digitalization, the business models that were most sustainable directly opposed the features that the technology offered and the utopia promised. Centralized services created additional business value through monetization of information and through making services and applications more convenient. Exactly this centralization that was needed for the success of the business models ended the marvelous utopian ideas created by the technology in its early days.

Innovation adoption is often solution oriented and influenced by the cognitive bias that involves over-reliance on a familiar tool, which is illustrated by Maslow's hammer [1]. With the advent of innovation, society has a new tool but needs an ecosystem and infrastructure around it. New technology needs a legal setting that supports it and education on how to handle it. All these processes need time and, ideally, a debate within society. Eventually, then, these debates and processes will decide whether or not the feature

[3] https://www.eff.org/cyberspace-independence.

that a technology offers can be implemented. The invention or discovery of technology itself does not at all mean that its features can or will be used.

Digitalization of communities—social media

If the Internet was the digitalization of data, social media is the digitalization of communities and community building. In analogy to the first wave of digitalization disintermediation has several effects. With the advent of social media communities do not have to physically gather or use physical intermediaries or tools to interact. Again, the process of disintermediation allows a physical decentralization of communities by mapping communities and their interaction digitally. The acceleration that this disintermediation allows is more convenient and easier to access than traditional community building tools. It generates added value for users and the owners of these digital tools through network effects. Social media tools were perceived as an enabler for bottom-up movements that can change society. However, again, this utopia ended up in dystopian frustration with social media being used and abused for controlling and influencing society. Further, because these tools are based on a business model that incorporates a central intermediary, it also means that users have to both trust, and finance this intermediary third party directly or indirectly.

When looking at community-building tools, one of the key issues is the identity of community members and the role of an identity. In the digital world, this translates to access control and digital identity that still is a challenge in our digital world. In a large part of the world, users delegate many processes, including identification and payment services to the "Big Five" Google, Apple, Amazon, Microsoft and Facebook, and we accept them as trusted third parties when we interact with one another digitally. By doing so, users give away power and means of control over the data and metadata they create, and they give away power over the structure of the digital community.

Also, we still have the problem generated from the first wave. The problem of continuity, origin and authenticity of digital data. Combined with this second wave of digitalization we create a new challenge. An army of bots, algorithms and virtual digital identities influence us and our perception of the world, and even influence processes in our society like voting.

With the internet and social media, we now have two tools for society to act in the digital realm. Looking at the problems mentioned above, society might need a third tool that solves all the problems that were created in the first place. A tool that brings back control over data, a tool that can create and ensure transparency and authenticity, a tool that can bring privacy in interactions.

What if we could think of a digital interaction and cooperation tool that puts the user in the center? A tool that enables users to interact with one another securely without the need of a trusted third party and without the need to trust each other. A tool that puts the users in the center and not the platform or the company that runs the platform. A tool or model that empowers the user and not the organization running the platform, or even a model, where the users cooperate to run the platform. A model where users have control and ownership over value they agree on, over data they generate and transact, as well as control and sovereignty over their digital identity.

Welcome to the Blockchain—Utopia, which brings back to us what we lost when the Internet and social media were created.

Digitalization of value—blockchain technologies

While the Internet was about digitizing information and social media was about digitizing communities, the origins of blockchain technology were about digitizing value and creating a decentralized network to map and transfer value independent of trusted third parties. Blockchains address a very specific problem: the problem of digital continuity. Before blockchain technology, digital continuity could not be guaranteed without the help of a trusted third party. Anyone could copy and paste data and multiply it; there was no way a user could keep track of data changes. With the advent of blockchain technology, this is now possible. This will have many implications that we will only slowly start to understand. It means, for instance, that digitized assets could become independent economic agents. We can now keep track of data alteration and create transparency in dataflows. It enables something that simply could not be done before the advent of this technology.

Blockchain technology triggers similar utopian dreams to the two first waves of digitalization. Unfortunately, in the blockchain technology adoption, we see a similar backlash to in the two other digitalization waves. The real innovation of the new technology wave—empowerment of participants through disintermediation—is undermined by additive layers and business models that create centralized structures.

Digitalization challenges

There seems to be little awareness for the fact that digitalization created new and relevant challenges on a global level. There is no or little awareness that digitalization resulted in an accumulation and centralization of power and influence beyond most people's imagination. The centralization of the digitalization processes brings certain standards, which, in turn, brings a certain level of convenience.

Further, because convenience seems to be the tradeoff for handling value, it is not clear whether mass adoption of blockchain technologies is possible to implement on a peer-to-peer level in society.

The fact that society needed trusted third parties in the non-digital world when cooperating on a large scale created two things. It created large organizations that did accumulate a certain level of trust on the one hand. Moreover, it created large communities that are used to interacting with one another other through trusted third parties. As consequence, many ecosystems could be established with checks and balances that ensured sustainability for these trusted third-party ecosystems. On a personal and human level, the need to trust someone might be a fundamental human trait or need that cannot be overcome by rational arguments. As consequence, the concept of trusting an algorithm or a technology might be hard to communicate, hard to explain and hard to do as a human.

Users or organizations might be incapable of handling the technology and need third-party support and/or education. Third-party support is usually structured and organized by a business, which, in turn, is interested in keeping a certain degree of control and power.

In order to avoid that, you have to handle responsibility, and as a user or organization you need three things. Knowledge and knowhow to understand and handle the technology, a sense of responsibility and the willingness to act upon it and, finally, a user or an organization needs a certain frustration tolerance to cope with the efforts that come with behavioral changes associated with changes of paradigms.

Probably the perpetual conflict and the balance-finding process between decentralized tendencies versus cooperative processes that drift towards a certain degree of centralization will always be dynamic. However, for now, we find ourselves rather on the dystopian side of a centralized digital world. We have to suffer the consequences of poorly-managed centralism, which include massive data breaches, crippling financial crises, digital propaganda and global tax competition.

Digitalization summary

All three digitalization waves above started with the use of a technology infrastructure designed for a decentralized, digital world that would empower users. We see tendencies that the digital tools built on the infrastructure still quite often function in a centralized manner. Since we are still in the very early phase of the third digitalization wave, we still have the opportunity, and I may say responsibility, to actively decide on how to adopt the blockchain technology. Probably, the challenge is not in the technology but in educating the user, in creating a sense of citizenship, in provoking an active decision and avoiding the scenario, where the desire for convenience prevents knowledge about the technology and prevents a conscious decision.

Or as Paul Goodmann [2] puts it: “Throughout society, the centralizing style of organization has been pushed so far as to become ineffectual, economically wasteful, humanly stultifying, and ruinous to democracy. The only remedy is a strong admixture of decentralism. The problem is where, how much, and how to go about it” [2].

What is blockchain?

Despite normative and descriptive efforts in defining terms and definitions, the consensus about “blockchain” and “distributed ledger technology” (DLT) taxonomies and ontologies is only slowly being reached. However, when comparing literature [3–5] a wide range of coexisting inconsistent definitions can be observed. Since there is an economic potential in blockchain and DLT applications, the dispute within several standardization authorities is being fought very fiercely.

What is important to understand is that a blockchain is nothing but a data structure. The problem the bitcoin whitepaper tried to solve was the digitalization of cash. That leads to several problems. If you have a digital representation of some kind, a dataset, a picture for example, you can easily duplicate that dataset and have two digital representations, two identical datasets, two pictures. Obviously, this is something you want to avoid when digitizing cash. So, how can you ensure that you cannot simply copy and paste a bitcoin and multiply the number of bitcoins on your computer?

Of course, a central authority could run a ledger listing all transactions and ensure that participants do not tamper with the dataset. However, the bitcoin whitepaper was written as a reaction to the 2008 financial crisis. The ambition of the anonymous author publishing the whitepaper under the pseudonym Sathoshi Nakamoto was to create a peer-to-peer electronic cash system that is independent of trusted third parties like banks. If you had a centralized solution of some kind, you would need to trust that party. The consequences of trusting a third party are manifold. A centralized solution creates a honeypot for attackers, and you would need confidence that the trusted third party can handle security issues because a centralized solution is also single point of failure. Furthermore, a centralized solution cannot prevent that a payment is reversed, and the central authority would receive the power to control, stop and or alter transactions. Finally, a centralized solution has the power over decisions like creating or reducing inflation by regulating the amount of cash in the system. You could argue that a centralized solution creates higher costs because of the intermediaries and would put users at the mercy of decisions of centralized institutions like central banks.

The technology that solves this trust problem is a peer-to-peer network with a decentralized distributed ledger. It delegates trust from trusted third parties to algorithms, and the control over the infrastructure is in the hands of the peers. The decentralized distributed ledger ensures that a peer does not spend more than he has or that he does not reverse a transaction. Because every participant has a copy of the ledger, and the ledger lists the history of all transactions, a single peer cannot pretend to own more bitcoins that he has. To oversimplify again: copy/paste of a bitcoin is not possible because if a bad actor tries to cheat in his ledger, all the other peers with a copy of the ledger would disagree with the bad actor's version of the ledger. The majority of the peers decide what version of the ledger is the valid history of transactions.

The decentralized peer-to-peer database solution solves the trust problem. However, it creates other problems. How do the peers decide which transactions are stored? What majority-decision processes would you implement? How can you ensure that bad actors do not create majorities by simulating additional peers representing the bad actor's version of the ledger? How do you incentivize peers to run a such a digital ledger? Most importantly: how does the network ensure consensus is reached within a group of decentralized peers in a digital network that is not synchronized by a central authority? Bitcoin solved all these problems using a consensus algorithm called Proof of Work. This algorithm is designed to force participants to act fairly in order to participate at all. The mechanism itself relies on cryptographic techniques. For the scope of this text, it is not important to understand the mechanism of this algorithm. What is important is that this algorithm takes time and consumes energy. What is also important to understand is that there are other consensus algorithms that do not consume that much energy or time. However, in order to implement them, you would need to make certain tradeoffs.

Three generations of distributed ledgers

As was mentioned before, there are different ways to solve problems that are created by decentralization. On the metalevel, three elements are relevant when we look at distributed ledgers. First, we need some kind of structure as a database. Second, we need some instructions for handling that database structure, usually referred to as consensus mechanism. Finally, we need some motivational tool for peers to participate. That motivational tool can be a native intrinsic asset like a bitcoin reward for successful cooperation. However, the motivational aspect for participation can be well outside the database construct itself. In the scope of this text, we will not look into the motivation tool aspect at all.

The database structure can be a chain of blocks that registers valid transactions. However, a database can have a different structure, so blockchains are only a subset of the family of distributed ledger's technologies. For the sake of simplicity, I will refer to blockchains or blockchain technologies to indicate community consensus-based distributed ledgers in general. So, blockchains can be designed in different ways and can be adopted to the problem they are trying to solve. The development of designing different blockchains can be structured in three generations. The first generation of blockchains enabled digital cash, with bitcoin being the first cryptocurrency. Several variations were created, each of them being a decentralized peer-to-peer network solving digital currency-related problems.

The second generation of blockchains created the possibility of having a programmable blockchain that enabled the implementation of business logic into a decentralized system. You could oversimplify again and say: Why should we have a multitude of networks all creating slightly different cryptocurrencies? Is it not possible to have one network, one infrastructure that allows programming, and one of these program types could be a cryptocurrency?

Programs that run on a blockchain are often called smart contracts. The idea was first mentioned by computer scientist and former law professor Nick Szabo in 1997 [6]. It is a term that became popular when Vitalik Buterin, the inventor of Ethereum, adopted it. Smart contracts are agreements enforced not by law, but by hardware or software. However, translating the term "smart contract" into "program on a blockchain" can help to read certain literature but hides the revolutionary fact that in a smart contract, the contract itself, as well as the elements of the contract are in the same channel. If you compare this to the analogue world, it is not possible, you cannot have an executing piece of paper, you always need to involve trusted third parties for execution or enforcement. In the digital world, contracts either fully execute or do not fulfil the agreed conditions and do not execute. With smart contracts you can prevent two situations: first, the situation that only one party fulfils the contract, and second, you can prevent that either party only fulfils the contract partially. So, for objects that can be represented digitally, smart contracts significantly reduce litigation costs and increase confidence.

The third generation of distributed ledgers moves away from chains of blocks as a database. Typically, third-generation blockchains allow better scalability in terms of transaction volume and/or the possibility that certain branches of the distributed ledger are disconnected. It is important to know that first and second-generation blockchains do not scale very well. Bitcoin and Ethereum, for example, currently allow an order of 10 transactions per second, 7 for bitcoin and 15 for Ethereum. If blockchain technology wants to solve problems generated by the first and second wave of digitalization, blockchain technology needs to scale significantly.

However, in order to enable the scalability in numbers of transactions or to allow the disconnection of branches, there is a tradeoff. Typically, third-generation blockchains are not chains of blocks but are rather structured like meshes or tangles.

On the metalevel, there are protocols that try to connect several blockchains with each other and allow interoperability of several blockchains. A functional and user-friendly interoperability of blockchains would bring enormous benefit to the blockchain industry. Since blockchains have scalability issues and are typically designed for a specific problem, interoperability would allow both an extremely high scalability and flexibility in implementation.

Various types of blockchains

We have seen that there are three fundamentally different generations of blockchains. However, each type can have different specifications again. In order to understand the diversity of blockchain tools it is helpful to look at a few varieties of designs. We will establish a terminology and potential advantages and disadvantages of the specific designs along the way.

Brewer's theorem, or the CAP theorem, is a central element in theoretical computer science, which states that in a distributed system of nodes, it is impossible to simultaneously have more than two out of the three following guarantees or properties [7]:

- *C: consistency*—At any given time, all nodes in the network have exactly the same (most recent) value.
- *A: availability*—Every request to the network receives a response, though without any guarantee that returned data is the most recent.
- *P: partition tolerance*—The network continues to operate, even if an arbitrary number of nodes are failing.

Blockchains and DLTs are distributed system and, hence, have to cope with the CAP theorem. Blockchains are systems where nodes can fail at any time because users cannot be forced to participate. Of the two remaining options, bitcoin chooses availability over consistency. Yet there are other ways to design a blockchain that have other priorities. What is important to understand is that there are different ways to design a blockchain system. Depending on the design, we have different advantages and bottlenecks.

In the bitcoin blockchain, all participants can read the entire ledger; all participants are allowed to validate transactions and write them into the ledger, the transaction history is visible. However, other settings are possible. Here is a list of terms that seem to have some validity and consistency in the blockchain community:

- *Permissioned:* restricted write OR restricted validation permission
- *Permissionless:* public write permission AND public validate permission
- *Public:* public permission to participate in the consensus process
- *Private*: restricted permission to participate in the consensus process
- *Privacy:* obfuscated traceability.

Sometimes, terms like federated blockchain or consortium blockchain, are used, but these terms have a wider definition-variability within the blockchain community. They often refer to permissioned blockchains.

Between scalability, security, and decentralization, a tradeoff needs to be made. Public blockchains have sacrificed scalability, whereas permissioned blockchains sacrificed decentralization for security and scalability. Privacy blockchains allow us to create transactions that cannot be traced back to addresses.

Depending on the problem that is addressed, a thorough evaluation of advantages and risks needs to set up a blockchain or use an existing blockchain infrastructure.

Two ways to look at blockchain

No matter how we define the technological design and the applications of blockchain technology, the technology creates two shifts in paradigm: from "trusting humans and third parties" to "trusting machines, process design, and mathematics" and, second, from "centralized" to "decentralized" control [8]. As a consequence, there are two ways of looking at a blockchain. The technology has a disruptive potential from both an information and communication technology (ICT) perspective and an institutional perspective.

First, it can be seen as an ICT to record data and provide access to it. This data can be any data really. Being decentralized, the disruptive potential of it lies in the fact that participants do not need to trust the owner of the database. The reason is very simple. Participants themselves organize and run both: the database and the governance of the database. Each participant has a copy of a database, a so-called node. Hence, participants need only to trust their ability to handle data or code. Moreover, as mentioned earlier: when users do not need to trust a third party and reclaim control and ownership over data and data governance, they also inevitably regain responsibility. On the one hand, not everyone wants that and, on the other hand, not everybody can handle it. Very often, the price of responsibility is convenience, and ultimately, this could be an important threshold for acceptance.

Second, it can be seen it as an “institutional technology” to decentralize governance structures used for the coordination of people and economic decision-making as well as for enforcing transparency [9]. From this perspective, the interpretation of data and the interaction of the digital data with the physical world is relevant. Here, the blockchain technology is about rights/obligations arising from agreements that can be mapped with blockchain data. It is about interpretation of data as a mapping process of what happens in society. Here, we talk about currency aspects, ownership rights, copyrights, digital identities, credit exposures, and tracking of goods and services. This second aspect of blockchain, however, creates other relevant problems, namely:

The oracle problem: how do we get data from outside the blockchain into the blockchain? When we do have a digital representation of some sort within the blockchain, the question remains: how that representation entered the digital world, what authority, person, organization, or interface created the digital representation within the blockchain. There are several approaches to cope with the oracle problem that are not elaborated here.

The enforcement problem: what if we do have consensus within the data, consensus in the interpretation of the data but nobody to enforce the consequence of the interpretation? Again, here we see a dilemma when looking at decentralized, disintermediated digital structures in a physical word, a digital representation of a society with a local jurisdiction and structure that needs some sort of eventually central coordination.

The interpretation problem: what if we have consensus within the data but miss consensus about the interpretation of the data? This problem is more subtle but will be of greater importance when blockchain technology becomes more prevalent.

Blockchain narratives

Storytelling and religion both shape society. They both help establish group identity, moral code, and social norms in a community. Nobel prize winning economist Shiller [10] shows 2017 how epidemiology of narratives influences economic fluctuations. He shows that stories motivate and connect activities that spread and have an impact, like the Depression in the 1920s, the Great Depression in the 1930s, and even the so-called Great Recession of 2007–2009. Being humans, our actions are not solely rational, as we will see in a separate section about game theory. The sum of the nonrational actions can be interpreted by looking at the storytelling aspect.

Technological developments are usually wrapped in stories or narrative structures that are not only technical. The technology in the narration also encompasses collective ideas of how we should build our society, our institutional reality. Blockchain technology, however, is not merely used as an object in narrations. Peer-to-peer technologies do more: according to Agre [11] they configure the narratives through which we understand our social reality.

The storytelling aspect of blockchain

Looking at the genesis of bitcoin we have excellent elements of a good story. An interesting setting: the world on the brink of collapse during the 2008 financial crisis. An interesting character: Satoshi Nakamoto, the anonymous genius author that gifts the world with this new technology. An ongoing complex plot with conflicts between the community and outsiders, as well as conflicts within the community.

The narrative aspects of cryptocurrencies are elaborated in detail by Reijers and Coeckelbergh [12]. The most compelling point in the narration, as well as in the technology itself, is that cryptocurrencies and blockchain technology are “weapons in the new control society” [13]. This is interesting when comparing the cryptographic software developed in the early 1990s, the cypherpunk movement context, and the fact that also the cryptographic software PGP was legally considered as a weapon in the United States of America and investigated for *munition export without a license*.

Storytelling allows the propagation of a complex technology, it allows us to propagate the effects and the power of the technology, and creates a community sense for the adopters. The community bonds in the blockchain community can be very strong; they can be so strong that we could look at it from a religious aspect.

Religious aspects of blockchain community

Similarities Both blockchain and religion create irrational disputes; their respective followers will make considerable sacrifices. There is often a powerful sense of belonging to a movement with its own code. There are heroes and leading figures, followers, and even a glorified messiah—the story with the mysterious Satoshi Nakamoto and the barely 18-year-old prodigy child, Vitalik Buterin, who invents an entirely new, programmable blockchain concept. Another example of a savior-like character is Roger Ver, the promoter of Bitcoin Cash, who speaks of himself as Bitcoin Jesus. There are sacred texts in the blockchain community, like the bitcoin whitepaper. Ritual celebrations, events, and gatherings like the bitcoin pizza day, the reward-halving events in bitcoin, and the developer conferences that are important moments in the lives of blockchain followers. Then, like in religions, there are schisms also in blockchain communities. Typically, what happens in a blockchain-schism is that a change to a blockchain code is not accepted by the entire community. So, both the community and the blockchain code split. This digital blockchain-splitting process is called fork. Of course, what is important for the marketing and storytelling momentum of the blockchain movement and religions are the miracles. Stories would gravitate about the idea that a miracle happened with blockchain, everybody has seen and experienced it. In the perception of many inside and outside the blockchain world, money has been created out of thin air. There is a monetary value and a market for decentralized projects without a company or juristic body running it, and only the community keeps it alive. Another miracle story is propagated: that of the powerless individual person or oppressed organization that has received a tool to be independent, empowered. A storytelling element that is dominant in the WikiLeaks history. Every organization and even ordinary people with a computer can be part of a bigger picture, possibly a revolution that helps setting up a new order of the world. So much for the conceptualized mindset of the blockchain enthusiast in the last decade.

Criteria There are several ways to define what a religion is. There is the functional definition. It does not focus on what a religion is, not on beliefs and practices for example. However, it focuses on the effect of the religion, on what the religion does for the individual, the group, and on the needs the religion fulfils. A religion, for example, can provide or contribute to bonding, identity comfort, and security. "A system of beliefs and practices by means of which a group of people struggles with the ultimate problems of human life." [14, p. 7].

On the other hand, there are substantive definitions that are more narrow and closer to common-sense perception and to what ordinary people mean when they talk about religion. An example of such a substantive definition would be "Religion, then, consists of beliefs, actions, and institutions which assume the existence of supernatural entities with powers of action, or impersonal powers or processes possessed of moral purpose" [15p. ix].

Synthesis Similar to football, blockchain and its community offer social and psychological functions that would allow football and blockchain to fall under the functional definition of religion. Yet the question remains, whether being active in the community is a religious activity. Personally, I think both the storytelling and the religious aspect that can be observed in the blockchain community are a reaction to the technical complexity. The simple, irrational explanations based on feelings and opinions in combination with the power of storytelling elements allows us to cope with a technically complicated tool that is designed to empower an individual. Furthermore, storytelling allows us to propagate blockchain technology as a tool without using blockchain, without fully understanding it and without the effort of crossing the adoption threshold that is induced by the powershift to the individual. With storytelling and community experiences users can contribute to the idea and be supportive of it without the strenuous act of either understanding it intellectually or handling it on their devices.

Added value to society and potential impact

Blockchain technology can be understood and interpreted as a tool in the context of digitalization processes. It can enable the decentralization of digital processes and, consequently, both provide privacy and enforce transparency. It guarantees immutability of datasets and can enhance security.

The decentralization it enables is only one feature that this technology offers. Even though technology is being appropriated by ideological movements and shows aspects of a functional religion, it offers unique opportunities. The main opportunity is that it enables us to debate openly on what kind of digitalization process we want to have. How we want to handle the disintermediation that is created because of the technology. Before the advent of blockchain technology, we could not debate the position we want to represent in the centralization versus decentralization spectrum, because the utopian promises of both the Internet and social media ended up in dystopian realities. Now we do have a chance to actively and consciously debate. Also, we have the opportunity to educate the next generation in our society, so that they can decide consciously how they want to map the digital representation of society and what kind of tradeoffs they are willing to make.

The impact of the technology itself is that it provides a fundament for creating peer-to-peer networks for sharing, exchanging, and trading digitized information, goods, and assets. This can have an impact on services, business, and regulation. Instead of a comprehensive list claiming completeness, I will focus on a few use cases that I encountered as an active member of the blockchain community.

Blockchain adoption and game theory

In certain settings, information asymmetry or intransparency is a desired state because it creates profits. If we look at a set of incumbent market participants that have either few pain points, or are not agile or are highly regulated, then there is little opportunity for change in any direction. In an industry of established, incumbent players, nobody needs to act and offer transparency as an added value, as long as nobody else acts. If we are in an industry or community where agile players have high regulatory thresholds, blockchain adoption is unlikely. Game theory findings show that there are some fundamental challenges to the adoption of blockchain technology [16]. In essence, game theory describes how actors rarely make decisions based only on reason and rational thinking. So, if the technology is being marketed with the rational aspect ignoring the human elements, then the adoption threshold is higher. Trust is a highly emotional element based on experience within a community. So, the shift from this highly emotional experience to machines and algorithms is a challenge. There is the possibility that trust is not something that needs to be delegated in some societies. Also, being a decentralized coopetition structure, it is hard to find a central authority organizing common efforts. A fractionalization of the community is likely to happen and can currently be observed in the blockchain space.

Operational efficiency through immutable and distributed record keeping

Depending on the particular use case, blockchain technology can reduce costs and improve efficiency. Blockchain challenges the logic of information silos between market participants, and it can reduce the need for interfirm reconciliation. If assets are represented digitally, smart contracts have both the content of the contract as well as the contract itself in the same channel. They either execute both or none. Also, they only execute fully and never partially. As a consequence, as seen before, litigations can be reduced drastically provided that the content of the contract can be represented on a blockchain.

Blockchain introduces the concept of proof-of-existence or proof-of-nonexistence over events. Since blockchains provide a historically structured database that has community consensus, they can certify the authenticity of documents, digital or digitalized assets, licenses, or certificates at a given moment. This feature will have an impact on notarial services. Of course, the impact of such tools will only be relevant in a regulatory setting that accepts digital identities, digital representation of ownership rights, and digital signatures that are accepted by the authorities.

Blockchain technology not only provides an infrastructure for creation, transfer, and programming of digital value, it can also protect digital value from counterfeit, illegal, and/or uncontrolled duplication. This could be game changing in sectors like the arts, where digital artefacts were hard to protect before the advent of blockchain technology. Evidently, the finance industry must rethink its role. The postal service did not disappear with the invention of emails, but it had to reconsider its role. Similarly, blockchain technology will not make banks or insurances disappear, but they have to reconsider their role and position, as well as the value proposition they offer to clients and society.

According to the World Bank, emittances from abroad are major economic assets for some developing countries. By being more efficient, faster, and less costly blockchain technology can improve the situations of people in the world that need it most and can create additional benefit in places in the world where small changes have a big impact.

Based on the decentralized peer-to-peer infrastructure combined with low transaction costs financial inclusion is easier to implement with blockchain technology. Globally, 1.7 billion people are still unbanked, yet two-thirds of them have access to a mobile phone[4]. The upside is an undeniable potential for improvement, particularly because financial intermediaries in developing countries often have trust issues. On the downside, there is an ongoing debate whether the blockchain approach does not widen the already existing gender gap in financial inclusion [17]. The approaches enable to blockchain-based financial inclusion are complex but can be structured in four dimensions

- Inclusion through blockchain-powered economic identity
- Blockchain-powered services for refugees and migrants
- Inclusion through blockchain-powered remittance services.
- Blockchain-powered digital identity for citizens in poverty

The most obvious reasons why blockchain-based applications make sense in this context are the low costs and fast settlement. There is no need for branches of a financial intermediary, the payments are digital, the fees involved are low, and most importantly, micropayments are possible.

Information symmetry through transparent record keeping

Information asymmetry has an influence on trade and negotiations. It is causing problems like moral hazard and adverse selection. Historically, such problems were solved by central authorities that are a single point of control in good times and have the potential of failure in bad times. Furthermore, there are challenges in traceability and little transparency in accounting, which increases the need for regulatory intervention. Introducing a shared transparent ledger should increase the cooperation between regulators and regulated partners. It has the potential to move from post-transaction monitoring to immediate or on-demand monitoring and intervention. The reliability and rep-

[4] https://www.worldbank.org/en/news/press-release/2018/04/19/financial-inclusion-on-the-rise-but-gaps-remain-global-findex-database-shows.

utation of actors in a system can be verified and monitored. Ideally, we will see a convergence of industry and government interests, which will reduce regulatory compliance costs.

Using blockchain technology to create transparency and efficiency in the donation or philanthropy market seem to be a use case. However, from what we see in the donation space, donors still seem to care about trusted institutions and brands that vouch for projects and guarantee project delivery. Donors and philanthropists seem to be willing to pay a premium to delegate trust to an institution. The transformational effort for an NGO from a value redistribution organization to a knowledge handling organization that is associated with a blockchain-based value sharing infrastructure seem too-big a threshold tackle. However, there is an opportunity for donation-dependent entities to streamline their processes. Today, as little as 20% of donations actually make it to the intended recipients. The rest is eaten up by the infrastructure and the intermediaries in the value transfer chain.

Decentralized corporations and governance

Our societies are centralized to a certain extent, and institutional hierarchies govern the activities of our socio-economic life. With the automation potential of the second-generation blockchain and the scalability potential of the third-generation blockchain, we can now think of new processes. We can imagine unique and secure digital processes that can self-execute, self-enforce, self-verify, and self-regulate in a decentralized network. This allows the creation of new organizational structures in a society. It allows the creation of self-organized economies and decentralized autonomous organizations (DAO) that enable new models of nonhierarchical cooperation, coopetition, and governance, where decision-making happens throughout the network and not in the center. An important DAO that was created on the Ethereum Blockchain, its failure, the debates about the role of code, the role of community, and the legal aspects, as well as the consequences it had on code and community are worth further reading [18].

Aligning incentives to reduce the tragedy of the commons

Smart contracts have the power to lock value and only release it if certain conditions are met. This allows an alignment of stakeholders with nonaligned interests. If the stakeholders do not cooperate or find compromises, if they do not align their value to a certain extent, then the value locked in the smart contract is lost. Creating such settings allows the incentivization for competitors to cooperate in areas where they disagree and ideally create an antifragile system. The concept of coopetition is where a community has more to gain when they approach a status that has both elements of competition and cooperation. Examples of coopetition could be drivers of carsharing services like Uber. The drivers compete, but the service is only useful and chosen by consumers because there are so many of drivers. Another example of coopetition is participants running a mining node in a bitcoin network. They compete to generate the next block and receive a mining reward. The value of the network and the reward heavily depends on the size of the network.

Environmental and natural resource problems are associated with overexploitation or underprovision of public goods. This is an important area where blockchain-supported coopetition incentivization can offer a solution space. Overfishing, excessive air pollution, unwarranted extraction or diversion of ground or surface water, and extreme depletion of oil and gas reservoirs are examples of the tragedy of the commons where private decision-makers do not consider or do not internalize social and environmental benefits and costs in their investment or production action. The gap between social and private net returns results in externalities that have harmful effects on third parties.

Blockchain technology is key when there is a need for a transparent, decentralized and/or distributed coopetition setting where stakeholders need to cooperate in the long term in order to create a sustainable and stable system.

In Romania in the Carpathian Mountains, a blockchain-based ecosystem is planned to implement exactly such a solution. It is a cooperation between WWF Panda Labs, the Swiss Porini Foundation, and ETH Zürich. They are planning a blockchain-based circular economy that tries to create a coopetition setting around the project. In the center of the project are the communities in the ecosystems affected by the rewilding of the wisent, the European bison.

Blockchain support for public services and digital citizenship

The world is being digitalized more and more, and we have to accept the fact that many countries have a critical information technology (IT) infrastructure that needs to be protected, a critical infrastructure that is accessible in a world populated with objects that are directly connected to the Internet. These IoT (Internet of Things) devices are vulnerable to cyberattacks. Centralized IT structures add to the danger, since they may a have singular points of failure or represent honeypots that make such targets even more attractive.

On the other hand, looking at blockchain from a governmental perspective most probably involves a fear of loss of power moving away from central authorities to self-governed peer-to-peer structures.

However, new forms of participation and relationship between government and civil society are possible and could add or create value for both, the society and the government. Blockchain technology could manage, handle, and trace processes and operations like identity management, tax collection, service distribution, national digital currencies, and any type of government register. At a first glance, advantages could include the simplification and standardization of internal processes, the reduction of transaction costs, more reliable interactions and exchanges of data with other organizations and governments, and greater protection against errors and falsifications. Yet most probably, the relevant value creation would be realized when new radical tools or processes are enabled. Let us imagine a digital citizen-centric identity solution, where the state is still the authority confirming identity, but the citizen has the control over the digital representation. A bit like he has over a physical passport, really. Or let us try to imagine more granular or even liquid government services. A process like liquid voting or a hide-and-reveal voting process, where a citizen can check whether his or her vote counted. To decentralize and digitize services through blockchain technology does not mean dismissing the state, but to promote good governance. In doing so, civil society could protect its interests more effectively. Now both blockchain technology and the digitalization of political processes are in the very early stages. In many cases, legal ownership of assets is connected to a physical piece of paper. Probably it is too early to think about blockchain-enabled liquid voting. It probably is a smart strategic decision to wait until some incubation time has passed so the technology itself can be tested and users can learn to handle the technology.

Adoption challenges and scenarios

Blockchain started with bitcoin, a decentralized peer-to-peer payment system as a reaction to the financial crisis in 2008. Now, 10 years later, we may ask two questions: can blockchain technology provide an alternative payment solution? Here the answer clearly is positive. The more interesting question, however, might be: is there a need for an alternative payment system? Looking at Occam's razor, the problem-solving principle that favors the simplest solution, we may doubt whether blockchain payment systems will reach widespread adoption. On the other hand, if a new financial crisis should appear, there are now new alternative tools that can be used. Maybe, the fact that there is an alternative technology available for independent value transfers, creates the antifragile situation that keeps the traditional financial system more stable.

In the first 10 years of blockchain technology many use cases were developed. One of the major reasons why the technology has not been widely adopted yet is very simple. The creation of laws is always delayed compared to the technological innovation. However, since blockchain technology is about digitalization of value, and the regulation density in financial markets is very high, there is much insecurity involved in the adoption of this technology. This insecurity creates high risks that incumbent organizations are not willing to take in the short term, and agile organization cannot take in the short term, because of the regulatory thresholds.

Adoption through blockchain ecosystem networks

Optimistic blockchain experts predict that the concept of blockchain ecosystem networks will be the new dominating business model of the future. The average lifespan of S&P500 companies has dropped from 61 to just 18 years in past 60 years[5]. Nowadays, 50% of global annual company revenues come from products launched within just the past 3 years. How can we interpret these figures? Before the year 2010, dominant businesses used to make business on resources, products, and services. However, new business models have drastically changed the landscape. Platform businesses started taking over in 2010, where value is generated facilitating exchange between interdependent groups, usually consumers and producers. Later, around 2015, exponential organizational models like Uber and Airbnb took the lead. These exponential organizations create output from owned assets in a sharing economy. The next level that includes blockchain technology will be ecosystem economies. A realistic scenario envisions companies that start the integration core business functionalities with both third-party networks and platforms. This allows them to be more agile, scale faster and on-demand, and benefit from economies of scale beyond their organizational boundaries though accessing and aggregating otherwise inaccessible resources.

These blockchain ecosystem networks can be either centralized, where a central company runs a blockchain or DLT infrastructure, they can be federated and would typically be run by a consortium, or they can be decentralized. The adoption optimistic argumentation would include the following points: in the long term, there will be a migration from centralized to federated and eventually decentralized networks, participants do not want that a competitor to run the ecosystem. Consortia take a lot of time. Also, possibly, some data-privacy scandals have accelerated adoption. An intensive, detailed, and very differentiated scenario-based analysis of this process can be found in the work of Andranik Tusmanian et al. [19].

[5] https://www.cnbc.com/2017/08/24/technology-killing-off-corporations-average-lifespan-of-company-under-20-years.html.

Sceptics focus on the challenges of adoption, which can be structured into three inherent problem categories. First, the underestimation of the human aspect, second the missing and potentially impossible balance between decentralization and centralization, and third, a set of six business challenges. As was mentioned earlier, this text focuses on the opportunities, which is why these sceptics' arguments are not elaborated at all.

Underestimation of the human aspect

By focusing on removing trust, most blockchain architectures do away with central human elements that turn a reluctant person into a devout user. Possibly this is why the religious aspect of the blockchain community comes back into play and why it is celebrated within the community to various extents. Behavioral economics studies show how people reach and stick to at-first-sight irrational decisions. Numerous experiments prove that we are not fully and strictly rational agents. By trying to communicate, design, and implement a setting that focuses on logic and reason, the adoption threshold rises, since the human aspect, which is relevant and key to human action, is missing.

The blockchain utopia comes with a price and might have negative ethical and political implications. The potential that it brings for emancipation are undermined by the fact that interactions become rigid. If social or transactional relations need to be rigid, like in the context of financial services or property registers, then blockchain technology is an advantage. In the context of human care and education, you may want to have relations that are not rigid. Particularly for social contexts in which there is a necessity for human freedom and responsibility in shaping and reshaping social interactions, the application of blockchain technologies will probably be very undesirable or complicated to design and implement.

If blockchain technology will be used by authorities to enforce a totalitarian and fully transparent society, then the technology will be a Trojan horse and destroy all the libertarian dreams together with the blockchain utopia.

Challenging balance between decentralization and centralization

With the decentralized peer-to-peer approach, Satoshi Nakamoto started a bottom-up movement by introducing a radical innovation. Why did Satoshi Nakamoto decide to stay anonymous? Let us look at some possible answers to have an idea of the challenge of implementing decentralization tools in a large society.

Possibly, Satoshi Nakamoto decided to remain anonymous in order to avoid leadership in a peer-to-peer system. As a de-facto leader and inventor, users would possibly trust the person not the bitcoin data structure and by doing so undermine the idea of trust shift. Moreover, any communication by Satoshi would likely have been interpreted in an investment context independently, whether or not the creator hoped for such an outcome. As a peer, the inventor of bitcoin remained nothing else but a peer. This way, a myth was created, and a great storytelling aspect was born. However, it also shows that a delicate balance even on the macrolevel of the blockchain ecosystem needs to be established to foster adoption.

Another reason for anonymity is the highly regulated aspect of financial markets. By creating a tool like bitcoin, you are a potential threat to very powerful forces in a society. Looking back at the legal debate about the encryption software PGP, labeled as munition in the 1990s, Satoshi Nakamato's decision is understandable.

Business challenges

According a report[6] from one of the Big 5 tax firms, the challenges that need to be overcome before business adoption are related to performance, complexity, regulation, cooperation, and interoperability. Another topic that seems to be relevant is privacy and security concerns, but that needs to be seen in an educational context. Typically, blockchains are not designed to store large amounts of data. There are other peer-to-peer systems that are designed to do that.

Also, since the technology is also about digitalization of value, it can be a driver in the financial industry and is often associated with it. The actions of regulators and policymakers are taking a long time to implement technological development. Consequently, there is an insecurity in the industry and with incumbent organizations. On the other hand, small and agile organizations run risks with an early adoption and might not have a lobby powerful enough to educate decision-makers.

What companies can invest sums large enough to run a team to develop an innovation project, a proof of concept solid enough to be tested and business conditions? It is companies that are very large and have a big-enough research and development budget. If blockchain as a technology is not on a strategic map of top-level management, it typically must go through processes within the organization for the next step of implementation. Blockchain as a tool designed to work in decentralized peer-to-peer environment ends up in the hands of very large organizations that have a hierarchical structure. In this setting, it might be difficult to deploy the originally intended potential.

The important questions for decision-makers in business should be: how does my business look like in a blockchain

[6] https://www.cnbc.com/2018/10/01/five-crucial-challenges-for-blockchain-to-overcome-deloitte.html.

enabled world? What is the price we are willing to pay not to evaluate that question? Only if these questions are addressed can a sustainable business development be guaranteed with arriving new technologies that can have disruptive impact.

Summary

Switzerland is a federated, highly decentralized society with a high stake in the financial market sector. It runs on the forefront when it comes to regulatory efforts and political support of blockchain technology. Both the decentralization aspect and the fact that this third wave of digitalization is about sharing and exchanging value makes it both easy to understand and an opportunity to adopt for Swiss society.

The technology itself offers the possibility to rethink some fundamental decisions that a society must take when it starts digitalizing information, communities, and value. If individuals, organizations, or parts of society want to regain ownership over digital data, blockchain technology can help. However, with regaining ownership over data comes a price. The price is responsibility, which involves some tradeoffs.

In order to be able to make a decision about how to handle the incoming third wave of digitalization it remains of pivotal ethical and political importance to educate users, policymakers, and decision-makers about the potential of the technology. Only when we know the potential of the technology, when we know the potential implications of the use of the technology, and when we know about human aspects handling technology, can we decide on how we want to regulate and apply it. Education is key when a society faces a new technology and decisions on how to handle the technology.

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HAUPTBEITRAG

Schlusspunkt: Demokratie in Zeiten des Internets – (m)ein Reifungsprozess

Edy Portmann[1]

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Zusammenfassung
Die Digitalisierung erfordert von uns verstärkt die Fähigkeit, innovative soziotechnische Systeme zu gestalten. Herbert A. Simon begriff in diesem Zusammenhang Gestaltung als eine «Wissenschaften vom Künstlichen». Als gestaltungsorientierter (Wirtschafts-)Informatiker beschäftigte ich mich als Förderprofessor der Schweizerischen Post bis vor kurzem vor allem mit postalischen Innovationen. Um zu innovieren, arbeitet die Post nämlich über (Disziplins-)Grenzen hinweg mit meinem Human-IST Institut zusammen, um für die Schweizer Bürger zukünftige, intelligente Produkte und Dienstleistungen zu entwickeln. In diesem Sinne gehe ich hier zuerst auf diese transdisziplinäre Zusammenarbeit ein, die wegen ihrer technischen Ausrichtung meine bisherige Sicht auf die Demokratie prägte. Danach stelle ich dieser Ansicht jedoch meine heutige Sicht entgegen, die sich (in Zeiten des allgegenwärtigen Internets) mit einer demokratischen (und föderalistischen) Digitalisierung unserer Netzwerkindustrie beschäftigt. Abschliessend befasse ich mich mit dem Auslöser meines Reifungsprozesses, sprich einer regulatorischen Komponente für künstliche Intelligenz, mit welcher sich auch die Europäische Union zum Schutz ihrer Bürger immer stärker beschäftigt.

Als ich mich vor einem Jahr mit Joël Vogt und Patricia Hetter Kelso, meinen Mitherausgebern dieses Schwerpunkts zu „Demokratie in Zeiten des Internets“, unterhielt, wusste ich noch nicht, welche Entwicklung ich selbst diesbezüglich durchleben würde. Als Informatiker kümmerte mich nämlich bis dato die Regulation des Internets relativ wenig – das hat sich aber über das letzte Jahr verändert. In diesem Essay möchte ich Ihnen meinen Reifungsprozess, welcher während der Anfertigung dieser Ausgabe der *Informatik Spektrum* zustande kam, nicht vorenthalten.

Als Professor für gestaltungsorientierte (Wirtschafts-)Informatik und Förderprofessor der Schweizerischen Post am Human-IST Institut der Universität Freiburg (Schweiz) beschäftigte ich mich bis vor kurzer Zeit vor allem mit technologischen Innovationen für die Schweizerische Post. Diese arbeitet eng mit dem Human-IST Institut zusammen, um mit und für die Schweizer Bürger die Zukunft postalischer Produkte und Dienstleistungen zu erforschen und (weiter) zu entwickeln. Auf diese Zusammenarbeit, die wegen ihrer technischen Ausrichtung meine damalige Sicht auf die Demokratie in Zeiten des Internets prägte, gehe ich im ersten Abschnitt ein. Im zweiten Abschnitt präsentiere ich meine heutige Sicht, die sich mit einer holistischen Digitalisierung der Netzwerkindustrie beschäftigt, bevor ich im dritten und letzten Abschnitt auf den Auslöser dieses Reifungsprozesses, sprich eine regulatorische Komponente für künstliche Intelligenz, eingehe.

✉ Edy Portmann
edy.portmann@unifr.ch

[1] Human-IST Institut, Universität Freiburg, Boulevard de Pérolles 90, 1700 Fribourg, Schweiz

Die (Wirtschafts-)Informatik als Brückenbauerin

Die allgegenwärtige Digitalisierung in Politik, Gesellschaft und Wirtschaft erfordert Experten, welche die Fähigkeit besitzen, innovative soziotechnische Systeme zu gestalten und zu implementieren. Eine gestaltungsorientierte (Wirtschafts-)Informatik stellt dazu Konzepte bereit. Sie entwickelt Methoden zur Abstimmung von politischen, gesellschaftlichen und wirtschaftlichen Anforderungen mit technischen Systemen und greift auf Modelle aus der Informatik und der Wirtschaftswissenschaft sowie verwandter Disziplinen zurück.

Da die Digitalisierungswelle auch vor Demokratien nicht haltmacht, bemühen sich in der Schweiz die hiesigen, nationalen Netzwerkindustrien wie die Schweizerische Post, die Schweizerische Bundesbahn (SBB) und das Schweizerische Telekomunternehmen (Swisscom) um die Digitalisierung unserer Infrastruktur. Projekte reichen weit und umfassen unter anderem mit Blockchain- und Smart-Governance-

Projekten auch die Digitalisierung der Demokratie und des Föderalismus. So drehen sich beispielsweise wesentliche Projekte bei der Schweizerischen Post und Swisscom mit Smart-Voting und -Health und bei der SBB um intermodale Mobilität. Mit unserer Art der (Wirtschafts-)Informatik unterstützt das Human-IST Institut die Netzwerkindustrien dabei.

Mit meiner Berufung an die Universität Freiburg entwuchs dieses als transdisziplinäres Institut der gleichnamigen Forschungsgruppe des Departments für Informatik. Das Human-IST Institut widmet sich der Erforschung und (Weiter-)Entwicklung von „Human-centered Interaction Science & Technology". Weil es immer entscheidender wird, wie sich unsere Bedürfnisse in Technologie widerspiegeln, entwickelt das Institut intelligente Schnittstellen, die für die Anwender nutzbar, zweckvoll, ansprechend und mit hochstehenden Nachhaltigkeitskriterien vereinbar sind. Zu diesem Zweck betreibt es gestaltungsorientierte Forschung zur Konzeption, Implementierung und Bewertung neuer Paradigmen der Interaktionswissenschaft und -technologie in Zusammenarbeit mit den Netzwerkindustrien.

Basierend auf Internettechnologien bauten multinationale Technologiekonzerne wie Google, Facebook & Co ihre KI-Geschäftsmodelle auf, welche auf Sammlung und Nutzung großer Datenmengen beruhen. Mit unseren Daten entwickeln sie nicht nur ihre KI-Modelle weiter, sondern lernen unsere Gewohnheiten, Einstellungen und Werte kennen. Sogar von Überwachung der Benutzer, also von uns Bürgern, ist die Rede. Aufgrund ihrer vermeintlichen Gratisangebote haben wir hierbei die Hoheit über unsere Daten längst verloren. Heute werden die Dienste von Google, Facebook & Co von Drittfirmen finanziert, die wir nicht kennen. Diese Drittfirmen versprechen sich einen Vorteil aus den Informationen, die sie aus unseren Daten extrahieren können. Das hat nicht nur Auswirkungen auf unsere Märkte, sondern eben auch auf unsere Demokratie, da diese Firmen anhand unserer Daten nicht nur uns sehr gut kennen, sondern immer mehr versuchen, uns mit datengetriebenen KI-Systemen zu manipulieren.

In unseren Demokratien geht die Regierung durch politische Wahlen aus der Bevölkerung hervor. Da die Macht dabei von uns Bürgern als Allgemeinheit ausgeübt wird, ist unsere Meinungs- und Pressefreiheit zur politischen Willensbildung zentral. Mit Blick auf den Facebook-Skandal rund um das Analyseunternehmen Cambridge Analytica, das 2016 während des amerikanischen Wahlkampfs Daten über Millionen von Wählern sammelte, um damit durch Microtargeting, also mit persönlich zugeschnittenen Informationen, das Wählerverhalten zu manipulieren, ist die Demokratie meines Erachtens stark in Gefahr. Mit einer einseitigen Digitalisierung unserer Demokratie verkommen wir Bürger immer mehr zum Produkt, welches die Technologiekonzerne in ihren Serverfarmen bewirtschaften. Diese Erkenntnis führte mich zu einer neuen Sicht auf Demokratien im Zeitalter des Internets. Bis zu diesem Zeitpunkt begriff ich Demokratie nämlich vor allem als analoges System, in welchem Macht vom Bürger beziehungsweise dem Volk ausgeht. Das Zusammentreffen mit dem Internet verändert aber auch dieses herkömmliche System komplett. Im nächsten Abschnitt zeige ich deshalb meine heutige Sichtweise auf.

Eine transdisziplinäre Brücke als Lösung?

Um die Gefahr für unsere Demokratien, die von Technologiekonzernen rund um Google und Facebook ausgeht, einzudämmen, braucht es einerseits Brücken zwischen unterschiedlichen (Forschungs-)Disziplinen, andererseits aber auch solche zwischen Forschung und Praxis. Um unsere komplexe Zukunft mit intelligenten Systemen zu gestalten, müssen wir uns mit neuen, transdisziplinären Methoden beschäftigen. Dieser Brückenschlag, der Disziplinen verbinden will, muss robust gebaut werden.

Um so eine robuste Brücke zu bauen, leben wir am Human-IST Institut eine über die Grenzen des universitären Wissenschaftsapparats hinausgehende Forschung. Um gemeinsam mit unseren (Forschungs-)Partnern Ziele anzupacken, beziehen wir nichtakademische Teilnehmer als gleichberechtigt in unseren Forschungsprozess mit ein. Während interdisziplinäre Forschung Wissen aus bestehenden unterschiedlichen Disziplinen synthetisiert, verbindet unser transdisziplinärer Ansatz alle Disziplinen aus Forschung und Praxis zu einem kohärenten Ganzen. Die Dinge ganzheitlich zu betrachten und die Interdependenz zwischen ihnen zu berücksichtigen scheint für uns ein Weg, die wirklich harten und komplexen Probleme unserer Gesellschaft zu adressieren.

In ausgiebigen transdisziplinären Diskussionen stellten wir nun fest, dass heute dieselben Mechanismen, die 2008 unsere Wirtschaft bedrohten – und vermutlich zum globalen Börsencrash führten – über (Internet-)Technologie ihren Weg wieder in unsere Gesellschaft zurückfanden. Die binären Modelle der Technologiekonzerne, deren KI-Modelle eine Person zum Beispiel entweder als kreditwürdig oder -unwürdig klassifizieren, führen zu einer reduktionistischen Sichtweise, die wir von unseren Demokratien fernhalten müssen. Dies verdeutlicht uns etwa auch das erwähnte Beispiel des Analyseunternehmens Cambridge Analytica, welches Wähler in binäre Klassen von [0,1] aufteilte, um diese zum Wählen (= 1) respektive Nichtwählen (= 0) zu animieren.

Was heißt das? All unsere heutigen Systeme, von der Wirtschaft bis zu künstlicher Intelligenz und maschinellem Lernen, basieren unter anderem (noch) auf binären Methoden der Logik und „objektiven Fakten". Unsere Kunst, unser Herz und unsere Biochemie – also all das, was eine ganzheitliche Sicht ausmacht – findet in dieser Argumentation und Bewertung häufig keinen Platz. Diese „weichen Faktoren" (z. B. kann natürliche Sprache mit vagen Wahrnehmungen wie Subjektivität, Objektivität, Emotionen, Werte, Erinnerungen etc. ganzheitlich umgehen) sind aber für eine nachhaltige, demokratische Entscheidungsfindung zentral.

Die Binärmethoden setzen, als komplexitätreduzierendes Hilfsmittel, auf Wahrscheinlichkeit, obwohl diese Mittel wohl eher nicht angemessen oder repräsentativ dafür sind, wie wir Menschen denken. Denn die Prinzipien, nach denen unsere menschliche Intelligenz funktioniert, die diese Methoden repräsentieren, beruhen nicht auf der Wahrscheinlichkeit. Vielmehr sind für unser Gehirn scheinbar harte Fakten graduell, unscharf, unbestimmt und bis zu einem gewissen Grad ungenau. So können wir nie mit Sicherheit sagen, ob Gras grün ist, Atome schwingen und die Zahl der Seen gerade oder ungerade ist. Um damit umgehen zu können, runden wir solche Behauptungen lieber auf oder ab, und so sind diese entweder alle richtig oder alle falsch, weiß oder schwarz, 1 oder 0.

Diese binäre Logik spiegelt sich heute in all unseren Digitalisierungsbestrebungen, wie etwa im maschinellen Lernen, wo wir uns des Auf- und Abrundens bedienen und dieses Verhalten erhalten, verbreiten und verstärken. Beispiele hierfür sind etwa die Bedienung von Applikationen mit Wischen nach rechts (1) oder Wischen nach links (0) und bei Facebook „Gefällt mir" klicken oder „Gefällt mir" nicht klicken. Problematischer wird es aber dann, wenn unsere komplexen Emotionen entweder als positiv oder negativ eingeteilt werden. Das Problem liegt darin, dass solche Einteilungen keine Räume lassen, wie und warum wir etwas gegenüber etwas anderem gewählt haben. Denn unsere Entscheidungen werden selten aufgrund einer hundertprozentigen Sicherheit gefällt. Wir entscheiden uns für etwas (oder gegen etwas), wenn wir (unsere Wahrnehmung und Einstellung) zu einem großen Teil davon überzeugt sind (oder eben nicht). Die Maschinen registrieren nur, dass wir eine Wahl getroffen haben – und arbeiten mit dieser absoluten Aussage weiter, ohne die Zwischenwerte in Betracht zu ziehen.

Aus diesem Grund begnüge ich mich mittlerweile nicht mehr nur mit technologischen Post-Innovationen, sondern beschäftige mich auch mit einer ganzheitlichen Regulation zum Wohle von uns Bürgern. Und so beteiligte ich mich an einem „Austausch über unsere Zukunft mit künstlicher Intelligenz" der Europäischen Union. Um das binäre Winner-takes-it-all-Problem der großen Technologiekonzerne bei der Wurzel zu packen, schlug ich in einer Anhörung vor, eine europäische Regulation zu gestalten, die ganzheitlich auf uns Bürger eingeht und dabei eine biologieorientierte Dezentralisierung begünstigt.

Während unserer Diskussionen haben wir auch festgestellt, dass das Internet eine immer wichtigere Rolle in der Informations- und Machtasymmetrie spielt, die zwischen den Organisationen, die diese Dienste anbieten, sowie den Nutzern, die sie befüllen, besteht.

Wir dürfen nicht vergessen, dass unsere Internetarchitektur zwar dezentral ist, aber Dienste sich wegen des inhärenten Netzwerkeffekts, sehr schnell auf ein paar wenige Plattformen konzentrieren. Die gesammelten Daten und Informationen der Technologieunternehmen sind Macht, sie ermöglichen oder zerstören ein demokratisches System. Daher ist es von höchster Bedeutung, sich mit dieser Thematik auseinanderzusetzen und die Zukunft der Schweiz (und Europas) mitzugestalten. Im Folgenden, letzten Abschnitt möchte ich zum Schluss auf den Reifungsprozess im letzten Jahr eingehen.

Über die Grenzen unserer heutigen Wissenschaft

Es gibt Alternativen zum binären Denken unserer heutigen Wissenschaft. Beispielsweise basieren östliche Philosophien auf einem natürlicheren Gleichgewicht von Kohärenz und Integration: Die Natur kodifiziert Wissen in der DNA als eine Koexistenz von X- und Y-Chromosomen, und auch in der Quantenforschung zeigt sich, dass Teilchen überlagerte Zustände, sogenannte Superpositionen, haben können, in denen sie gleichzeitig 0 und 1 sind. Auch die Natur lässt sich also nicht in Binärcodes einteilen. Dem besser gerecht wird etwa das Yin-Yang-Konzept, das eine gleichwertige und symbiotische Dynamik sowohl in uns wie auch in unserem Universum betont.

Von diesen Alternativen geleitet, versuchen wir am Human-IST Institut unterdessen eine ganzheitlichere Sicht einzunehmen, damit unsere künftigen soziotechnischen Systeme das Farbbild unseres Lebens erfassen. Wir erkannten in unseren transdisziplinären Gesprächen nämlich, dass, wenn wir dies nicht tun, nicht nur unsere Wirtschaft, sondern auch immer mehr unsere Gesellschaft und ihre Demokratien gefährdet werden. Mit dem Weggeben unserer persönlichen Daten vergeben wir auch unsere demokratische Stimme. Facebook und Cambridge Analytica etwa hätten wohl nicht unsere Daten für politische Zwecke missbrauchen können, wären wir „Kunden" auch Inhaber unserer eigenen Daten.

Mit unserem transdisziplinären Forschungsvorgehen packen wir am Human-IST diese Herausforderung an und versuchen mit der Entwicklung von smarten Schnittstellen, die unsere verschiedenen menschlichen Sprachen, Denkweisen, Werte, Kulturen, Qualitäten und Verhaltensweisen wiedergeben, diese Probleme anzugehen. Hierzu erforschen wir das Rechnen mit Wahrnehmung, einer Umsetzung der Theorie des Rechnens mit Worten nach Lotfi A. Zadeh.

Seine Theorien dienen uns bei unserer (subjektiven) Urteilsbildung.

Das Rechnen mit Wahrnehmung sowie das Rechnen mit Worten haben ihren Ursprung in einer unscharfen Logik, die wir als biomimetische Systemtheorie begreifen lernten. Aus diesem transdisziplinären Blickwinkel nahmen wir eine immer stärkere Reduktion unserer Gesellschaft und Demokratie auf Effizienz wahr. Diese Reduktion, welche mich in meinem Reifungsprozess begleitete, wurde mir nach dem ersten Gedankenaustausch mit meinen Mitherausgebern Joël Vogt und Patricia Hetter Kelso immer mehr bewusst.

Für unsere Demokratie sind wir verantwortlich. Daher liegt es auch in meinem Interesse, (biologieorientierte) Systeme für unsere kollektive Intelligenz zu schaffen, die den Bürgern zugutekommen. In diesen dezentral organisierten (Öko-)Systemen, die eine große Anzahl von Individuen umfassen können, kommunizieren die Individuen nicht unmittelbar, sondern nur indirekt miteinander, indem sie ihre lokale Umgebung modifizieren. Das gemeinsam Erstellte wird gleichsam zum Auslöser von Anschlussaktivitäten und zur allgemeinen Anleitung dafür, wie mit dessen Erstellung fortzufahren ist. Wir könnten uns doch für die Regulation an solchen natürlichen Konzepten orientieren – oder nicht?

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AKTUELLES SCHLAGWORT

Memristoren für zukünftige Rechnersysteme

Dietmar Fey[1] · Wolfgang Karl[2] · Theo Ungerer[3]

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Zusammenfassung

Als Memristor bezeichnet man eine Klasse von elektronischen Zweitor-Bauelementen, deren Strom-/Spannungsverlauf eine zumeist durch den Nullpunkt verlaufende eingeklemmte Hystereseschleife aufweist („if it's pinched it's a memristor"). Memristoren bieten interessante Eigenschaften wie hohe Speicherdichten, niedrige elektrische Leistungen beim Schreiben und Lesen, Multibit-Speicherfähigkeit und CMOS-kompatbile Herstellungsprozesse. Darüber hinaus lassen sich Memristoren nicht nur zum Speichern, sondern auch zum Verarbeiten von Daten nutzen. Memristoren werden zukünftige Rechensysteme auf verschiedenen Ebenen verändern und mehr in Richtung hin zu Speicher-zentrierten Architekturen verschieben. Dies beginnt bei noch weitgehend konventionellen Ansätzen wie den sog. Storage-Class-Memories, einer neuen Speicherhierarchieebene zwischen Arbeits- und Hintergrundspeicher, und endet bei unkonventionellen Near- und In-Memory-Architekturen, in denen Speichern und Verarbeiten von Daten ohne räumlich große und damit vergleichsweise Energie-intensive Datenbewegen zwischen Prozessoren und Speicher stattfindet. Ferner bieten Memristoren durch die dem biologischen Vorbild eines neuronalen Netzes weitgehend entsprechende direkte Nachbildung von Synapsen als flexible elektrische Widerstände vielfältige Möglichkeiten zur Realisierung neuromorpher und biologisch-inspirierter Schaltkreise und Architekturen.

Memristoren, d.h. *resistive Speicher*, bezeichnen eine für die Rechen- und Speichertechnik stetig wichtiger werdende Klasse von Technologien mit nichtflüchtigem Speicher (Non-volatile Memory, NVM). Memristor ist ein Begriff, der von Leon Chua geprägt wurde. Ein Memristor hat einen elektrischen Widerstand, der nicht konstant ist, sondern von der zuvor angelegten Spannung V und dem daraus resultierenden Strom I an den beiden Elektroden eines Memristors abhängt. Eine Besonderheit ist, dass der eingestellte Widerstand beim Aus-und Einschalten der Stromversorgung erhalten bleibt. In diesem Sinne erinnert sich der Memristor an seine Geschichte. Zu der Klasse der memristiven Bauelemente gehören NVM-Technologien wie z.B. Resistive RAMs (ReRAMs), Phasenwechselspeicher (PCMs) und magnetische RAMs mit Spin-Torque-Transfer-Eigenschaft (STT-MRAMs).

Gegenwärtig ist NAND-Flash die gängigste NVM-Technologie. Memristoren bieten jedoch wesentlich schnellere Zugriffszeiten und eine höhere Lebensdauer als Flash. Darüber hinaus sind sie CMOS-kompatibel, d.h. sie können in CMOS-Fertigungsprozesse und mit CMOS-Transistoren integriert werden, um völlig neue Rechen- und Speicherschaltungen zu schaffen. Weiterhin bieten sie Multi-Bit-Fähigkeit, d.h. es können mehr als zwei Bits in einer physikalischen Speicherzelle gespeichert werden. Dies eröffnet völlig neue Möglichkeiten für Computerarchitekturen, z.B. Multi-Bit-Caches oder ternäre Datenverarbeitung zur Unterstützung von übertragsfreien Addierern. Da man mit Memristoren nicht nur speichern, sondern auch rechnen kann, sind sie für zukünftige In-Memory-Computing-Konzepte besonders attraktiv. Das Speicherelement wird hierbei zu einem inhärenten Bestandteil des Verarbeitungsschritts, was energieintensive Datenübertragungen vermeidet.

Ein großer Forschungszweig im Bereich des Memristiven Rechnens zielt darauf ab, die Memristor-Widerstandswerte zur Realisierung von Gewichten in analogen neuromorphen Schaltkreisen gezielt zu modifizieren und zu speichern. In der Literatur wurden viele Vorschläge für neuronale Verarbeitungsschemata mit Memristoren zur direkten Nachahmung von Prozessen im Gehirn gemacht, z.B. bei der zeitabhängigen Verarbeitung (Spike Time Dependent Processing, STDP).

✉ Dietmar Fey
dietmar.fey@fau.de

[1] Friedrich-Alexander-Universität Erlangen-Nürnberg, Erlangen-Nürnberg, Deutschland

[2] Karlsruhe Institute of Technology (KIT), Karlsruhe, Deutschland

[3] Universität Augsburg, Augsburg, Deutschland

Gegenwärtig werden Memristoren bereits in Prozessoren eingesetzt, z. B. als Firmware-Speicher in Mikrocontrollern oder als Alternative zu eingebettetem SRAM. Außerdem werden sie als Kandidat für die nächsten Cache-Realisierungen betrachtet. Die Memristor-Technologie wird die Speicherhierarchie von Computersystemen in naher Zukunft immer stärker beeinflussen. Memristoren werden eine Schlüsselrolle in nichtflüchtigen Prozessoren spielen, die aufgrund ihrer nichtflüchtigen Eigenschaft und ihrer prinzipiellen KI-Berechnungsmöglichkeiten ohne Batterie oder externe Stromversorgung für IoT- oder Embedded-Anwendungen betrieben werden können.

Die Memristor-Technologie wird dazu beitragen, die Unterscheidung zwischen Arbeitsspeicher und Hintergrundspeicher ein großes Stück weit aufzuheben, indem sie neue Datenzugriffsmodi und Protokolle ermöglicht, die beide Speicherarten bedienen. Solche Techniken werden mittel- bis langfristig radikale Veränderungen in unserer Computer- und IT-Landschaft mit sich bringen. Memristoren werden einerseits viele attraktive qualitative Verbesserungen für neue Computerarchitekturen bieten, andererseits aber auch neue Herausforderungen hinsichtlich der Sicherheits- und Datenschutzrisiken mit sich bringen, die sich aus der Nichtflüchtigkeit von Memristor-Technologien ergeben. All diese Dinge und die immer besser werdende und sich immer weiterverbreitende Technologie der Memristoren machen es mehr denn je notwendig, diesen Prozess von der Seite der Informatik aus zu begleiten und zu gestalten.

Ein kürzlich veröffentlichter Bericht mit dem Titel „Implications of Memristor Technologies for Future Computing Systems“ geht genau auf diese Fragen ein. Er gibt einen Überblick über die verschiedenen Memristor-Technologien sowie über ihre zukünftige Verwendung in der Speicherhierarchie, für die Datenverarbeitung und verschiedene Anwendungsbereiche. Er wurde von einer Expertenarbeitsgruppe innerhalb des GI-/ITG-Fachausschusses ARCS (Architecture of Computing Systems) erarbeitet und im Juni 2019 fertiggestellt. Der Bericht kann unter https://fb-ti.gi.de/ heruntergeladen werden.

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FORUM

Gewissensbits – wie würden Sie urteilen?

Otto Obert[1] · Debora Weber-Wulff[2]

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Mit den ethischen Leitlinien der GI haben wir es uns zur Aufgabe gemacht, Diskurse zu ethischen Problemen der Informatik zu initiieren und zu fördern. Mitglieder der Fachgruppe „Informatik und Ethik" der GI stellen jeweils ein hypothetisches, aber realistisches Fallbeispiel vor, das zur Diskussion anregen soll. Die Fälle können jeweils von Interessierten im Blog der Fachgruppe auf der GI-Website https://gewissensbits.gi.de kommentiert und diskutiert werden.

Fallbeispiel: „Safety first"

Zeit: In der nahen Zukunft

Paula und Viktor arbeiten nach ihrem gemeinsamen Studium seit einigen Jahren schon bei der Flying-Bavaria, einem noch recht jungen und aufstrebenden Unternehmen im Bereich Transport- und Passagierdrohnen. Paula ist als Informatikerin in einem interdisziplinären Team eingesetzt, das sich unter Hochdruck mit der Weiterentwicklung von autonomen Flugtaxis hin zur Marktreife beschäftigt. Viktor hingegen ist als Assistent des Vertriebsvorstandes und Schnittstelle zum Produktmanagement eingesetzt. Paula und Viktor sind sehr eng befreundet, können sich alles anvertrauen und treffen sich regelmäßig mindestens einmal pro Woche beim Badminton-Betriebssport, wo auch viele andere Kollegen aus den verschiedensten Unternehmensbereichen mit Freude teilnehmen und sich alle am Rande gerne privat wie beruflich rege austauschen.

Flying-Bavaria und ihr Hauptkonkurrent Air-Hang haben vom Bundesministerium für Verkehr und digitale Infrastruktur einen Millionenzuschuss bekommen, um möglichst rasch Prototypen für Passagierdrohnen zu erstellen. Sie wollen möglichst kostengünstig und vor allem noch vor Air-Hang einen Zweisitzer herausbringen. Laut Prognosen wird dies das mit Abstand umsatzstärkste Marktsegment. Daher leuchtet es ein, dass Flying-Bavaria einfach ihr bei Transportdrohnen bereits verfügbares und bewährtes Antriebsaggregat im Prototypen verwenden soll. Was schon läuft, kann nicht ganz verkehrt sein.

Allerdings ist dieses Antriebsaggregat für die Anwendung in einem Zweisitzer leicht unterdimensioniert. Bei ungünstigen Wetterbedingungen neigt es zur Instabilität. Es werden einfach zwei Aggregate verwendet, die aber miteinander kommunizieren müssen. Daher wurde das Team um Paula damit beauftragt, ein *Maneuvering Characteristics Augmentation System* (MCAS) zur Koordinierung der Flugstabilisierung zu entwickeln. In der Marketingliteratur wird man den eingeführten Begriff „Künstliche Intelligenz" verwenden, obwohl das System nur auf Basis der Daten von Lagesensoren und mit dem Einsatz eines einfachen Algorithmus die Flugtauglichkeit garantieren soll. Schließlich sind es einfache, physikalische Zusammenhänge, die hier berechnet werden.

Paula empfindet das geplante Design des Zweisitzers als Fehlkonstruktion. Sie ist sich des Stellenwertes des MCAS bewusst und spricht mit ihrem Team über die Methodenauswahl. Im Scherz meint Andreas, ihr Maschinenbauingenieur: „Lasst uns doch KI einsetzen, wenn die Werbefritzen das schon schreiben!". Eigentlich ist das keine schlechte Idee, meint Paula, und das Team beginnt, sich mit überwachten, selbstlernenden Verfahren zu beschäftigen. Allerdings brauchen sie noch sehr viel Zeit, sich in das Thema einzuarbeiten und die Algorithmen entsprechend zu trainieren.

Da es noch kein System gibt, das die geeigneten Trainingsdaten erzeugen kann, bietet sich Andreas an, diese zu generieren. Er habe ja im Studium gelernt, wie gut man mit einfachen Prototypen Daten generieren kann und dann mit einem weitverbreiteten Rechensystem durchrechnen lässt, wie die Parameter zu setzen sind. Er macht sich mit Begeisterung auf den Weg, einige neue Modelldrohnen für seine Experimente zu kaufen. Er erstellt „Menschendummys" für die Drohnen, die vom Gewicht her passend sind,

✉ Debora Weber-Wulff
weberwu@HTW-Berlin.de

1 Karlstadt, Deutschland

2 HTW Berlin, Berlin, Deutschland

und zeichnet fleißig Flugdaten auf. Dann fragt er beim Team nach, das ein Einsitzermodell baut, ob sie nicht auch Daten haben, damit er überprüfen kann, ob er bei seinen Werten vielleicht einen Overfitting erzeugt hat oder nicht. Bei einer *true positive rate* von 97 % ist er schwer begeistert – es funktioniert!

Aber Paula ist nicht so glücklich über die Lerndatengenerierung. Es gab ein sehr komplexes Umrechnungsmodell, angepasst an die Architektur des Zweisitzermodells, mit den Daten aus dem Transport- und dem Einsitzermodell. Sie ist sich nicht sicher, ob Algorithmen aus Modellen und Einsitzer sich beliebig auf Zweisitzer anwenden lassen. Paula weiß nur zu gut um Verzerrungen (Bias) in den Lerndaten, die man nie ganz in den Griff bekommen kann.

Nach einem Badmintonabend sitzen Paula, Viktor und weitere Kollegen auf ein Bier zusammen und kommen auch auf das aktuelle Thema, den Zweisitzer, zu sprechen. Viktor berichtet aus einer Vorstandssitzung, in der aufgrund des erneut gestiegenen Zeitdrucks weitere Maßnahmen vorgestellt wurden, mit der Bitte um Erarbeitung der Für und Wider. In der nächsten Vorstandssitzung in ein paar Tagen soll dann nach Diskussion und Abwägung entsprechend beschlossen werden.

Unter anderen ist man im Vorstand aktuell der Überzeugung, bei der Meldung für die Zulassung an die Flugaufsichtsbehörde das MCAS als nicht systemrelevant einzustufen. Auch sollen alle Möglichkeiten genutzt werden, die Entwicklungszeit zu verkürzen, was auch Paulas Methodenentscheidung revidieren würde. Viktor berichtet aus Gesprächen mit dem Produktmanagement über technische Konsequenzen aus der Einstufung des MCAS als nicht systemrelevant. Dadurch könnte man nun formal auf die redundante Auslegung des Lagesensors verzichten, um wieder etwas Kosten und besonders Zeit zu sparen. Immerhin waren selbst ein paar Kollegen im Produktmanagement dagegen, sie haben es als unverantwortlich kommentiert. Paula hatte mit mindestens vier Lagesensoren gerechnet, ist insgesamt entsetzt und möchte handeln.

Fragen

1. Sollte Paula mit dem Vorstand über ihre Sorgen sprechen? Sie bringt damit Viktor in eine schwierige Situation, denn die Diskussionen im Vorstand sollen geheim bleiben, bis eine Entscheidung getroffen worden ist.
2. Sollte der Vorstand bei technischen Fragen nicht besser direkt mit den Fachabteilungen sprechen?
3. Hätten Paula und ihr Team bei der Methodenauswahl schon aus ethischen Gründen maschinelles Lernen ablehnen sollten?
4. Als Maschinenbauer hat Andreas nicht besonders viel Erfahrung mit den Algorithmen, die sein Rechensystem anwendet. Er vertraut einfach auf die Ergebnisse, die geliefert werden. Hat er die Verantwortung für die Ergebnisse oder das gesamte Team?
5. Ist eine *true positive rate* von 97 % bei Systemen, die Menschenleben tangieren, wirklich ein guter Wert?
6. Sollte Viktor seine Rollen als Direktreport zum Vertriebsvorstand und Schnittstellenkoordinator nutzen, um auf Vorstandsebene auf das Risiko des Gesamtkonstruktes aufmerksam zu machen?
7. Wie bewerten Sie die Art und Weise der Lerndatengenerierung?
8. Welche Einzelaspekte führen zu der Gesamtsituation und wie würden Sie diese vom Gewicht ihrer negativen Auswirkung aus betrachtet bewerten?
9. Welche Gegenmaßnahmen entsprechend dieser Einzelaspekte würden Sie vorschlagen? Wie beurteilen Sie deren Wirksamkeit, um die Gesamtsituation zu verbessern?

Die Fachgruppe ist unter http://www.fg-ie.gi.de/ erreichbar. Unser Buch „Gewissensbisse – Ethische Probleme der Informatik. Biometrie – Datenschutz – geistiges Eigentum“ ist im Oktober 2009 im Transkript-Verlag erschienen.

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FORUM

Incident Response und Recht

Ursula Sury[1]

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Cyberkrieg und Cryptotrojaner

Ein Mail kommt und wird vom Mitarbeiter geöffnet ... und schon ist es passiert. Dieser und viele andere Wege sind möglich, um auf ganz einfachem Weg Zugang zur IT eines Unternehmens zu bekommen und grundsätzlich zu schädigen. Schäden reichen von Verschlüsselungen verbunden mit Erpressungen über Datenklau und Datenzerstörung bis hin zur Übernahme von Steuerungen von Anlagen und Maschinen.

Insbesondere durch die immer weitere Einbindung von IOT und generell weitere Verknüpfungen mittels Digitalisierung werden die Angriffsflächen und somit die Risiken immer größer, Opfer einer Cyberattacke zu werden.

Incident Response und Governance

Unter Incident Response (Vorfallreaktion) versteht man die notwendige Reaktionsweise nach einer erfolgreichen Cyberattacke. Dazu benötigt man in aller Regel die Unterstützung durch eine spezialisierte Unternehmung. Es braucht technisches Spezialwissen, um Schäden zu beheben oder auch nur möglichst einzuschränken.

Solche Incidents möglichst zu verhindern, ist Aufgabe des Riskmanagements und der Informationssicherheit, organisatorisch bei den jeweiligen Verantwortlichen CISO (Chief Information Security Officer) zu verorten.

Im Rahmen der Unternehmensführung gehört die Vorwegnahme/Planung eines möglichen Incident Response zur Business Continuity Management (BCM) Planung.

✉ Ursula Sury
ursula.sury@hslu.ch

1 Luzern, Schweiz

Verantwortung für Incident Management

Wer für Unternehmensführung verantwortlich ist, muss mit sämtlichen Risiken, die damit verbunden sind, adäquat umgehen. In der digitalisierten Welt sind Risiken hauptsächlich rund um den Umgang mit Informationen zu finden. Diese Informationen werden eben über IOT-Schnittstellen übertragen, sind in Servern gelagert, über Rechner verbunden. Kurz und gut: Sie sind digital vorhanden.

Das bedeutet, dass der Verwaltungsrat und die Geschäftsleitung sich mit diesen Hauptrisiken und damit verbundenen Maßnahmen vornehmlich beschäftigen sollten. Es müsste somit alles daran gesetzt werden, das Eintreten der Risiken zu vermeiden und ein realistisches BCM zu planen.

Nach Eintritt eines Incident ist unbedingt auch die Arbeit und Sensibilisierung des Verwaltungsrates und der Geschäftsleitung zu überprüfen.

Rechtsverletzungen rund um Daten und Informationen

Die beschädigten oder gestohlenen Informationen bei einem Incident könnten verschiedene Rechtsaspekte und somit auch verschiedene Personen betreffen.

Es können Geheimnisse publik gemacht oder gestohlen werden. Zu denken ist hier zum Beispiel an Fabrikationsgeheimnisse, wie Rezepte oder Geschäftsgeheimnisse und Umsatzzahlen. Dies kann strafbar sein und es muss mit Strafuntersuchungen gerechnet werden, insbesondere falls nicht genügende Sicherheitsmaßnahmen ergriffen wurden. Waren angemessene Sicherheitsmaßnahmen installiert, ist aber nicht mit einer Verurteilung der verantwortlichen Führungspersonen zu rechnen.

Falls Personendaten betroffen sind, kann es notwendig sein, nach DSGVO oder nach dem kommenden Datenschutzgesetz der Schweiz eine Meldung (DSGVO Art. 33 und Art. 34) bei der Datenschutzaufsichtsstelle zu machen. Der Art. 33 der DSGVO (Datenschutzgrundverordnung) sieht vor, dass bei einer Verletzung des Schutzes von personenbezogenen Daten der Verantwortliche dies innerhalb von 72 h der zuständigen Aufsichtsbehörde mitzuteilen hat. Zur Erleichterung der Mitteilung haben die Aufsichtsbehörden umfangreiche Eingabemasken eingerichtet, die online bearbeitet werden können. Kommt es beim Betroffenen, Kunden oder Nutzer zu einer schwerwiegenden Verletzung, hat eine Information durch den Verantwortlichen gem. Art. 34 DSGVO sofort zu erfolgen. Bei einer entstandenen Datenpanne hat jeder Beteiligte zu wissen, an wen sich dieser im Ernstfall wenden muss. Schließlich handelt es sich um ihre persönlichen Informationen und somit Persönlichkeitsanteile, die durch die Malwareattacke betroffen und somit verletzt sind.

Auch hier ist mit möglichen Sanktionen als Konsequenz zu rechnen, wenn die Meldepflicht bei der Aufsichtsbehörde oder die Informationspflicht bei den Betroffenen durch die Verantwortlichen unterlassen wird.

Vererben der Probleme

Ist die Malware über Mail eingeschleppt worden, jedoch auch sonst, besteht die Gefahr, dass mögliche Außenkontakte, wie Kunden, Lieferanten, eigene Subunternehmen oder auch Dritte infiziert werden. Deshalb ist es auf dem Hintergrund der Schadenminderungspflicht wichtig, sofort zu reagieren und die Betroffenen zu informieren und nach Möglichkeit zu unterstützen, damit keine größeren Probleme entstehen.

Verhinderung an eigener Leistungsmöglichkeit und AGB

Wenn wegen des Informationsverlustes die eigene Leistungsmöglichkeit eingeschränkt oder unmöglich geworden ist, müssen die Kunden umgehend informiert werden. Da das Problem im Machtbereich des Lieferanten aufgetreten ist, ist zu klären, ob es sich um entschuldbare Verhinderung – analog höhere Gewalt – handelt, oder ob für Verspätungen oder Nichterfüllungen gehaftet werden muss. Grundsätzlich wird das Verschulden bei Unmöglichkeit vermutet. Hier wären sicher noch entsprechende Hinweise und Enthaltungen für Fahrlässigkeit in den Allgemeinen Geschäftsbedingungen von Vorteil.

Zusammenfassung

Unter Incident Response versteht man die Reaktion eines Unternehmens auf einen IT-Angriff. Dazu zählen alle organisatorischen und technischen Maßnahmen zur Abwehr und schnellen Eindämmung des Cyberangriffs, damit der Schaden möglichst gering bleibt. Digitale Angriffe sind heute so vielfältig wie die digitale Welt selbst. Dies macht die Ausarbeitung eines umfassenden Incident Response Plans für viele IT-Abteilungen auch zur Herausforderung. Mit einem ausgereiften und erprobten Incident Response Plan lassen sich der Erfolg der Angreifer und die Höhe des angerichteten Schadens jedoch minimieren.

Ursula Sury ist selbständige Rechtsanwältin in Luzern, Zug und Zürich (CH) und Vizedirektorin an der Hochschule Luzern – Informatik. Sie ist zudem Dozentin für Informatikrecht, Datenschutzrecht und Digitalisierungsrecht.

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FORUM

Einsichten eines Informatikers von geringem Verstande

Bär, Bulle, Dachs und Co

Reinhard Wilhelm[1]

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Das Schreiben dieser Glosse hat meinen Schreibstress auf ein Jahreshoch gebracht; ich habe die 200-Tage-Linie gerissen, keine Unterstützung nach unten gefunden, aber oben einen massiven Pull Back erlitten, alles deshalb, weil ich die Glosse unbedingt komplett in Börsendummdeutsch schreiben wollte. Leider musste ich feststellen, dass die Ausdrucksfähigkeit dieser Sprache beschränkt ist auf Fahrten nach Norden oder Süden oder auf Seitwärtsbewegungen – unklar, ob nach Ost oder nach West –, alle oft zögerlich; dann werden Linien getestet, Unterstützungen gesucht, aber Widerstände gefunden, im Rahmen von technischen Überschussbewegungen werden Verlaufshochs erreicht, aber im Verlauf wieder abgegeben, das ganze über kurz oder lang, genauer Short oder Long.

Liebe*r Leser*in, Sie vermuten richtig, in dieser Glosse geht es um den Einsatz von Rechnern im Börsengeschehen und bei der Verwaltung von Vermögen. Wenn man nicht genau hinhört, könnte man andererseits auch meinen, dass es um den Rechnereinsatz in unseren Zoos ginge. Denn es wird von Bär, Bulle, Dachs und Co geredet, und wie sie sich unter Rechnereinsatz verhalten.

Aber Spaß beiseite! Der Einsatz von Rechnern an den Börsen ist eine unglaubliche Erfolgsgeschichte. Das hat zu tun mit der Geschwindigkeit, mit der aktuelle Information produziert wird und mit der auf sie reagiert wird. Musste sich Herr Rothschild angeblich noch auf die Geschwindigkeit von Brieftauben verlassen, welche ihm den Ausgang der Schlacht bei Waterloo schneller meldeten als berittene Boten dem Rest der Welt, sodass er mit einem geschickten Täuschungsmanöver einen Riesengewinn machen konnte – er verkaufte britische Aktien, als ob Napoleon gewonnen hätte, und verleitete dadurch alle Welt dazu, ihm nachzueifern, um anschließend heimlich und billig massenhaft britische Aktien zu kaufen –, so sind vernetzte Rechner in der Lage, in Millisekunden Millicent-Preisunterschiede von Aktien an verschiedenen Börsen auszunutzen. In New York wurde sogar schon ein Rechenzentrum näher an die Wallstreet gerückt, um (vermutlich) im Mikrosekundenbereich Nanocent-Unterschiede auszunutzen. Aber wir wissen ja, Nanovieh macht auch Micromist.

Aber nicht nur im Hochgeschwindigkeitshandel werden Rechner eingesetzt. Börsianer können viele verschiedene Arten von Orders abgeben, welche bei Vorliegen einer Bedingung durch Rechner ausgeführt werden. Eine Stop-Loss-Order löst beim Unterschreiten des angegebenen Kurses eine Verkaufsaktion aus. Da Börsianer und auch Fondmanager Herdentiere sind, kann dies den Bär, der von Haus aus weder eine hohe Grundgeschwindigkeit noch gute Beschleunigungswerte hat, ganz ordentlich in Schwung bringen. Das kostete z. B. 2010 dem Standard & Poor's 500-Index auf einen Schlag 5 % Börsenwert. Ähnlich kriegt der Bulle eine beschleunigte Aufwärtsbewegung, sobald halbwegs verlässlich eine positive Tendenz sichtbar wird. Die ultrakurzen Reaktionszeiten der Rechner führen also zu starr gekoppelten Systemen, und wie schön starr gekoppelte Systeme Fehler fortpflanzen, kann ein Physiker leicht erläutern. Das erfährt auch der von Verspätungen geplagte Bahnfahrer, wenn er zum Beispiel mit einem schwer beladenem E-Bike verspätet in Mannheim ankommt, noch 3 min verbleibende Zeit hat, in dieser Zeit runter vom Bahnsteig in den Untergrund, wieder raus aus dem Untergrund rauf auf den Bahnsteig muss, um dort festzustellen, dass er am falschen Ende des Bahnsteigs heraufgekommen ist und ans andere Ende des Bahnsteigs muss.

Aber was wäre eine Glosse ohne tiefes Lernen? Geht doch gar nicht! Tiefes Lernen muss in jeder Glosse gelobt werden, weil es in letzter Zeit so beeindruckende Erfolge gezeigt hat. Voraussetzung für ziemlich alle Beispiele von erfolgreichem tiefen Lernen ist eine große Datenbasis, auf welcher man lernen kann, – für das Börsengeschehen leider nicht gegeben – und – das wird sich noch als wichtig her-

✉ Reinhard Wilhelm
wilhelm@cs.uni-saarland.de

[1] Saarland Informatics Campus, Saarbrücken, Deutschland

ausstellen – statische Verhältnisse im Lernbereich. Beides ist erfüllt für die Bilderkennung von Tumoren, von Tieren, Sportarten und Verkehrszeichen. Wann entdeckt man schon mal eine neue Krebsart? Ähnlich stabil ist die Lage bei Verkehrszeichen. Es gibt selten mal ein neues Verkehrszeichen. Na ja, bis auf das PKW-Maut-Kennzeichen. Kaum entworfen und bestellt, wurde es schon wieder aus dem Verkehr gezogen.

Das Problem der Marktdynamik sah schon der Vorsokratiker Heraklit voraus. Er stellte fest, *Panta rhei*, auf Deutsch, *Man investiert nicht zweimal in denselben Markt*. Tiefe Erkenntnisse, aus *einem* Markt gelernt und auf einen veränderten Markt angewendet, werden diesen im Allgemeinen nicht besonders gut voraussagen. Aber nicht nur das! Jede tief gelernte Erkenntnis, wenn erst einmal angewendet, verändert den Markt. Sobald sich die Erkenntnis verbreitet hat, löst sie einen Herdentrieb aus, alle wenden sie an, und sie produziert statt Gewinne eher Verluste. Das ist wie bei den Geheimtipps für Touristen, erst einmal per Bestseller bekannt gemacht, und schon wünscht man sich, dort angekommen, nach Malle oder Antalya.

Die Automatisierung der Vermögensverwaltung beruht auf der Identifizierung von *Faktoren*, welche einen relevanten Einfluss auf die Performance eines Unternehmens oder eines Marktes gehabt haben. Sind sie identifiziert, nutzt man sie zur Verbesserung der Voraussage. Da stellen sich gleich mehrere wichtige Fragen: Sind der Intelligenzquotient, der Narzissmus und das angespannte Verhältnis zur Wahrheit des amerikanischen Präsidenten oder britischer Spitzenpolitiker solche Faktoren? Schließlich haben sie in den vergangenen Jahren den größten Einfluss auf die weltweiten Aktienmärkte gehabt. Und wenn sie solche Faktoren sind, hätte ein tief gelerntes System sie identifizieren können? Dem steht entgegen, dass tief gelernte Systeme bisher nicht durch rassistisch verzerrte Ergebnisse gegen weiße alte Männer aufgefallen sind. Nein, ganz im Gegenteil!

Trotzdem ist zu beobachten, dass beim Umfang der rechnerverwalteten Bonds der Rechnereinsatz nur bullische Tendenzen kennt, keine Seitwärtsbewegung und als Haltelinie nur 100 % des Volumens.

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FORUM

Datentrickserei im Taxigewerbe – von Abzocke bis Steuerbetrug

Hans-J. Lenz[1]

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Jeder, der im Inland oder Ausland erstmals ein Taxi in einer ihm nicht bekannten Stadt oder Gegend benutzte, hat mit ziemlicher Sicherheit dieselbe Erfahrung gemacht: Er ist übers Ohr gehauen worden, also einem Abrechnungsbetrug „schwarzer Schafe" unterlegen. Dazu gehören dreiste Fantasiepreise wie Fahrtkosten in Höhe von 300 €/10 km, die einem ortsunkundigen Touristen im Jahr 2014 abgefordert wurden, siehe [1]. Zu solcher Abzocke rechnen weltweit unnötige Umwege zum Ziel zu nehmen oder nachts während der Fahrt vom Flughafen zum Hotel den vereinbarten Preis überraschenderweise schlicht zu verdoppeln und zu drohen, bei Nichtzahlung den Fahrgast, während ein dunkles Waldstück durchquert wird, unverzüglich samt Koffer an die frische Luft zu setzen, wie es dem Autor außerhalb von Sofia in den neunziger Jahren passierte.

Nimmt es da Wunder, dass entsprechende Witze im Volksmund grassieren, wie zum Beispiel der: Fahrgast: „Sind Sie frei?", Taxifahrer: „Ja, aber nur auf Bewährung!". Wer da glaubt, mit derartigen Tricksereien lassen sich alle typischen Mogeleien im Taxigewerbe beschreiben, irrt sich sehr. Die Spanne des Abrechnungsbetrugs im Taxigewerbe war und ist groß. Sie wird finanziell bedeutsam, wenn man gewerbesteuerliche, wettbewerbs- und sozialpolitische Sichten mit einbezieht. Dann kommen Polizei, Zoll, Steuer- und Sozialbehörde mit ins Spiel. Das beginnt mit strafrechtlichem Fehlverhalten betrügerischer Taxibetriebseigner, wenn die Einnahme aus einem Beförderungsfall in die schwarze Kasse wandert, was der Volksmund „Schwarzfahrerei" nennt, oft zu erkennen daran, dass der Fahrer keine Quittung ausstellt oder – bislang – keine Kredit- oder EC-Karten akzeptierte. Es geht weiter mit dem Fälschen der jährlichen operativen Betriebsdaten, wie u. a. durch Verschwindenlassen von Tankquittungen u. ä. In Großstädten rechnen dazu weiterhin Lohnbetrug und amoralische Ausnutzung gesetzlich-sozialer Spielräume und endet da, wo selbst Taxi-Konzessionen verkauft werden, die gesetzlich gar nicht verkauft werden dürften. Einen schnellen Überblick über die Bandbreite der Manipulationen gibt die unten stehende Abb. 1.

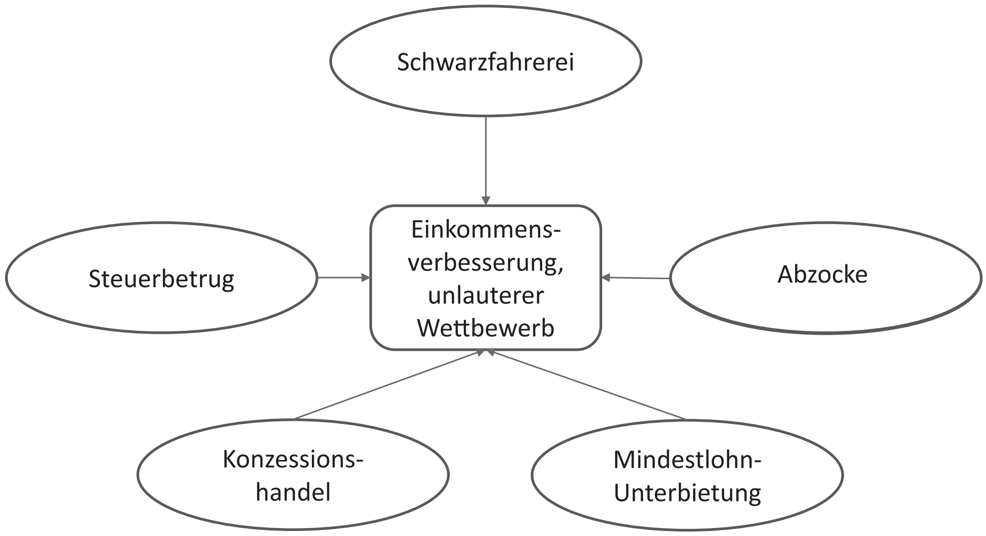
Beginnen wir mit der Schwarzfahrerei. Obwohl eine reguläre Personenbeförderung vorliegt, wird diese Fahrt nicht abgerechnet und damit als Leerfahrt eingestuft. Das erhaltene Entgelt wandert in die schwarze Kasse und erhöht die steuerfreien, illegalen Jahreseinkünfte des unredlichen Gewerbebetreibenden. Der Dumme dabei ist Vater Staat, also alle Bürger, aber auch die ehrliche Konkurrenz, die dadurch Wettbewerbsnachteile erleidet. Die Erfahrung in Großstädten wie Berlin, Frankfurt, Hamburg, Köln, München usw. in den letzten Jahren hat gezeigt, dass es nicht gelingt, diesen Abrechnungsbetrug den schwarzen Schafen mit herkömmlicher Abrechnungstechnik nachzuweisen. Selbst wenn man für eine Abrechnungsperiode wie ein Kalenderjahr die laut Tachometerstand ableitbare Jahresfahrleistung JFL km/a, den fahrzeugtypischen, regional topologisch unterschiedlichen Verbrauch V l/100 km und den mittleren Anteil q von Betriebsstrecke/(Betriebs- und Leerfahrtenstrecke) je nach Gemeinde oder Stadt kennen würde, um die abrechnungseffektiven Treibstoffkosten $JTK = q \times JFL \times V$ abzuschätzen, hätte das bis 2017 in etwa 40 % der Fälle nicht zum Ziel geführt [2]. Nicht nur waren die digitalen Tachometer manipuliert, sondern auch die fahrtproportionalen Betriebsdaten selbst. Dazu rechnen Kosten für Reifenwechsel, Reparaturen, Inspektionen, Treibstoff usw. Ein betrügerischer Taxibetrieb „fabriziert" eine individuelle Daten-Scheinwelt. „Fermi"-Abschätzungen mittels Modellgleichungen mit Fehlern in den Variablen sind nur dann brauchbar zur Aufdeckung von Manipulationen, wenn unter den Messwerten überwiegend solche ohne systematische Messverzerrung (Bias) und mit „normaler" Messgenauigkeit (Präzision) sind [3]. In einem betrügerischen Umfeld hätte es auch nicht geholfen, die Anzahl der Betriebsprüfer zu erhöhen. Sie hätten dasselbe Problem gehabt, den manipulierten Istwerten korrekte Sollwerte gegenüberzustellen.

Um den Betrügereien ein für alle Mal ein Ende zu bereiten, haben die Steuerbehörden der Bundesländer mit der gesetzlichen Vorschrift, in alle Taxis beginnend ab 2017 an-

✉ Hans-J. Lenz
Hans-J.Lenz@fu-berlin.de

1 Freie Universität Berlin, Berlin, Deutschland

Abb. 1 Datentricksereien im Taxigewerbe



lässlich der Neukonzessionierung Fiskaltaxameter auf Kosten des Konzessionärs einzubauen, das Problem an der Wurzel gepackt. Nach derzeitigen Erkenntnissen gelten diese als manipulationssicher. In Berlin werden in Kürze die letzten Taxis nachgerüstet werden, da mit der alle fünf Jahre fälligen Konzessionserneuerung der Einbau erzwungen wird. Damit ist Steuerbetrug auf diese Art unmöglich geworden, und auch Abzocke wird erschwert. Vielleicht sollten ortsunkundige Taxikunden auch mehr zu der App „taxi.eu" greifen, mit Google Maps die Länge der vermutlichen Strecke vorweg bestimmen oder auf ihren Smartphones einen Tracker mitlaufen lassen, um Fahrtkosten abzuschätzen.

Wenden wir uns nun dem Konzessionshandel zu, der u. a. in Köln im Jahr 2011 publik wurde [4]. Fest steht, dass ein gesetzliches Verbot besteht, eine Taxikonzession überhaupt zu verkaufen. Dies wurde in dem oben geschilderten Fall wie folgt umgangen. Der Verkauf wird als angebliche Unternehmensveräußerung getarnt, allerdings ohne Autoübergabe. Vor rund zehn Jahren lag der Anschaffungspreis bei rund € 80.000, 50 % wurden schwarz bezahlt. Der Vertrag über die Unternehmensveräußerung führt offiziell höchstens € 40.000 auf. Die Investition von € 40.000 sind Schwarzgeld und können nur über weitere Betrügereien reingeholt werden. Nehmen wir Folgendes für eine Großstadt an. Die monatlichen fixen Betriebskosten liegen bei € 2000. Es wird zum Schein ein Aushilfsfahrer eingestellt, der aber de facto nachts voll als Alleinfahrer eingesetzt wird und bei 60–70 h/Woche bis zu € 5000 Umsatzerlös monatlich erzielt. Er wird mit 2000 €/Monat entlohnt. Dem Betrieb verbleibt ein Schwarzgeldanteil zur Tilgung etwa in derselben Höhe, wobei zu beachten ist, dass die Umsatzerlöse manipuliert sind und Sozialabgaben eingespart werden.

Während der illegale Konzessionshandel einiger Betriebe ganz überwiegend ein „langfristig orientiertes Geschäftsmodell" darstellt, bieten sich im täglichen operativen Geschäft weitere Möglichkeiten der Datenmanipulation, die das Ziel illegitimer Gewinnerhöhung haben. Dazu werden Lücken in der Sozialgesetzgebung genutzt, eine Situation ähnlich zu den Steuerschlupflöchern und sog. Steuersparmodellen bei Weltkonzernen. Dies ist nicht illegal, aber verstößt gegen Sitte und Moral. Die Tricks sind denkbar einfach. Die Betriebe müssen nur die Hartz-IV- und Wohngeld-Regelung geschickt in Verbindung mit Halbtagsjobs bringen, beispielsweise als Nachtfahrer wie oben beschrieben. Die Mindestlohnregelung im Gewerbe kann dabei locker umgangen werden, indem die Fahrer über die Totmannregelung um Pausen geprellt werden, d. h. Wartezeiten einfach zu Lasten der Fahrer abgerechnet werden [4].

Wie schon beim bandenmäßigen Einkommensteuerbetrug und der Vermögensverlagerung in Steueroasen haben die deutschen Bundes- und Landesbehörden dem Treiben eines Teils des Taxigewerbes zu lange untätig zugesehen. Es war allerhöchste Zeit, die Pandorabüchse von Abrechnungsbetrug durch Schwarzfahrerei mittels Fiskaltaxametern zu schließen. Seit dem 1. Januar 2017 müssen die Berliner Taxibetriebe mit verstärkten Kontrollen insbesondere hinsichtlich der Ordnungsmäßigkeit ihrer Buchführung rechnen. Die Steuerverwaltung führt zusätzlich Umsatzsteuer-Nachschau und unangekündigte Kontrollen durch [5].

Warum allerdings geht das Treiben von einigen Taxiunternehmen bloß weiter, die Pausen- und Mindestlohnregelungen zu umgehen?

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REZENSION

Walter Hower: Informatik-Bausteine – Eine komprimierte Einführung

Springer Vieweg, Wiesbaden 2019, ISBN: 978-3-658-01279-3

David A. Plecher[1]

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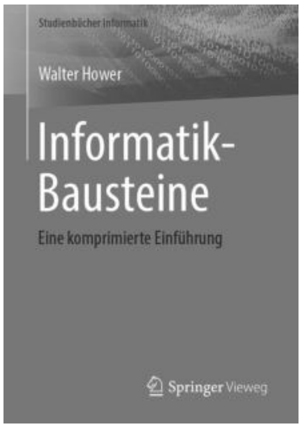

Nomen est omen – so beschäftigt sich der Inhalt des neuen Buches von Prof. Walter Hower mit den theoretischen und mathematischen Grundbausteinen der Informatik. Gerade diese Grundlagen sind nicht selten bei Studienanfängern gefürchtet und führen gerne zu Problemen. Diese zu lösen oder am besten von Beginn an zu vermeiden, hat sich der Autor zur Aufgabe gemacht.

Bereits die Widmung wendet sich direkt an den Leser und lässt diesen erkennen, dass es sich hier nicht um irgendein „trockenes" Lehrbuch handelt. Es ist stattdessen ein Buch, welches dem Leser auf lockere Weise, kompetent, aber mit Augenzwinkern bzw. Smileys, ein positives und bestärkendes Gefühl auf den Weg zur Meisterung der Bausteine mitgibt. Geschickt wird auch der interessierte Anfänger abgeholt und durch die Themen der Diskreten Mathematik, der Theoretische Informatik, der Algorithmik bis hin zu Einblicken in die Künstliche Intelligenz geführt.

✉ David A. Plecher
plecher@in.tum.de

[1] Lehrstuhl für Informatikanwendungen in der Medizin & Augmented Reality, Technische Universität München, München, Deutschland

Ohne zu viel zu verraten, möchte ich auf einige Highlights der einzelnen Kapitel hinweisen. Im Bereich Kombinatorik, genauer gesagt Ziehen mit Zurücklegen, wird als erklärendes Beispiel eine Situation des Alltags in einer Bar gewählt. Man beschäftigt sich hierbei natürlich nicht mit der Frage, was man bestellen möchte, sondern wie viele verschiedene Kombinationen aus Bier, Wasser, Wein, Saft, Sekt und Cocktail in einer Bestellung von bestimmten Gästen möglich sind. Die Lösung wird „fließend" erklärt und kann auf jedem Bierdeckel nachvollzogen werden. Den Abschluss des ersten Themenblocks bildet die Kryptologie. Hierbei wird der Leser mit dem RSA-Algorithmus und somit auch mit der Eulerschen φ-Funktion und dem kleinen Satz von Fermat konfrontiert oder – wie der Autor es ausdrückt – zum „Kielholen" geschickt.

Kap. 2 beschäftigt sich u. a. mit Komplexitätstheorie, Formalen Sprachen und Automatentheorie, welche am Ende in einem der Lieblingsthemata des Autors, nämlich der Unberechenbarkeit bzw. Unentscheidbarkeit, zusammengeführt werden. Es findet sich hier auch eine nachvollziehbare Beschreibung eines der Milleniumprobleme. Dieses wird auch für diejenigen nachvollziehbar dargestellt, welche P-NP bisher für die Passauer Neue Presse hielten.

In Kap. 3 über Algorithmik werden die klassischen Bauelemente Iteration und Rekursion erklärt. Es folgen daraufhin die Funktionsweise von Such- und Sortieralgorithmen sowie die Bestimmung der jeweiligen Komplexitäten. Dies wird am Beispiel des MergeSort-Algorithmus via Rekurrenz-Relation anschaulich gemacht.

Der Themenkomplex Künstliche Intelligenz bildet den abschließenden Block des Buches. Um z. B. autonomes Fahren zu ermöglichen, sind neben maschinellem Lernen natürlich auch Entscheidungen aufgrund riesiger Datenmengen in kurzer Zeit nötig. Dies ist unter Zuhilfenahme von Heuristiken, welche ausführlich am Beispiel der diskreten kombinatorischen Optimierung – dem Forschungsgebiet des Autors – beschrieben werden, möglich. Das Kapitel schließt wie die vorherigen mit beträchtlichen Hinweisen auf weiterführende Literatur.

Jeder, der Prof. Hower bei einem Gespräch oder einem Vortrag „live“ erlebt hat, erkennt sofort seinen Witz und seine mitreißende Art wieder, mit welchen er schwierige Zusammenhänge einfach und nachvollziehbar erklärt; denn Kompliziertes muss nicht kompliziert erklärt werden. Dieses Buch sei jedem/r Studienanfänger/in der Informatik empfohlen, denn es liefert eine schnelle und prägnante Einführung in die mathematischen bzw. theoretischen Grundlagen und somit die Bausteine für einen erfolgreichen Start ins Studium.

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REZENSION

Roger A. Grimes. Cryptography Apocalypse: Preparing for the Day When Quantum Computing Breaks Today's Crypto

John Wiley & Sons Inc., Hoboken, N.J., USA, 2020. ISBN 978-1-119-61819-5

Frank J. Furrer[1]

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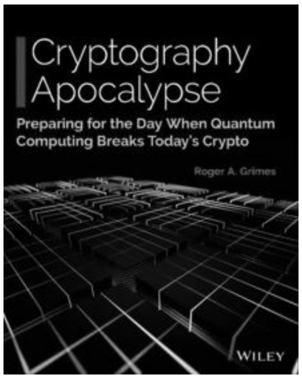

> *The post-quantum world will be full of weakened and utterly broken cryptography* (Roger A. Grimes)

Unsere heutige Informatikwelt beruht auf Kryptografie: Kryptografie ist eine kritische Infrastruktur geworden – wie die Stromversorgung, die Telekommunikation, die Bahnen usw. Sichere Kommunikation, zuverlässige Authentifizierung, rechtssichere Archive, alle Typen von Blockchains, elektronische Aktien, Kryptowährungen, PKI, digitale Zertifikate, Bluetooth, NFC – und viele mehr – beruhen auf *sicheren, kryptografischen Verfahren*. „Sicher" bedeutet hier, dass das unterliegende Verschlüsselungsverfahren nicht von einem Gegner (**ohne** Kenntnis des angewendeten Schlüssels) gebrochen werden kann. Die Wissenschaft vom Brechen von Verschlüsselungsverfahren heißt *Kryptoanalyse* und beruht auf mathematischen Erkenntnissen und großen Rechenleistungen.

✉ Frank J. Furrer
frank.j.furrer@bluewin.ch

1 Fakultät für Informatik, Technische Universität Dresden, Dresden, Deutschland

Zur Zeit existiert eine große Anzahl von „sicheren" kryptografischen Verfahren (vgl. Dietmar Wätjen [1]). Diese Verfahren beruhen auf mathematischen Konstrukten – wie z. B. die Primzahlzerlegung oder diskrete Logarithmen. Es ist (weitgehend) bewiesen, dass diese Verfahren auch mit brachialer Rechenleistung nicht in nützlicher Zeit gebrochen werden können. Also zur Zeit eine komfortable Situation.

Diese Lage wird sich in der Zukunft radikal ändern: Quantencomputer und Quantenalgorithmen werden viele der heute großflächig eingesetzten kryptografischen Verfahren unsicher werden lassen! Dies betrifft auch das RSA-Verfahren, diskrete Logarithmen und die elliptischen Kurven – auf denen heute fast die gesamte, elektronische Transaktionswelt beruht. Obwohl diese Entwicklung wahrscheinlich noch länger dauert (ein Jahrzehnt?) wird sie einen Tsunami über die gesamte ICT-Welt (Information, Communication, Technology) auslösen. Da passt der Buchtitel „Quantum-Apokalypse" tatsächlich.

> *When the forthcoming quantum crypto break happens, the world will change forever. There will be the world's history before and the world's incredible future after* (Roger A. Grimes)

Das Buch hat 248 Seiten, gegliedert in vier Hauptteile:

1. Eine *Einführung* in die Quantentheorie und Quantumrechner (Kap. 1 und 2);
2. *Analyse* der Auswirkung der Quantenalgorithmen auf die bestehenden kryptografischen Verfahren (Kap. 3–5);
3. „Quantum-resistente" kryptografische Verfahren (Kap. 6–8);
4. Eine *Anleitung*, wie sich die Welt auf den kommenden Krypto-Tsunami vorbereiten kann (Kap. 9).

Einführung in die Quantentheorie und Quantumrechner (Teil 1)

Dieser Teil ist eine Mischung aus Wissenschaftsgeschichte und Grundlagen: Hier werden die Überlagerung (*superposition*), der Beobachtungseffekt (*observer effect*) und die Verschränkung (*entanglement*) eingeführt. Dieser Teil ist wirklich gut und sorgfältig erklärt und der Autor versucht, ohne Mathematik auszukommen (Für eine schöne Erklärung des mathematischen Formalismus vgl. Chris Bernhardt [2]).

Als nächstes wird das grundlegende, technische Element der Quantencomputer präsentiert: das *Qubit* (=Quantenbit). Im Gegensatz zu klassischen Bits – die immer entweder 0 oder 1 sind – haben Qubit-Systeme mehrere, probabilistische Zustände: Zwei klassische Bits können nur exakt 4 Zustände annehmen: 00, 01, 10 und 11. Zwei Qubits aber können alle 4 Zustände gleichzeitig einnehmen. 20 Qubits können 1.048.576 Zustände abbilden. Die Rechenleistung mit Qubits nimmt dank der Superpositionsfähigkeit exponentiell zu! Dies ermöglicht die hoch-parallele Verarbeitung in Quantencomputern. Quantencomputer lassen sich in einer Vielzahl von physikalischen Formen implementieren: Der Autor erklärt und beurteilt einige Technologien.

Analyse der Auswirkung der Quantenalgorithmen auf die bestehenden kryptografischen Verfahren (Teil 2)

Ein Quantencomputer braucht – wie jede andere Rechenmaschine – *Algorithmen*, die durch *Software* implementiert werden. Quantenalgorithmen unterscheiden sich fundamental von klassischen Algorithmen und haben ein neues, faszinierendes *Software-Forschungsgebiet* begründet. Einer der ersten Quantencomputer-Algorithmen – aus dem Jahre 1994, bevor Quantenrechner überhaupt möglich schienen! – ist Peter W. Shor’s Paper [3]. Dieses Paper ist didaktisch und inhaltlich großartig und lohnt die Lektüre.

Anschließend analysiert das Buch die Wirkung von Quantenalgorithmen auf die heutigen Kryptoverfahren: Das Resultat ist sehr bedenklich. Alle Verfahren, die auf Primzahlfaktorisierung (z. B. RSA), diskreten Logarithmen oder elliptischen Kurven beruhen, sind mit Shor’s Algorithmus (oder Weiterentwicklungen davon) in Minuten zu brechen! Dadurch ist unsere gesamte *Krypto-Infrastruktur* (RSA, https:, TLS, ...) hochgradig gefährdet.

Weniger gefährdet sind die symmetrischen Chiffren und Hash-Verfahren (AES, SHA usw.): Hier verkürzen Quantenalgorithmen die Kryptoanalyse nur um 50 % – dies kann leicht mit längeren Schlüsseln aufgefangen werden. Der Autor präsentiert eine explizite, detaillierte und begründete Analyse der Quantumschwächen der heutigen Kryptoalgorithmen.

„Quantum-resistente“ kryptografische Verfahren (Teil 3)

Um RSA mit 2048 Bit Schlüssellänge zu brechen, braucht man 4099 stabile Qubits ($2 \times n + 3$). Um das 4096-Bit RSA System zu brechen, braucht man 8195 stabile Qubits. Verbesserungen des Shor-Algorithmus reduzieren diese Anzahl. Obwohl diese Anzahl Qubits heute (2019) noch bei Weitem nicht zur Verfügung steht, bleibt nicht mehr viel Zeit um eine Umstellung der Industrie zu planen und zu operationalisieren. Die US National Security Agency (NSA) hat bereits im Januar 2016 auf die nahende Gefahr aufmerksam gemacht (https://cryptome.org/2016/01/CNSA-Suite-and-Quantum-Computing-FAQ.pdf).

Glücklicherweise existieren heute *Quantum-resistente* Kryptoverfahren. Ein Quantum-resistentes Kryptoverfahren (oder: Post-Quantum-Verfahren) ist ein Verschlüsselungsverfahren, welches sowohl gegen klassische Attacken als auch gegen Quantum-Attacken resistent ist. Wie schon mehrfach in der Vergangenheit, ist in der westlichen Hemisphäre die NIST (US National Institute of Standards and Technology) federführend bei der Evaluation und Standardisierung der neuen Kryptoalgorithmen. In einer Ausschreibung wurden 2016 eine Anzahl vorgeschlagener Algorithmen für die Second Round Evaluation gewählt (https://nvlpubs.nist.gov/nistpubs/ir/2019/NIST.IR.8240.pdf). Es sind 17 Vorschläge für asymmetrische Kryptoverfahren und 9 Vorschläge für digitale Signaturen. NIST wird 2020/2021 einen neuen Kryptostandard definieren, welcher für die Post-Quantum-Welt sicher ist. Das Buch gibt einen summarischen Überblick über alle vorgeschlagenen Algorithmen – heavy, aber informativ!

Vorbereitung auf die Krypto-Apokalypse (Teil 4)

> *I believe there is a strong chance that not only will the crypto break happen in the next one to three years, but most of the world will be unprepared for it* (Roger A. Grimes, 2020)

Wann kommt die Krypto-Apokalypse? – Die Krypto-Apokalypse ist ziemlich nahe! Es fehlt nur noch an der *allgemeinen Verfügbarkeit* von genügend leistungsfähigen Quantenrechnern. Diese ist aber nicht mehr weit weg, z. B. bietet IBM Cloud-Quantum-Computing übers Internet an (https://www.ibm.com/quantum-computing/technology/experience/). Damit wird der „Quantenrechner für den Hausgebrauch“ baldige Realität ... und eine neue Art von Hacker kann gezielt gesicherte Verbindungen angreifen. Diese Tatsache begründet eine dringende *Verantwortung* unserer Community für die Vorbereitung/Abwehr der Krypto-Apokalypse.

Die wichtigste Voraussetzung für diesen erfolgreichen Umbau der Krypo-Infrastruktur ist ein internationaler Post-Quantum-*Kryptostandard*. Dieser ist vom NIST auf 2020/2021 angekündigt.

Der Autor empfiehlt als Vorbereitung auf die Krypto-Apokalypse *vier Phasen* (A–D):

A. Die jetzigen Verfahren *verstärken*: Dies geschieht durch die Erhöhung der Schlüssellängen (z. B. RSA mit 4096 Bit – benötigt 8195 stabile Qubits für die erfolgreiche Kryptoanalyse, oder 8192 Bit – benötigt 16.387 stabile Qubits, Verdoppelung der Schlüssellängen bei symmetrischen Verfahren). Allerdings gibt es dabei zwei Hindernisse: (a) die Rechenzeit verlängert sich stark – was bei gewissen Applikationen zu unakzeptablen Antwortzeiten führen kann, und (b) nicht alle Implementationen erlauben diese größeren Schlüssellängen.
B. *Quantum-resistente* Kryptoverfahren einführen: NIST führt derzeit eine Ausschreibung für Quantum-resistente Algorithmen durch. Die Evaluation sollte 2020/2021 abgeschlossen sein. Die Empfehlung von NIST wird zu weltweiten, neuen Kryptostandards führen, die als Grundlage für die Sicherung von PKI, TLS, https:, SSH usw. führen werden. Eine wertvolle Hilfe für diese Transition gibt https://openquantumsafe.org/. Hier wird allerdings eine konzertierte Zusammenarbeit aller Lieferanten und Anwender notwendig sein, um diese Herkulesaufgabe zu erledigen!

Die zwei weiteren Phasen sind eher visionär:

Don’t get caught remaining in a pure binary world as the world starts to go quantum (Roger A. Grimes, 2020)

C. *Quantum-hybride* Kryptoverfahren einführen: Dies ergibt eine Mischung von echten Quantum-basierten Verfahren (z. B. für die Schlüsselgenerierung und den Schlüsselaustausch) und klassischen Verfahren (für die Kanalverschlüsselung).
D. *Quantum-basierte* Kryptoverfahren einführen: Die Migration zu Quantum-Kommunikation (heute: im experimentellen Stadium) ist ein langfristiges Ziel (20 Jahre). Die Überlegungen des Autors dazu sind aber bedenkenswert.

Das Buch ist sehr präzise, gehaltvoll und in einem sehr persönlichen Stil (Ich-Form) geschrieben. Nach der Lektüre – und der geistigen Verarbeitung – des Inhalts weiß man so ziemlich alles über die Krypto-Apokalypse, ihre Folgen und die Antworten darauf.

Es liegt in der heutigen *Verantwortung* von Universitäten, Industrie und Politik, sich *rechtzeitig* auf diese disruptiven Herausforderungen einzulassen: Dieses Buch ist eine fundierte, lesenswerte Anleitung dazu!

Frank J. Furrer, Dezember 2019

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MITTEILUNGEN

Mitteilungen der Gesellschaft für Informatik 261. Folge

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Aus Vorstand und Präsidium

Wahlen zu Vorstand und Präsidium beendet: das Ergebnis

Am 6. Dezember 2019 endete die Wahl zu Vorstand und Präsidium der GI für die Wahlperiode 2020–2021 (Vorstand) und 2020–2022 (Präsidium).

Für die kommenden zwei Jahre wird die GI geführt von

- Prof. Dr. Hannes Federrath von der Universität Hamburg als Präsident sowie
- Dipl.-Inf. Alexander von Gernler (genua GmbH)
- Prof. Dr. Michael Goedicke (Universität Duisburg-Essen)
- Prof. Dr. Ulrike Lucke, Universität Potsdam als Vizepräsidentin, bzw. Vizepräsidenten.

Darüber hinaus sind sechs – zum Teil neue, zum Teil bereits erfahrene – Mitglieder gewählt. Für die Wahlperiode 2020–2022 kommen ins Präsidium:

- Prof. Dr. Nadine Bergner
- Prof. Dr. Ira Diethelm
- Dr. habil. Andrea Herrmann
- Dr. Judith Michael
- Prof. Dr. Simon Nestler
- Alexander Rabe

Insgesamt haben 2631 GI-Mitglieder gewählt. Darunter sind 23 Briefwahlstimmen. Die Wahlbeteiligung beträgt 16,82 %.

Wir danken allen Kandidatinnen und Kandidaten sowie den Kandidatenfindungskommissionen unter der Leitung von Prof. Dr. Arndt Bode (Vorstand) und Prof. Dr. Daniel F. Abawi (Präsidium), gratulieren den Gewählten und freuen uns auf die gemeinsame Arbeit.

Aufruf zur Kandidatur für das GI-Präsidium

Liebe GI-Mitglieder,

auf der Ordentlichen Mitgliederversammlung der GI am 24. September 2019 in Kassel wurde ich als Sprecherin der Kommission zur Findung von Kandidatinnen und Kandidaten für das Präsidium ab 2021 eingesetzt. In dieser Rolle und im Namen aller Mitglieder der Kommission bitte ich Sie, die Kommission in ihrer Arbeit zu unterstützen und mir Vorschläge für die Kandidatur bei den Wahlen zum Präsidium der GI zu nennen. Mit dem Jahr 2020 wird die Amtszeit von *drei* Präsidiumsmitgliedern auslaufen, deren Positionen neu zu besetzten sind.

Die von Ihnen vorgeschlagenen Personen müssen GI-Mitglieder sein und sollen besonderes Engagement für die GI und die Informatik zeigen. Sinnvolle Arbeit im Präsidium erfordert Zeit und den Wunsch, etwas für die Weiterentwicklung der GI zu bewegen. Daher sucht die Kommission Menschen, die bereit sind, sich über die Sitzungstage hinaus aktiv in die Präsidiumsarbeit einzubringen. Kenntnisse der GI-internen Strukturen und Abläufe sind hilfreich und wünschenswert, aber nicht zwingend.

Am Ende des Findungsprozesses soll die Liste das gesamte Spektrum der Informatik in Theorie und Praxis, von Forschung, Hochschulen und Industrie repräsentieren, denn das Präsidium spielt für die Weiterentwicklung der GI eine wesentliche Rolle. Deshalb sollten dort alle Mitgliedergruppen angemessen vertreten sein.

Bitte senden Sie Ihre Vorschläge formlos bis zum 30. April 2020 per E-Mail direkt an mich daniela.nicklas@uni-bamberg.de.

Außer Ihrem Namen und Ihrer eigenen GI-Nummer sollte Ihr Vorschlag Name und GI-Nummer der vorgeschlagenen Person sowie eine kurze Begründung Ihres Vorschlags enthalten. Bitte vergewissern Sie sich, dass die vorgeschlagene Person auch tatsächlich bereit ist zu kandidieren. Sie sollte ihre Zusage zur Kandidatur ebenfalls an zuvor genannte E-Mailadresse schicken.

Für die Ordentliche Mitgliederversammlung 2020 in Karlsruhe wird von der Kommission eine vorläufige Liste aus Ihren Vorschlägen erstellt und den versammelten Mitgliedern zur Ergänzung und Verabschiedung vorgelegt. Vielen Dank für Ihre Unterstützung!

Mit freundlichen Grüßen
gez. Prof. Dr. Daniela Nicklas
Sprecherin der Kandidatenfindungskommission, Dezember 2019

Ergebnisse der Vorstandssitzung vom 15. November 2019

Beschluss

Folgend Personalien werden bestätigt:
Nominationsausschuss für den GI-Dissertationspreis

- Prof. Kai Römer von der Uni Graz (Nachfolge Molitor)

Leitung des Beirats Wissenschaftlicher Nachwuchs

- Dr. Kerstin Lenk (Sprecherin)
- Dr. Mario Gleirscher (Stv. Sprecher)

Aus der Geschäftsstelle

GI-Junior-Fellows: Ausschreibung 2020 gestartet

Seit 2013 kürt die GI herausragende junge Leute zu GI-Junior-Fellows, die Besonderes für die und in der Informatik geleistet haben. GI-Junior-Fellows sitzen mittlerweile im Vorstand und im Präsidium, sind Chefredakteur des GI-Radars, Mitstreiter beim GI-Mentoringprogramm, Initiatoren von GI-Ministeriumsprojekten und und mischen die GI auf. Wir sind auf der Suche nach weiteren spannenden, jungen Leuten, die einerseits in ihrer Tätigkeit herausragend sind, sich aber auch in der GI ins Zeug legen wollen. Wir freuen uns auf viele Bewerbungen bis zum 15. Mai. https://gi.de/junior-fellows/ausschreibung-zum-gi-junior-fellowship/.

Personalia

Oliver Günther, Ralf Herrtwich, Katharina Morik und Günter Müller zu GI-Fellows ernannt

Mit dem GI-Fellowship zeichnet die Gesellschaft für Informatik e. V. (GI) jährlich Informatikerinnen und Informatiker aus, die sich in besonderer Weise um die Informatik und die GI verdient gemacht haben. Mit Prof. Oliver Günther, PhD, Prof. Dr. Ralf Herrtwich, Prof. Dr. Katharina Morik und Prof. Dr. Günter Müller würdigt die Gesellschaft für Informatik dieses Jahr gleich vier herausragende Persönlichkeiten. Die Ehrung fand in festlichem Rahmen auf der GI-Jahrestagung INFORMATIK 2019 in Kassel statt (Abb. 1).

Mit Oliver Günther ehrt die GI einen Wissenschaftler und Hochschullehrer, der in seiner beruflichen Laufbahn als Brückenbauer zwischen Wissenschaft, Wirtschaft und Politik wirkt. Er hat in der Informatik, der Wirtschaftsinformatik sowie der Geo- und Umweltinformatik herausragende Forschungsbeiträge geleistet. Mehrere seiner Forschungsarbeiten führten zu Unternehmensgründungen. Seine früheren Mitarbeiterinnen und Mitarbeiter sind als Hochschullehrer, Unternehmensgründer und Praktiker international wirksam. In der GI hat sich Oliver Günther durch seine Ämter als Finanzvorstand und Präsident ausgezeichnet. Zusätzlich vertritt er die GI im Stiftungsrat der Werner-von-Siemens-Stiftung.

Mit Ralf Herrtwich zeichnet die GI einen Wissenschaftler aus, dessen Arbeiten zur Vernetzung von Fahrzeugen und zur Entwicklung autonomer Fahrfunktionen auf der Basis künstlicher Intelligenz Pionierleistungen darstellen. Damit hat er, zum großen Teil auch in verantwortlicher Position, einen wesentlichen Beitrag zur Anwendung der Informatik in der Automobilindustrie geleistet. In der GI hat Ralf Herrtwich bei der Organisation von Tagungen, im Kuratorium von Schloss Dagstuhl – Leibniz-Zentrum für Informatik sowie in der Fachgruppe Kommunikation und verteilte Systeme mitgewirkt.

Mit Katharina Morik würdigt die GI eine Wissenschaftlerin, die als Pionierin das maschinelle Lernen mit einer Vielzahl grundlegender Beiträge auch international entscheidend vorangetrieben und beeinflusst hat. Ihre Arbeiten zeichnen sich insbesondere durch die Zusammenarbeit mit anderen Disziplinen, wie beispielsweise der Astronomie und der Medizin, und ihre hohe Anwendungsrelevanz aus. In der GI hat Katharina Morik die Fachgruppe Maschinelles Lernen gegründet und sich im Herausgebergremium des Informatik Spektrums engagiert. Hier lag ihr die Rubrik „Students Corner“ mit Berichten von Informatikstudierenden besonders am Herzen.

Mit Günter Müller ehrt die GI einen Wissenschaftler, der bereits in den 1980er und 1990er Jahren maßgeblich an der Entwicklung moderner Telekommunikationsinfrastrukturen beteiligt war und damit wesentlich zur Realisierung des Internet beigetragen hat. Beim Thema Sicherheit ist er mit dem Konzept „Mehrseitige Sicherheit“ richtungsweisend; die von ihm mit erstellten Schutzziele sind im bis heute geltenden ISO/IEC-Standard enthalten. In der GI hat er sich breit aufgestellt und in sich so verschiedenen Fachgliederungen wie „Kommunikation und verteilte Systeme“, „Informatik und Gesellschaft“, „Wirtschaftsinformatik“ und „Sicherheit – Schutz und Zuverlässigkeit“ nachhaltig engagiert.

Abb. 1 Die neuen GI-Fellows Günther, Müller, Morik und Herrtwich mit GI-Präsident Federrath (*von links*). (Foto: Nicolas Wefers)

Konrad-Zuse-Medaille: Dorothea Wagner erhielt höchste Informatik-Auszeichnung

Prof. Dr. Dorothea Wagner vom Karlsruher Institut für Technologie ist auf der INFORMATIK 2019 in Kassel für ihre Verdienste um die Informatik mit der Konrad-Zuse-Medaille der Gesellschaft für Informatik e. V. (GI) ausgezeichnet worden. Mit der alle zwei Jahre vergebenen Konrad-Zuse-Medaille zeichnet die Gesellschaft für Informatik seit 1987 herausragende Persönlichkeiten aus, die sich in besonderer Weise um die Informatik verdient gemacht haben. Mit Prof. Dr. Dorothea Wagner erhielt zum ersten Mal eine Frau die höchste Auszeichnung der deutschsprachigen Informatik-Community (Abb. 2).

Prof. Dr. Dorothea Wagner studierte Mathematik an der RWTH Aachen, wo sie 1986 auch promovierte. Nach ihrer Habilitation an der TU Berlin folgten Professuren an der Martin-Luther-Universität Halle-Wittenberg und der Universität Konstanz. Seit 2003 ist Dorothea Wagner Professorin für Informatik am KIT in Karlsruhe. Im Verlauf ihrer Karriere erhielt sie zahlreiche Ehrungen, darunter das GI-Fellowship (2008), den Google-Focused Research Award (2012) und die Werner Heisenberg-Medaille der Alexander von Humboldt Stiftung (2018). Im Jahr 2016 wurde sie darüber hinaus in die acatech – Deutsche Akademie der Technikwissenschaften gewählt.

Presse- und Öffentlichkeitsarbeit der GI

Internet Governance Forum 2019: 100 internationale Jugendliche erarbeiten ihre Forderungen an Politik und Unternehmen (28.11.2019)

Wem gehört das Internet? Und wie können wir es zum Wohle aller gemeinsam gestalten? Fragen zu Partizipation, Teilhabe und der ökonomischen, ökologischen und sozialen Nachhaltigkeit des Internets mobilisieren junge Menschen auf der ganzen Welt. Auch auf dem erstmalig in Deutschland stattfindenden Internet Governance Forum der Vereinten Nationen (IGF) und dem vorgelagerten Youth Summit spielten sie eine zentrale Rolle.

Im November 2019 trafen sich in Berlin 100 junge Menschen zwischen 18 und 30 Jahren aus fast 40 Ländern, die in der Netzpolitik tätig sind. Ziel des Gipfels war es, die Forderungen der Jugend an die Politik auszuarbeiten – die Positionen einer Generation zu digitalpolitischen Fragen rund um Bildung, Nachhaltigkeit, Grundrechte und öffentliches Gut. Daran wurde in den letzten drei Monaten im Rahmen von Webinaren mit mehr als 100 Beteiligten gearbeitet. Im Rahmen des Youth Summit IGF am Sonntag wurden schließlich 11 Forderungen final abgestimmt und vorgestellt.

GI kürt „10 KI-Newcomer*innen" im Wissenschaftsjahres 2019 (29.11.2019)

Die Gesellschaft für Informatik e. V. (GI) hat im Rahmen des Projektes #KI50: Künstliche Intelligenz in Deutschland

Abb. 2 Medaillenpreisträgerin Wagner und GI-Präsident Federrath. (Foto: Nicolas Wefer)

– gestern, heute, morgen 10 herausragende Newcomerinnen und Newcomer der deutschen KI-Forschung gekürt.

Im Rahmen des Projektes *#KI50: Künstliche Intelligenz in Deutschland – gestern, heute, morgen* im *Wissenschaftsjahr 2019 – Künstliche Intelligenz* des Bundesministeriums für Bildung und Forschung (BMBF) hat die Gesellschaft für Informatik 10 herausragende junge KI-Forscherinnen und Forscher ausgezeichnet. In den vergangenen Monaten wurden bereits die „10 prägenden Köpfe der KI", die „10 KI-Erfindungen", „10 popkulturelle Phänomene der KI" sowie vergangene Woche die „10 KI-Zukunftsfragen" gekürt.

Mit den KI-Newcomer*innen werden nun 10 junge KI-Nachwuchsforscherinnen und -forscher in fünf Kategorien geehrt, die bereits heute die KI-Entwicklung in Deutschland und darüber hinaus vorantreiben. Neben der Informatik stammen die Ausgezeichneten aus den Lebenswissenschaften, den Gesellschafts- und Sozialwissenschaften, den Naturwissenschaften sowie den Technik- und Ingenieurswissenschaften.

Die KI-Newcomer*innen wurden in einem offenen Online-Abstimmungsprozess ausgewählt, bei dem mehr als 11.000 Stimmen abgegeben wurden. Zuvor wurden die 30 Jungforscherinnen und -forscher bereits aus über 100 Bewerberinnen und Bewerbern in die engere Vorauswahl gewählt. Anschließend konnte die breite Öffentlichkeit sowie Mitglieder unterschiedlicher Forschungsgemeinschaften ihre Favoriten online wählen.

Den kompletten Text der Pressemitteilungen finden Sie unter https://gi.de/aktuelles/presse/.

Aus den GI-Gliederungen

Auszeichnungen der besten Abschlussarbeiten zum Thema Bildungstechnologien im Jahr 2019

Im Rahmen der DeLFI & GMW 2019 wurden am 18. September 2019 die besten Abschlussarbeiten zum Thema Bildungstechnologien des Jahres 2018 ausgezeichnet (Abb. 3).

Die Auszeichnung für die beste Bachelorarbeit ging an Herrn Elias John von der Humboldt-Universität zu Berlin (im Bild 3. v. l.) für seine Arbeit mit dem Titel „Serious Game basierter Ansatz als Hilfe für Programmieranfänger". Der Einsatz von Serious Games zur Motivationssteigerung ist zwar ein bekanntes, aber nach wie vor hochrelevantes Thema. Gerade beim Erlernen von Programmierkenntnissen verspricht die erhöhte Motivation einen leichten und niedrigschwelligen Zugang zum komplexen Thema. Herr John untersucht in seiner Bachelorarbeit, ob dieses Versprechen gehalten werden kann. Er untersucht dafür umfassend existierende Serious Games und stellt deren Schwächen aus Grundlagen etablierter Lerntheorien und didaktischer Konzepte dar. Diese Untersuchung führt ihn zur Entwicklung der eigenen Lernplattform PLAY & CODE – einem adaptiven Lernspiel zum Erlernen von Programmiersprachen. Nahezu vorbildlich durchläuft Herr John dabei sämtliche Entwicklungsphasen vom Design über die Implementierung bis hin zur Evaluation der Plattform. Als besondere Stärken der Arbeit haben die Gutachter die sehr gute Einbindung des aktuellen, interdisziplinären Forschungsstands, die gründliche Evaluation mit vielen Nutzern und den für eine Bachelorarbeit überaus umfangreichen Aufwand bei der Bearbeitung des Themas hervorgehoben. Insgesamt entstand so eine qualitativ herausragende und interdisziplinäre Arbeit, die das Forschungsfeld der Serious Games voranbringt.

Die Auszeichnung für die beste Masterarbeit ging an Herrn Kevin Duss von der Universität St. Gallen (im Bild 2. v. l.) für seine Arbeit mit dem Titel „Creation and evaluation of a computer-based formative assessment and feedback tool designed to support higher-level learning". Insbesondere Massenlehrveranstaltungen leiden unter unzureichenden Feedback-Möglichkeiten zu den Aktivitäten der Studierenden. Gerade hier können Informationstechnologien einen wichtigen Beitrag zur Verbesserung leisten. Die Masterarbeit von Herrn Kevin Duss nahm sich dieser Herausforderung an, indem sie die Entwicklung eines Tools für das Geben gewinnbringender Feedbacks in großen Vorlesungen zum Ziel hatte. Dabei griff Herr Duss auf die Methode des Action Design Research (ADR) zurück, und schuf auf Basis einer fundierten Literaturanalyse eine partizipativ und methodisch sauber entwickelte Feedback-Lösung, den „Learning Objective and Outcome Manager" (LOOM). Dieser wird abschließend umfassend evaluiert und diskutiert, wodurch schon eine hohe Praxisrelevanz ersichtlich

Abb. 3 Preisverleihung auf der DeLFI & GMW 2019 (*von links*: Raphael Zender, Kevin Duss, Elias John und Christoph Rensing). (Quelle: Katarzyna Biernacka, Humboldt-Universität zu Berlin)

wurde. Die Gutachter betonten vor allem die wissenschaftlich solide, fundierte Literaturanalyse sowie die sehr gute Methodik bei der Gestaltung des Feedback-Systems. Auch der Umfang der aus zwei Designzyklen bestehenden Arbeit ist bemerkenswert. Insgesamt hat Herr Duss eine herausragende wissenschaftliche Arbeit angefertigt, die dem Forschungsfeld der digitalen Feedback-Systeme neue, spannende Impulse gibt und dieses maßgeblich voranbringt.

Beide Abschlussarbeiten können Sie als PDF auf den Internetseiten der Fachgruppe Bildungstechnologien herunterladen: https://fg-bildungstechnologien.gi.de/nachwuchsfoerderung/beste-abschlussarbeit

GI-Fachgruppe Betriebssysteme verleiht Absolventenpreis 2019

Herr Tobias Landsberg ist beim Herbsttreffen der GI-Fachgruppe Betriebssysteme am 21./22. November 2019 an der Universität Osnabrück mit dem Absolventenpreis der Fachgruppe ausgezeichnet worden. Seine Arbeit *„Analyzing and Optimizing TLB-Induced Thread Migration Costs on Linux/ARM“* ist an der Leibniz Universität Hannover entstanden und überzeugte die Gutachter durch Umfang, Praxisbezug sowie technische Qualität. Die Arbeit widmet sich der Migration von Threads zwischen den Kernen moderner Multi-Core Prozessoren und beleuchtet Optimierungsstrategien unter Linux. Durch die weiter zunehmende Verbreitung von Mehrkernsystemen in vielen Anwendungsdomänen, stellt dieses Thema eine komplexe und noch immer weitgehend offene Forschungsfrage mit hoher Aktualität dar (Abb. 4).

Mentoring-Programm des GI-Beirats für den wissenschaftlichen Nachwuchs und der GI-Junior-Fellows: Mentorinnen und Mentoren sowie Mentees gesucht

Ob im Studium, im ersten Job, oder in der wissenschaftlichen Qualifikationsphase: ein guter Rat und eine zweite Meinung sind oft Gold wert! Um dem Nachwuchs in der Informatik eine Hilfestellung zur Erreichung der persönlichen Karriereziele zu bieten, wurde von einigen GI-Mitgliedern aus dem Beirat für den wissenschaftlichen Nachwuchs sowie einigen GI-Junior-Fellows ein Mentoring-Programm ins Leben gerufen.

Unsere Mentorinnen und Mentoren stehen anderen GI-Mitgliedern als Anlaufstelle zur Verfügung, mit einem offenen Ohr, als Anwältin oder Anwalt, Coach, positives Rollenmodell oder zur Unterstützung von Selbstwertgefühl und Selbstvertrauen. Das Mentoring ist dabei zunächst auf die Dauer eines Jahres angesetzt. Informationen und Anmeldung zum Mentoring-Programm finden Sie unter https://mentoring.gi.de/. Bei Fragen stehen wir per E-Mail (mentoring@gi.de) zur Verfügung.

Was erwartet Sie als Mentee? Studieren Sie aktuell einen informatiknahen Studiengang? Haben Sie gerade den Berufseinstieg in der Informatik hinter sich? Arbeiten Sie als Promovend oder Post-Doc in der Forschung? In diesem Fall können Sie als GI-Mitglied von dem Mentoring-Programm durch Unterstützung bei der Planung und Umsetzung konkreter Vorhaben und Karriereschritte profitieren. Gemeinsam mit Ihrer Mentorin oder Ihrem Mentor erarbeiten Sie einen Aktionsplan für den Mentoring-Zeitraum. Des Wei-

Abb. 4 Preisträger Tobias Landsberg (*in der Mitte*), Marcel Baunach (*links*) und Olaf Spinczyk (*rechts*). (Foto: Nils Aschenbruck, Uni Osnabrück)

teren entwickeln Sie Ziele und Umsetzungsstrategien. Ob es sich um die Begleitung im Studium oder Berufsleben handelt, das Aufzeigen von Wegen in die Praxis oder Wissenschaft oder die Beratung zu Karrierezielen – als Mentee können Sie nun auf den vertrauensvollen Rat einer erfahrenen Kollegin oder eines erfahrenen Kollegen zählen.

Mentorin oder Mentor werden. Die konkrete Ausgestaltung Ihres Engagements legen Sie selbst fest. Die Anzahl der gemeinsamen Treffen, gerne auch elektronisch, stimmen Sie direkt mit der beziehungsweise dem Mentee ab. Die GI lebt vom Engagement Ihrer Mitglieder. Über Ihre Unterstützung beim Anschub des Mentoring-Programms – gerne auch erst einmal nur zeitlich begrenzt – freuen wir uns daher sehr!

Neues Sprecherteam im Fachbereich Technische Informatik

Auf der Sitzung des Leitungsgremiums des Fachbereichs Technische Informatik am 18. Oktober 2019 in Frankfurt/ Main fanden turnusmäßig die Wahlen zum Sprecher und stellvertretenden Sprecher statt. Als neuer Sprecher des Fachbereichs Technische Informatik wurde Prof. Dr. Wolfgang Karl (Karlsruher Institut für Technologie, KIT) und als stellvertretender Sprecher Prof. Dr. Uwe Brinkschulte (Goethe-Universität Frankfurt) gewählt.

Ebenfalls wurde vom Leitungsgremium des GI/ITG Fachausschusses ARCS, dessen Sitzung zusammen mit der des Fachbereichs stattfand, als Sprecher Prof. Dr. Christian Hochberger (TU Darmstadt) und als stellvertretender Sprecher Prof. Dr. Sven Tomforde (Universität Kiel) gewählt.

Neuer Report zum Thema Memristor aus dem Fachbereich Technische Informatik

Fachleute aus verschiedenen Gliederungen des Fachbereichs Technische Informatik haben einen Report zum Thema „Implications of Memristor Technologies for Future Computing Systems“ erarbeitet, der im Juni 2020 unter https://fb-ti.gi.de/fileadmin/FB/TI/user_upload/Memristor_Report-2019-06-27.pdf erschienen ist. Der Report diskutiert die Bedeutung der Memristor-Technologie für zukünftige Rechensysteme. Ausgehend von einer Einführung in die technologischen Grundlagen dieser neuen nichtflüchtigen Speichertechnologie werden Memristors in der Sprecherhierarchie, Memristors in Near- and In-Memory Computing, der Einfluss von Memristoren auf Sicherheit und Privatheit diskutiert und die sich ergebenden neuen Forschungsthemen und Möglichkeiten aufgezeigt.

Tagungsankündigungen

ARCS 2020 in Aachen

Vom 25.05.–28.05.2020 findet in Aachen die 33. GI/ITG-Konferenz „Architecture of Computing Systems (ARCS 2020)“ statt. Die Konferenzreihe blickt auf eine über 30-jährige Tradition zurück und präsentiert neue Forschungsergebnisse auf den Gebieten der Rechnerarchitektur und Betriebssysteme. Der Konferenz sind Workshops der GI/ITG sind Workshops der Fachgruppen des Fachbereichs Technische Informatik angegliedert. Nähere Informationen findet man unter: http://arcs2020.itec.kit.edu/.

INFORMATIK 2020 – Ab nach Karlsruhe!

Wir laden Sie herzlich ein zur Jahrestagung INFORMATIK 2020, die vom 29. September bis 1. Oktober 2020 in Karlsruhe stattfindet.

Da es die 50. Jahrestagung ist, lautet unser Motto „Back to the Future“. Zum einen werden wir einen Rückblick auf die vergangenen Jahrestagungen wagen, wie aber auch eine Diskussion gegenwärtiger und zukünftiger Trends.

Das Programm ist in verschiedene Tracks eingeteilt und beleuchtet aktuelle Themen: von Anwendungen wie Robotik und autonomes Fahren bis hin zu Kernthemen wie Künstliche Intelligenz und Softwaretechnik.

Weitere Bestandteile des vielfältigen Programms sind Keynotes, mehrere Workshops, der Firmendialog inklusive Ausstellung sowie festliche Abendveranstaltungen im ZKM.

Informationen zur Tagung und Anmeldung finden Sie unter www.informatik2020.de.

Abb. 5 Das Panel „KI mit den Menschen – KI für die Menschen" mit den Teilnehmern (*v. l. n. r.*) Albrecht Schmidt, Dagmar Monett, Susanne Boll und Daniel Buschek. (Foto: Valentin Heller | Fraunhofer-Verbund IUK-Technologie)

Tagungsberichte

MCI-Symposium in Berlin

„Künstliche Intelligenz" wird die Art und Weise, wie wir arbeiten und leben grundlegend verändern. Die Forschung in der Informatik und den Ingenieurwissenschaften entwickelt zunehmend KI-basierte Systeme wie bspw. (humanoide) Roboter, Sprachassistenten und -schnittstellen, die auf „natürliche" Weise mit Menschen interagieren. Doch wie müssen diese Interaktionen gestaltet werden, dass sie funktionieren, akzeptiert werden und im Dienst der Menschen stehen? Das Symposium „KI für den Menschen" des Fachbereichs Mensch-Computer-Interaktion (MCI) der Gesellschaft für Informatik (GI) am 12. November 2019 in Berlin hatte zum Ziel zu zeigen, wie wichtig die Gestaltung der Benutzungsschnittstellen insbesondere für KI-Lösungen ist, und darüber zu diskutieren, wie KI-Systeme für den Menschen gestaltet werden müssen. Die Veranstaltung zog weit über 100 Teilnehmerinnen und Teilnehmer an und fand mit Unterstützung des Fraunhofer-Verbundes IUK-Technologie statt (Abb. 5).

Zum Einstieg berichtete Frau Prof. Dr. Dagmar Monett von der Hochschule für Wirtschaft und Recht Berlin darüber, wie sich die Definitionen von Intelligenz – beginnend bei Taine 1870 – und Künstlicher Intelligenz entwickelt haben, und wie divers verschiedene Definitionsversuche heute noch sind. Frau Leonie Beining von der Stiftung Neue Verantwortung ging im folgenden Impulsvortrag darauf ein, welche Probleme es dabei gibt, Algorithmische Entscheidungen für alle verständlich zu machen. Sie betonte dabei auch die Bedeutung und die Breite des Feldes, das weit über die reine Schaffung von Transparenz über das Zustandekommen der Entscheidungen hinausgeht. Prof. Dr. Michael Koch, Sprecher des Fachbereichs Mensch-Computer-Interaktion der Gesellschaft für Informatik (GI), führte aus Sicht des Fachbereichs aus, dass „KI ein MCI-Problem hat". Er machte dies an zwei bisher kaum erfüllten Forderungen (an komplexe lernende algorithmischen Systeme) fest: Transparenz und Erklärbarkeit der Ergebnisse/Vorschläge sowie Steuerbarkeit und Explorierbarkeit. Dies alles steht im Zusammenhang mit der Frage der Aufgabenteilung zwischen Mensch und technischem System, welche sich im Kontext von KI-Techniken auf neue Weise stellt, da zunehmend komplexere, geistige Aufgaben automatisiert werden können. Den Abschluss der Impulsvorträge lieferte Frau Ute Bernhardt, Leiterin des Referats „Künstliche Intelligenz" im Bundesministerium für Bildung und Forschung (BMBF), mit einem kurzen Überblick über die Nationale KI-Strategie mit vielen Bemerkungen zur konkreten Umsetzung im und durch das BMBF.

Im Panel wurden verschiedene Themen adressiert – von der Notwendigkeit oder Nicht-Notwendigkeit zu experimentieren bis zu fast technischen Aussagen wie: „Wenn ein Unternehmen Probleme hat Software zu erstellen, dann hat es erst recht Probleme, KI-Methoden in seine Produkte einzubringen".

Am Rande des Symposiums wurde die Neugründung einer Fachgruppe „Nutzerzentrierte Künstliche Intelligenz" im Fachbereich MCI bekannt gegeben. Gründungssprecher der neuen Fachgruppe ist Prof. Dr. Jürgen Ziegler von der Universität Duisburg-Essen, der sich über Kontaktaufnahme von Interessierten am Thema freut.

Gemeinsames Herbsttreffen des Fachbereichs SYS in Osnabrück im November 2019

Nach der erfolgreichen ersten Auflage im Frühjahr 2018 trafen sich auch in diesem Jahr die beiden Fachgruppen „Betriebssysteme" (BS) und „Kommunikations- und verteilte Systeme" (KUVS) des Fachbereichs SYS bei einer gemeinsamen Tagung. Insgesamt fanden knapp 70 Teilnehmer (teils mühsam) ihren Weg in den schönen Tagungsraum im Osnabrücker Botanischen Garten (Abb. 6).

Das Programm stand unter dem Motto „Von robusten Systemen zu robusten Netzen", verband also die Arbeiten beider Fachgruppen. Nach einem eingeladenen Vortrag von Prof. Georg Carle (TU München) zum Thema „Predictable High-Performance Networked Systems" folgten drei Vortragssitzungen zu den Themenkomplexen „Zuverlässige Systeme", „Zuverlässige Netze und verteilte Systeme" sowie eine themenoffene Sitzung mit aktuellen Arbeiten aus den Fachgruppen. Insgesamt wurden 13 Vorträge mit spannenden Themen präsentiert, die teils kontrovers diskutiert wurden. Das vollständige Programm inkl. Zusammenfassungen und Vortragsfolien findet man unter folgender

Abb. 6 Tagungsort im Bohnenkamphaus des Botanischen Gartens der Universität Osnabrück. (Foto: Nils Aschenbruck (Uni Osnabrück)

URL: https://www.betriebssysteme.org/aktivitaeten/treffen/2019-osnabrueck/programm/.

Zu vier Beiträgen gibt es zudem Papiere mit weiteren Details in der digitalen Bibliothek der GI (dl.gi.de). Die Publikation von Papieren ist ein Novum bei den Treffen unserer Fachgruppe. Das Interesse an diesem Angebot bei kommenden Treffen bleibt abzuwarten.

Abgerundet wurde die Tagung durch das Leitungsgremiumstreffen der Fachgruppe KUVS, die Mitgliederversammlung der Fachgruppe BS und die Verleihung des Absolventenpreises der Fachgruppe BS.

Wie üblich wurde für die Teilnehmer auch ein unterhaltsames Abendprogramm organisiert. Es begann mit einem Empfang durch die Bürgermeisterin im Friedenssaal des Osnabrücker Rathauses. Dort wurde im Jahr 1648 der Westfälische Friede ausgehandelt und somit der 30-jährige Krieg beendet. Danach ging es in die gemütlichen Räumlichkeiten des „Brauhaus Rampendahl“, wo ein schmackhaftes Buffet serviert wurde. Mit Bier zum selber Zapfen und anregenden Gesprächen klang der Abend angenehm aus.

Bundesweite Informatikwettbewerbe

Informatik-Biber 2019 verzeichnet Rekordteilnahme

So viele Teilnehmende gab es beim Informatik-Biber noch nie: 401.737 Schülerinnen und Schüler waren 2019 dabei. Dank der hohen Teilnehmerzahl ist der Informatik-Biber das Projekt mit der größten Reichweite im Bereich der Digitalen Bildung in Deutschland.

Wer ist im sozialen Netzwerk „Teenigram“ ein Superstar? Kann man aus den komprimierten Aufzeichnungen einer Überwachungskamera Rückschlüsse auf die beobachteten Ereignisse ziehen? Beim Informatik-Biber setzen sich Schülerinnen und Schüler mit altersgerechten informatischen Fragestellungen spielerisch auseinander. Die insgesamt 35 Aufgaben stammen aus 18 Ländern, darunter Indien, Vietnam, Thailand, Japan, Korea, Ukraine, Irland, Slowenien, Litauen und Belgien. Die Bearbeitung der Aufgaben ist vielfach interaktiv.

Gegenüber dem Vorjahr verzeichnet das Online-Quiz eine Steigerung um mehr als 28.000 Teilnehmerinnen und Teilnehmer, bzw. um mehr als 7 %. In der Jahrgangsstufen 3 und 4 liegt die Steigerung sogar bei rund 34 %, hier waren rund 12.000 Kinder dabei. Der Mädchenanteil liegt mit 177.370 Teilnehmerinnen bei rund 45 % (Abb. 7).

Die Zahlen des Informatik-Bibers belegen das große Interesse am Thema Informatik in den Schulen. Allein acht Schulen waren mit über 1000 Schülerinnen und Schülern dabei. Stärkster Teilnahmetag war der erste Donnerstag mit rund 53.000 Teilnahmen, davon knapp 10.000 Kinder und Jugendliche zwischen 8 und 9 Uhr.

GI-Veranstaltungskalender

18.02.2020–22.02.2020 – Furtwangen

meccanica feminale 2020
https://scientifica.de/bildungsangebote/meccanica-feminale/meccanica-feminale-call-for-lectures/

Abb. 7 Biber-Begeisterte. (Foto: BWINF)

19.02.20–19.02.20 – Stuttgart

7 Languages in 7 Months: Rust
JUGS
https://www.jugs.org/va2020/02-19.html

07.03.2020–07.03.2020 – Rostock

Landestagung der Informatiklehrer/innen MV
LT-MV2020
https://gi-ibmv.de/

17.03.2020–20.03.2020 – Göttingen

Jahrestagung des Fachbereichs Sicherheit Schutz und Zuverlässigkeit
GI Sicherheit 2020
https://www.uni-goettingen.de/de/603140.html

25.03.2020–26.03.2020 – Bonn

Agile Release & Change Management 2020
Agile Release 2020
https://www.agile-release-insights.com/

08.06.2020–10.06.2020 – Berlin

Languages & the Media 2020 13th International Conference on Language Transfer in Audiovisual Media
LM2020
https://www.languages-media.com/

18.06.2020–19.06.2020 – Kloster Seeon

European Conference of Software Engineering Education
ECSEE 2020
http://www.ecsee.eu

15.08.20–19.08.20 – Potsdam

Informatik-Sommercamp für Schülerinnen und Schüler
https://hpi.de/open-campus/schuelerakademie/schuelercamps/hpi-sommercamp.html

06.09.2020–09.09.2020 – Magdeburg

Tagung Mensch und Computer 2020
MuC2020
https://muc2020.mensch-und-computer.de

07.09.20–11.09.20 – Leipzig

Digital Operating Room Summer School
DORS
https://www.iccas.de/en/dors/

29.09.2020–01.10.2020 – Karlsruhe

INFORMATIK 2020: Jahrestagung der Gesellschaft für Informatik
INFORMATIK 2020
https://informatik2020.de/

08.03.21–12.03.21 – Dresden

19. Fachtagung für „Datenbanksysteme für Business, Technologie und Web“
BTW 2021
http://btw-konferenz.de/2021

08.09.21–10.09.21 – Wuppertal

GI-Fachtagung Informatik und Schule – Lehrerbildung
INFOS 2021
http://www.infos2021.de/